

Complex Matrix Multiplication Below 2.258 and Rectangular Bounds

OpenAI

September 24, 2026

Abstract

Over every field of characteristic zero, we prove that the square matrix-multiplication exponent satisfies $\omega < 2.258$, the dual exponent satisfies $\alpha > 0.465$, and $\omega(1, 0.709, 1) < 2.092$. The strict square and $k = 0.709$ rectangular bounds also hold over every field except possibly in one finite set of positive characteristics, in the arithmetic-operation model.

Contents

1	Introduction	3
1.1	Context and main result	3
1.2	Separating labels with an auxiliary tensor	4
1.3	The square bound: finite constructions and comparison	5
1.4	Rectangular bounds and finite-family optimization	6
1.5	Organization	6
2	Tensors and arithmetic exponents	7
3	Finite tensor constructions	9
3.1	Pairing conventions and the input rank	9
3.2	Supplying a label to its missing party	10
3.3	Exact types and a finite hierarchy of labels	13
3.4	An explicit four-component construction	14
3.5	Completion and the 75-component bound	15
3.6	Parity and conditional W constructions	16
3.7	Finite recipes and weighted laws	17
3.8	Physical orientations and canonical laws	18
3.9	Resolving and composing nested recipes	19
4	Banked constructions and attainable values	21
4.1	Canonical banks and preservation of unresolved tensors	21
4.2	Finite recipes and the closure of their values	22
4.3	Conditional pooling and monotonicity in the formal parameter	25
4.4	A direct matching lower bound	28
4.5	The five-component face	30

5	Arithmetic-progression restrictions and differential comparison	31
5.1	Locally readable target windows	32
5.2	Exact finite compilation	32
5.3	A lower test supplies inexpensive small drifts	35
5.4	A control that retains every path	35
5.5	The contradiction and the order of limits	38
5.6	Controlling the supporting-plane term	40
6	The three-state boundary seed	40
6.1	The affine chart and the baseline	41
6.2	A polynomial improvement and its domain	41
6.3	What is certified by the finite library	43
6.4	Arithmetic errors and exact distributions	44
6.5	The finite checks and the continuum conclusion	46
7	A rational certificate for the comparison bound	47
7.1	The chart and its exact coefficients	48
7.2	Uniform residual and polynomial comparisons	49
7.3	Geometry of the domain	51
7.4	The barrier and the interior inequality	52
7.5	The six-component boundary inequality	55
7.6	The positive inset and the comparison argument	59
8	From the certificate to the exponent bound	61
9	Conditional labels and entropy	61
10	Recursive completion trees	63
10.1	Two starting tensors	63
10.2	Completion and termination	64
11	An exact finite-family optimization	65
12	Three parameter certificates	67
12.1	A dual-exponent witness	67
12.2	A one-parameter rectangular curve	70
12.3	A finite-family square witness	71
12.4	Elementary interval verification	71
13	Comparison with ordinary combinations	72
13.1	A separator preserved by the baseline operations	73
13.2	The strict improvement interval	74
13.3	A sufficient test for further old estimates	75
14	Finite-exception characteristic transfer	76
A	Complete barrier coefficients	78
B	Reproducing the finite certificate	79

1 Introduction

Matrix multiplication asks for the bilinear map $(A, B) \mapsto AB$, where the two input matrices have compatible sizes. Its arithmetic complexity measures the number of field operations needed to compute all entries of the product. Except in the field-transfer results of Section 14, the field is $\mathbb{C}$, and the constants in an algorithm may belong to $\mathbb{C}$. For $0 \leq k \leq 1$, let

$$w(k) = \omega(1, k, 1)$$

be the infimum of exponents for multiplying an $n \times \lceil n^k \rceil$ matrix by a $\lceil n^k \rceil \times n$ matrix. The bound $w(k) \leq \tau$ means that for every $\varepsilon > 0$ these products can be computed using $O(n^{\tau+\varepsilon})$ arithmetic operations. The square exponent is $\omega = w(1)$, and the dual exponent is

$$\alpha = \sup\{k \in [0, 1] : w(k) = 2\}.$$

The output size gives $w(k) \geq 2$. Thus α measures how far the inner dimension can grow while multiplication retains the exponent of its output size. Square and rectangular multiplication are basic operations in algebraic algorithms; controlling the rectangular shape can matter even when a square bound is already available.

1.1 Context and main result

Strassen’s rank-seven algorithm for 2×2 multiplication gave $\omega \leq \log_2 7$, establishing subcubic arithmetic complexity [28]. Approximate bilinear algorithms and their conversion to exact algorithms [5], Schönhage’s asymptotic sum inequality [24], and Strassen’s laser method [30] made tensor powers and degenerations central to subsequent advances. Coppersmith and Winograd combined low-border-rank tensors with arithmetic-progression constructions [10]. Their extraction method restricts powers of a small tensor to direct sums of matrix products, whose dimensions enter the asymptotic sum inequality; their tensor-square analysis gave $\omega < 2.375477$ [10, Section 8]. Higher powers and more detailed analyses of these tensors led to the improvements of Stothers, Vassilevska Williams, and Le Gall [18, 27, 32].

Rectangular multiplication requires separate control of the three matrix dimensions. Early work of Coppersmith and of Lotti and Romani developed this direction [8, 20]; later constructions of Coppersmith, Le Gall, and Le Gall and Urrutia improved the available rectangular and dual bounds [9, 17, 19]. Recent advances include the refined laser method of Alman and Vassilevska Williams [1], the asymmetric hashing of Duan, Wu, and Zhou [14], and the square and rectangular analyses of Vassilevska Williams, Xu, Xu, and Zhou [33] and Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [2]. The four-author work establishes $\alpha \geq 0.321334$; these two papers also give detailed tables of rectangular estimates. Dupont et al. subsequently optimized the asymmetric framework at a larger scale to obtain $\omega < 2.371177$ [15], a bound incorporated into the updated table of [2].

The common extraction problem is to separate many candidate products that share variables, while accounting for the cost of every operation used to separate them. We adjoin auxiliary group tensors that both resolve shared labels and supply useful matrix-product factors. Their full rank cost enters the resulting inequalities. This construction gives the following bounds.

Theorem 1.1. *Over $\mathbb{C}$, the arithmetic matrix-multiplication exponents satisfy*

$$\omega < \frac{1129}{500} = 2.258, \quad \alpha > 0.465, \quad w(0.709) < 2.092.$$

Corollary 14.1 transfers the strict square and $k = 0.709$ inequalities to every field of characteristic zero or outside one finite set of positive characteristics, in the same arithmetic-operation model. It selects two fixed exact rank schemes, realizes their coefficients over one number field, and specializes their polynomial identities. The exceptional primes are not computed. Corollary 14.2 also gives the same dual-exponent bound over every field of characteristic zero. No dual-exponent transfer in positive characteristic is asserted.

The square bound also lowers the arithmetic exponent of basic linear-algebra tasks. Given $A \in \mathbb{C}^{n \times n}$ and $b \in \mathbb{C}^n$, one can compute $\det A$ and, when A is nonsingular, A^{-1} and the solution of $Ax = b$, using $O(n^c)$ field operations for some fixed $c < 2.258$, with exact zero tests for pivot selection. Choose $\omega < c < 2.258$ and apply the fast elimination reductions of Strassen [28] and Ibarra, Moran, and Hui [16, Theorem 2.1 and Corollary 2.1].

Two applications of the same separation lemma prove the theorem. The square bound uses finite tensor constructions, a differential comparison for their attainable values, and an exact numerical certificate. The dual and rectangular bounds come from explicit conditional-label trees and rational entropy certificates. We also optimize all leaf probability laws in a specified finite family of such trees, obtaining a bound for every aspect ratio in $[0, 1]$. The square and rectangular construction families, and the precise way a square estimate augments the latter, are described below.

1.2 Separating labels with an auxiliary tensor

A trilinear tensor is a sum of monomials xyz , with variables drawn from three finite coordinate sets X, Y, Z . Its rank is the least number of products of three linear forms needed to express it. A restriction substitutes linear forms separately on the three sides. A degeneration allows parameter-dependent substitutions and takes a leading coefficient. Border rank is the least rank of tensors approaching the given tensor. These operations are local: a coordinate substitution cannot depend on information visible only on another side.

Suppose that the supported monomials carry K labels, and that each of two sides can independently determine the label from its own coordinate. The third side may still share variables between labels. The pairwise separation lemma, Theorem 3.3, tensors the input with a finite abelian group-addition tensor of rank $K^{1+o(1)}$. Local projections then give a direct sum over the labels, with a size- K dot-product factor on the two original readers in every summand. The label is now visible on all three sides. Dot-product factors on the three possible reader pairs combine into a matrix multiplication tensor.

The construction combines Fourier diagonalization of the group tensor with tags of equal Euclidean norm and a separate constant-inner-product slice for each tag. Its geometry adapts Behrend's construction [4], following the progression-free constructions of Salem and Spencer [23]. The use of group algebras also has an established place in the approach of Cohn and Umans [7]; the local separation identity needed here is proved in full. It preserves every original coordinate and coefficient, including additional tensor indices correlated across occurrences. Consequently the same identity can be applied inside a previously restricted tensor. A separate support argument ensures that every summand counted in an application is nonzero.

Successive separations are organized by exact conditional counts. Once a preceding label is direct on all three sides, each side can locate the positions on which the next operation acts. Fixing the number of each possible continuation makes the next auxiliary tensor and the output dimensions common to all branches. The multinomial counts and entropy chain rule are the standard method-of-types tools [13, 25]; locality and the uniform tensor accounting are proved alongside them here.

1.3 The square bound: finite constructions and comparison

The square argument starts from the six-term Coppersmith–Winograd tensor

$$T = \sum_{\substack{i+j+k=2 \\ 0 \leq i,j,k \leq 2}} x_i y_j z_k,$$

which has border rank three [10, Section 7, equation (10)]. A five-term degeneration of T has three groups, each detectable on one designated side. A local degeneration of three copies retains 75 scalar components. Separating their successive labels gives the elementary estimate $\omega \leq 9 \log 3 / \log 75 < 2.290108$ in [Theorem 3.7](#). This example displays the auxiliary-rank accounting before the larger construction is introduced.

For a proposed exponent β , suppose a finite construction from n copies of T and an auxiliary tensor of border rank at most R produces L independent matrix products of common dimensions $a \times b$ by $b \times c$. Its normalized score is

$$\frac{\log L - \log R + (\beta/3) \log(abc)}{n}.$$

Schönhage’s asymptotic sum inequality implies that a score strictly greater than $\log 3$ proves $\omega < \beta$. Thus the square argument must produce a strict score gain while retaining the entire auxiliary charge.

We organize the finite constructions by their source distribution. There are six collections of original tensor occurrences, called *orientation banks*, one for each permutation of the three sides. The distribution records the same exact frequencies in the original coordinates of each bank. This keeps a physical permutation of a tensor distinct from a change of its recorded law. The square sections use α for this vector-valued source law; the dual exponent in [Theorem 1.1](#) is a scalar. A construction preserves arbitrary unresolved tensors attached to its original words, and its output labels identify distinct surviving words. Taking the closure of the finite law–score pairs gives a continuous concave attainable-value function. Conditional pooling and a new application of the group separator to each longer pool provide the composition properties needed in the comparison.

The key analytic step uses restrictions on sums of the local grades i, j, k . Each side can read its own sum. Windows placed at progression-free target indices force the three locally read targets to agree, creating another direct label. The progression-free construction and the matching argument have the ancestry of Behrend and Coppersmith–Winograd [4, 10]. To exploit the new labels, a finite-horizon control steers every word allowed by its controls into the assigned terminal window. This every-word guarantee is what permits all the resulting tensor summands to be counted. Exact rational prefix populations then translate the finite control tree into local tensor maps with common auxiliary supplies.

This construction forces a differential inequality on any smooth function touching the attainable value from below. The covariance of the local grades determines the weighted combination of second derivatives in this inequality. Smooth touching tests and strict comparison follow the viscosity-solution methodology of Crandall and Lions and of Crandall, Ishii, and Lions [11, 12]; the required inequality and comparison are proved directly from the finite tensor constructions. A rational polynomial chart and a polynomial barrier then give the strict score gain at an interior source distribution. Boundary constructions supply the initial lower estimates, while explicit residual, interpolation, and rounding bounds turn the finite integer checks into inequalities on the entire comparison domain.

Finally, a strict gain in the closed attainable set selects one actual finite construction with score greater than $\log 3$. Its law may be close to the chart’s central law. All control horizons, rational approximations, group sizes, and bank sizes are fixed before the asymptotic sum inequality is applied.

1.4 Rectangular bounds and finite-family optimization

The rectangular argument tracks the three matrix dimensions separately. A readable label tree partitions a tensor's support until each leaf is one monomial; at every node, conditional on the preceding labels, the next child is independently readable on two specified sides. Fix a rational probability law on its leaves and apply the tree to an admissible number of copies with those exact leaf frequencies; arbitrary leaf laws follow by approximation. The number of child words at a node is a multinomial coefficient. The product of these coefficients is exactly the number of final direct summands, so the leading auxiliary rank charge is canceled by this multiplicity in the equal-summand inequality. The three remaining logarithmic dimensions are the conditional entropies accumulated on the three reader pairs. This is [Theorem 9.2](#).

Explicit polynomial degenerations giving border-rank bounds two and three supply the initial trees. Their supports have three colors, with a designated side able to detect its own color. A *completion* takes several copies, chooses a center color, and retains color words containing at most one distinct noncenter color. Local color tests implement this leading-term degeneration and preserve the conditional trees. Deleting one color at the end gives a fully readable tree. The family $\mathcal{F}_7$ consists of both bases, at most seven completions of sizes two or three, all center choices, all pairs of retained colors, and all choices of a distinguished side.

For each member of this finite family, a three-state recurrence maximizes every nonnegative weighted sum of its incident and nonincident conditional entropies over all leaf laws. Its support function therefore describes the closed downward convex hull of the normalized rate pairs exactly. [Theorem 11.3](#) turns this description into a maximization over one real parameter for each aspect ratio. An available square estimate contributes one additional rate point. In particular, the square part of [Theorem 1.1](#) supplies the point for $p = 2.258$. This finite-family optimization is distinct from the square attainable-value construction with orientation banks and count windows.

Two explicit laws give the dual and rectangular conclusions. For the dual exponent, an exact conditional-independence identity preserves all the entropy of a uniform coordinate word on its incident reader pairs. This equality certifies exponent two, and the remaining entropy certifies the inner dimension. A small fiber histogram gives the rectangular point. Rational enclosures establish their strict numerical margins. The same finite family also gives an independent square bound below 2.267 by an integer leaf count.

[Section 13](#) compares the resulting bounds with the specified numerical tables of [2, 33], augmented by the same square point $p = 2.258$ and closed under the ordinary combination operations stated there. The new bounds strictly improve this table baseline for $0.321334 < k < 1$. This comparison concerns recorded upper estimates. Extensions to additional estimates retain their stated separator or envelope hypotheses, and strict improvement is asserted only where the enlarged baseline itself exceeds two.

1.5 Organization

[Section 2](#) establishes the tensor and arithmetic-exponent conventions. [Section 3](#) proves the shared separation lemma and the finite tensor constructions. The square argument begins in [Sections 4](#) and [5](#), which define the attainable values and prove the count-window comparison. [Section 6](#) establishes the three-state boundary estimate used by the numerical certificate in [Section 7](#). [Section 8](#) extracts the finite witness proving the square part of [Theorem 1.1](#). [Sections 9](#) to [11](#) develop conditional entropy, completion trees, and exact finite-family optimization. [Section 12](#) proves the remaining parts of [Theorem 1.1](#), and [Section 13](#) gives the baseline comparisons. [Section 14](#) proves the finite-exception field corollary from two exact rank witnesses and the characteristic-zero dual corollary using accuracy-dependent witnesses. The appendices give barrier coefficients and reproduction details.

2 Tensors and arithmetic exponents

We first fix the tensor conventions shared by the square and rectangular constructions, then relate rank and explicit polynomial degenerations to arithmetic exponents. The equal-summand inequality at the end of the section will convert separated tensors into rectangular bounds.

All vector spaces and tensors are over $\mathbb{C}$. A tensor on three finite coordinate sets X, Y, Z is identified with a trilinear form

$$T = \sum_{x,y,z} T_{xyz} xyz.$$

A restriction is obtained by a linear map on each of the three sides. In particular, a local zeroing-out sets selected variables on each side to zero. A degeneration $U \supseteq V$ permits these maps to depend polynomially or Laurent polynomially on a parameter and extracts the transformed tensor's first nonzero coefficient, after an overall rescaling. The maps may be composed finitely many times. Finitely many parameters can be replaced by sufficiently separated powers of a single parameter.

The rank $\text{rank}(T)$ is the minimum number of products of three linear forms whose sum is T . Its border rank $\underline{R}(T)$ is the least integer r such that tensors of rank at most r can approach T . Border rank is submultiplicative under tensor product and cannot increase under degeneration. The direct sum $U \oplus V$ uses disjoint variable spaces on all three sides. A claimed direct sum must therefore have both disjoint pieces and no additional terms joining their variables.

Our matrix-multiplication convention is

$$\langle a, b, c \rangle = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^c x_{ij} y_{jk} z_{ki}, \quad \text{Vol} \langle a, b, c \rangle = abc.$$

For positive integers l_1, l_2, l_3 , write $\mathcal{R}(l_1, l_2, l_3) = \text{rank} \langle l_1, l_2, l_3 \rangle$.

For $a_1, a_2, a_3 \geq 0$, let $W(a_1, a_2, a_3)$ denote the arithmetic exponent for the three dimensions $\lceil n^{a_i} \rceil$. The trilinear interpretation makes W invariant under permutations, including those implemented by transposition. In particular $w(k) = W(1, k, 1)$.

Lemma 2.1. *For $a_i \geq 0$,*

$$W(a_1, a_2, a_3) = \lim_{s \rightarrow \infty} \frac{\log \mathcal{R}(\lceil e^{sa_1} \rceil, \lceil e^{sa_2} \rceil, \lceil e^{sa_3} \rceil)}{s}. \quad (2.1)$$

Moreover, W is coordinatewise nondecreasing, homogeneous, subadditive, permutation invariant, and continuous. It satisfies

$$W(a_1, a_2, a_3) \geq \max_{i \neq j} (a_i + a_j).$$

Proof. Fix $s > 0$, put $l_i = \lceil e^{sa_i} \rceil$, and take a rank- r scheme for this triple. Each of the three flattenings has rank $l_i l_j$, so $r \geq \max_{i \neq j} l_i l_j$. The scheme applied recursively to blocks of dimensions l_i^{h-1} uses r smaller products and at most a fixed constant times $(\max_{i \neq j} l_i l_j)^{h-1}$ linear operations at each level. Its total cost is $O((h+1)r^h)$. All products keep a left-input block before a right-input block, so rectangular block multiplication is legitimate. Padding with $h = \lceil t/s \rceil$ proves the corresponding arithmetic upper bound for arbitrary large dimensions. The same padding and tensor-power argument shows that the limsup in (2.1) is at most the infimum over fixed $s > 0$. Hence the rank limit exists.

Conversely, a straight-line arithmetic computation of the bilinear matrix product gives rank at most a constant times its operation count. Introduce separate formal variables for

the two operands and retain bidegrees at most $(1, 1)$. Each gate introduces only a bounded number of new products of a linear form in the first operand and one in the second; the remaining mixed coefficients are linear combinations of earlier ones. For divisions we use Strassen's division-elimination method [29], in the following bilinear adaptation. If divisions are allowed, first translate to a generic constant input at which every denominator is nonzero and use the same truncated-series argument. The mixed coefficient of the output is still the desired bilinear map. This proves the converse inequality in (2.1).

Restriction and padding give monotonicity; tensor products give subadditivity; rescaling s gives homogeneity. These statements include coordinates equal to zero. For continuity, compare two triples through their coordinatewise maximum and use the naive bound on the nonnegative increment: the change in W is at most the sum of absolute coordinate changes. Finally the flattening bounds give the displayed lower bound. $\square$

All logarithms in this paper are natural. The properties in Theorem 2.1 imply that w is convex. For instance, subadditivity and homogeneity applied to $\lambda(1, k_1, 1) + (1 - \lambda)(1, k_2, 1)$ give its usual convexity inequality.

The approximate bilinear algorithms of Bini [5] and the asymptotic-sum method of Schönhage [24] motivate the following explicit form of a rank budget. We use polynomial border-rank witnesses. Saying that T has cost R means that T , up to a nonzero scalar, is the first nonzero coefficient of a polynomial or Laurent polynomial tensor family with a rank- R decomposition. This is a supplied witness and a certified upper bound, which need not be minimal. It implies $\underline{R}(T) \leq R$; it is not our definition of border rank. All uses of cost below have an explicit witness, so no converse assertion about arbitrary border-rank limits is needed. Coordinate restrictions and monomial leading-term degenerations preserve such witnesses and their budgets. Finitely many parameters can be combined into one by assigning sufficiently separated powers, so a composition of the operations used below again has a univariate witness.

The following interpolation argument is the border-to-exact transfer for polynomial approximate bilinear algorithms [5].

Lemma 2.2. *For a fixed tensor of cost R , its N -th tensor power has exact rank at most $O(N + 1)R^N$, where the implicit constant may depend on the fixed witness.*

Proof. Clear negative parameter powers. The degree of the N -th power of the witness is $O(N)$, and its desired coefficient is the N -th tensor power of the first coefficient. Univariate interpolation expresses that coefficient using $O(N + 1)$ evaluations, each of rank at most R^N . $\square$

A dot tensor on the pair Y, Z of size D is $x_0 \sum_{j=1}^D y_j z_j$. The product of dot tensors on the three pairs is a matrix-multiplication tensor. More precisely, if their lengths on XY, XZ, YZ are d_{XY}, d_{XZ}, d_{YZ} , respectively, their product is

$$\langle d_{XZ}, d_{XY}, d_{YZ} \rangle.$$

Thus the edge XZ supplies the index i , the edge XY supplies j , and the edge YZ supplies k in our convention. Dot sizes on the same pair multiply. A tuple of logarithmic edge rates listed in the order (XY, XZ, YZ) therefore gives matrix-dimension rates in the order (XZ, XY, YZ) . By the permutation invariance in Lemma 2.1, either order gives the same value of W . This accounts for the edge-rate order used in the later conditional-label entropy bound.

Lemma 2.3 (Equal rectangular summands). *Suppose a tensor of cost R restricts or degenerates to a direct sum of M copies of the matrix-multiplication tensor with dimensions l_1, l_2, l_3 . Then*

$$W(\log l_1, \log l_2, \log l_3) \leq \log R - \log M. \quad (2.2)$$

Proof. Write the left side as u . If $u = 0$, a flattening of the N -th power of the direct sum, together with [Theorem 2.2](#), gives $M^N \leq O(N+1)R^N$; thus $R \geq M$.

Otherwise choose an integer c with $cu > \log M$. A rank scheme for dimensions l_i^{cN} can be interpreted as requests for products of blocks with dimensions l_i^N . The N -th power of the direct sum supplies M^N such requests in one batch, at cost $O(N+1)R^N$. The requests may depend on common inputs: substituting those linear forms into the independent batch inputs is a restriction. Pad the last batch if necessary. Consequently

$$\mathcal{R}(l_1^{(1+c)N}, l_2^{(1+c)N}, l_3^{(1+c)N}) \leq O(N+1)R^N \left\lceil \frac{\mathcal{R}(l_1^{cN}, l_2^{cN}, l_3^{cN})}{M^N} \right\rceil.$$

By [Theorem 2.1](#), the expression inside the ceiling grows exponentially because $cu > \log M$. Take logarithms, divide by N , and pass to the limit to obtain $(1+c)u \leq \log R + cu - \log M$, proving the claim. $\square$

This is the equal-summand rectangular form of the asymptotic-sum principle [24]. The proof applies it to each fixed instance before varying that instance. Later auxiliary tensors may therefore grow with a type parameter without any uniformity assumption on the interpolation constant in [Theorem 2.2](#).

3 Finite tensor constructions

The elementary construction in this section already gives $\omega \leq 9 \log 3 / \log 75 = 2.290107196\dots$. Its main ingredient separates a label that two parties can read independently. The operation supplies the missing label to the third party and leaves a large dot product. We pay for the auxiliary tensor that performs this operation. At the exponential scale, the new direct labels compensate for that payment, while the dot product remains available for matrix multiplication. We first prove the operation, including its action on unresolved tensor factors, and then give the finite constructions used later for the boundary estimates.

3.1 Pairing conventions and the input rank

We use the tensor and matrix-dimension conventions of [Section 2](#). A pairing on X, Y of length m , with a scalar third side, is $\langle 1, m, 1 \rangle$; the other orientations are $\langle m, 1, 1 \rangle$ on X, Z and $\langle 1, 1, m \rangle$ on Y, Z . Their tensor products satisfy

$$\langle a, 1, 1 \rangle \otimes \langle 1, b, 1 \rangle \otimes \langle 1, 1, c \rangle \cong \langle a, b, c \rangle. \quad (3.1)$$

The logarithmic volumes of such factors therefore add.

The input is the six-term specialization of the Coppersmith–Winograd tensor [10, Section 7, equation (10)],

$$T = CW_1 = x_2 y_0 z_0 + x_0 y_2 z_0 + x_0 y_0 z_2 + x_0 y_1 z_1 + x_1 y_0 z_1 + x_1 y_1 z_0. \quad (3.2)$$

Write $\mathcal{S} = \{200, 020, 002, 011, 101, 110\}$ for its support, abbreviating (i, j, k) by ijk . At every position the three indices sum to two. The maps in our square construction for this input have algebraic coefficients. The general preservation identities below also allow arbitrary complex coefficients in the unresolved tensors.

Lemma 3.1. *The tensor T has border rank three. Consequently $\underline{\mathbf{R}}(T^{\otimes n}) \leq 3^n$ for every integer $n \geq 1$.*

Proof. Let $\zeta = e^{2\pi i/3}$, and put

$$X_r(t) = x_0 + \zeta^r t x_1 + \zeta^{2r} t^2 x_2 \quad (r = 0, 1, 2),$$

with analogous definitions for Y_r, Z_r . Filtering the degree modulo three gives the exact identity

$$\frac{1}{3t^2} \sum_{r=0}^2 \zeta^r X_r(t) Y_r(t) Z_r(t) = T + t^3 (x_2 y_2 z_1 + x_2 y_1 z_2 + x_1 y_2 z_2).$$

For every nonzero t , the left side has rank at most three. Taking $t \rightarrow 0$ proves the upper bound. In the flattening with the X side as rows, the three coefficients of x_0, x_1, x_2 are linearly independent: the monomials $y_0 z_2, y_0 z_1, y_0 z_0$, respectively, distinguish them. The flattening rank is therefore three, giving the reverse bound. Tensoring the upper-bound construction proves the last assertion. The displayed identity also supplies an explicit polynomial witness of cost three in the terminology of Section 2. $\square$

We use the following established theorem in exactly its border-rank form; see also the explicit statement in Ambainis et al. [3, Theorem 3.1].

Theorem 3.2 (Asymptotic sum inequality, Schönhage [24]). *If a tensor U degenerates to $\bigoplus_{\ell=1}^L \langle a_\ell, b_\ell, c_\ell \rangle$, with all dimensions positive integers, then*

$$\sum_{\ell=1}^L (a_\ell b_\ell c_\ell)^{\omega/3} \leq \underline{R}(U).$$

In particular, if $T^{\otimes n} \otimes \mathcal{A} \supseteq \bigoplus_{\ell=1}^L \langle a, b, c \rangle$ and $\underline{R}(\mathcal{A}) \leq R$, then

$$L(abc)^{\omega/3} \leq 3^n R. \quad (3.3)$$

The auxiliary rank R in (3.3) is part of the input cost. Thus, for a proposed exponent β , a finite realization with

$$\log L - \log R + \frac{\beta}{3} \log(abc) > n \log 3 \quad (3.4)$$

proves $\omega < \beta$. We never cancel an auxiliary tensor as a tensor identity; only its explicitly charged logarithmic rank will be offset by output multiplicity.

3.2 Supplying a label to its missing party

Suppose a tensor is partitioned into pieces whose X -variables are disjoint and whose Y -variables are disjoint. Each of these two parties can then determine the same piece label from its own variable. The Z -variables may still be shared. The next lemma resolves this last sharing without resolving the terms inside a piece. We call this operation GROUP.

The construction uses equal-norm shells, following the geometry of Behrend [4], together with affine slices that constrain two independently chosen auxiliary indices. Its auxiliary tensor is the multiplication tensor of a finite abelian group, linking the construction to the group-theoretic tensor approach of Cohn and Umans [7]. The identity and its full rank charge are proved below.

Lemma 3.3 (Pairwise label separation). *Let $U = \sum_{s=1}^K U_s$, where U_s uses X -variables in X_s and Y -variables in Y_s , and the families $(X_s)_{s=1}^K$ and $(Y_s)_{s=1}^K$ are each disjoint. The Z -variables may overlap. There are a finite abelian group G , its group multiplication tensor $\mathcal{A}_G$, and a positive integer m such that*

$$U \otimes \mathcal{A}_G \supseteq \bigoplus_{s=1}^K (U_s \otimes \langle 1, m, 1 \rangle). \quad (3.5)$$

The degeneration is a zeroing-out on auxiliary coordinates controlled by the original local variables. It preserves all original coordinates and coefficients. For $K \rightarrow \infty$, the parameters can be chosen with

$$\text{rank}(\mathcal{A}_G) = \underline{\text{R}}(\mathcal{A}_G) = |G|, \quad \log |G| = \log K + O(\sqrt{\log K}), \quad m = K. \quad (3.6)$$

The same statement holds for any other pair of knowing sides.

Proof. The case $K = 1$ uses the trivial group and $m = 1$, so no nontrivial auxiliary tensor is needed. For $K \geq 2$, set

$$d = \lceil \sqrt{\log K} \rceil + 3, \quad Q = \left\lceil ((d+1)K)^{1/(d-2)} \right\rceil, \quad m = K.$$

Since $dQ^2 + 1 \leq (d+1)Q^2$, these choices give $Q^d / (dQ^2 + 1) \geq K$. The grid $\{0, \dots, Q-1\}^d$ has Q^d points and at most $d(Q-1)^2 + 1$ possible squared norms. Pigeonholing supplies at least K distinct points $t_1, \dots, t_K$ on one Euclidean sphere. Assign t_s to label s .

For each s , the integer inner product $t_s \cdot u$, as u ranges over this grid, has at most $d(Q-1)^2 + 1$ values. A further pigeonhole argument supplies a set V_s of at least m grid points on one affine slice

$$t_s \cdot u = h_s.$$

Trim each V_s to exactly m points. Only the common inner product within a slice is needed; the points of a slice need not have equal norm.

Take

$$G = (\mathbb{Z}/(4Q)\mathbb{Z})^d, \quad \mathcal{A}_G = \sum_{a+b=c} x_a y_b z_c,$$

using separate auxiliary variables on the three sides. For completeness, character orthogonality gives a rank decomposition

$$\mathcal{A}_G = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \left(\sum_{a \in G} \chi(a) x_a \right) \left(\sum_{b \in G} \chi(b) y_b \right) \left(\sum_{c \in G} \chi(-c) z_c \right).$$

It has $|G|$ summands. The X-flattening has $|G|$ linearly independent rows, since the coefficient of $y_0 z_a$ distinguishes row a . This proves both rank equalities.

For an original X-variable in X_s , retain auxiliary coordinates $a = u \in V_s$. For an original Y-variable in Y_s , retain $b = t_s - u'$ with $u' \in V_s$. On the Z side retain only $c \in \{t_1, \dots, t_K\}$. Negative coordinates of $t_s - u'$ are interpreted modulo $4Q$. These are three local variable projections: each knowing side uses the label determined by its original variable, and the third side uses only the prescribed tag set.

A surviving term satisfies $c \equiv t_s + u - u' \pmod{4Q}$. Each coordinate of $c - t_s - u + u'$ has absolute value at most $2(Q-1) < 4Q$, so the congruence is an equality of integer vectors. The tag norms are equal and $t_s \cdot (u - u') = 0$, whence

$$0 = \|c\|^2 - \|t_s\|^2 = \|u - u'\|^2.$$

Thus $u = u'$ and $c = t_s$. Conversely, every $u \in V_s$ gives one surviving triple $(u, t_s - u, t_s)$. The auxiliary factor in piece s is precisely

$$\sum_{u \in V_s} x_u y_{t_s - u} z_{t_s},$$

a length- m pairing on the knowing sides. The pieces were already disjoint on those sides and now have distinct auxiliary tags on the third. The calculation also excludes every cross term, proving (3.5).

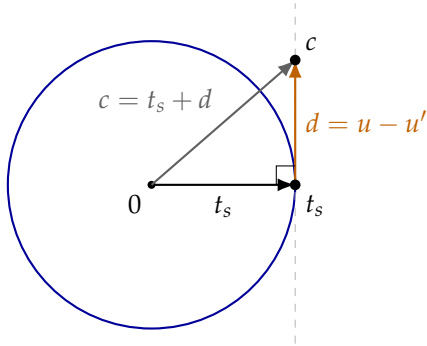
Finally, with $L = \log K$, the displayed choice of d, Q satisfies

$$d \log Q \leq \frac{d}{d-2} (L + \log(d+1)) + o(1) = L + O(\sqrt{L}).$$

The rounding error in $\log Q$ is exponentially small in $\sqrt{L}$. Since $K \leq Q^d$ and $|G| = (4Q)^d$, these inequalities and $m = K$ prove (3.6). $\square$

For repeated uses at different label counts, fix one such group G_K for each K , and write $A_K = \mathcal{A}_{G_K}$. This is the auxiliary-tensor notation used in the conditional-label entropy bound.

The geometry rules out a stronger collision than the usual three-term-progression condition: the two slice points u, u' are chosen independently. Their difference is perpendicular to t_s ; any nonzero such displacement from t_s lies outside its sphere. Requiring the endpoint to be another tag therefore forces that difference to vanish. Figure 1 shows this displacement argument.



The affine slice gives $t_s \cdot d = 0$.

The group equation gives $c = t_s + d$.

If c is another tag on the same sphere,

$$\|c\|^2 = \|t_s\|^2 + \|d\|^2 = \|t_s\|^2,$$

so $d = 0$ and $c = t_s$.

Figure 1: The separation identity in a two-dimensional slice. The dashed line represents the displacement hyperplane $t_s + t_s^\perp$, rather than the slice containing u, u' . A nonzero displacement along it leaves the tag sphere, so no other tag can survive. The argument applies in every dimension.

Lemma 3.4 (Preservation of unresolved fibers). *The identity in Lemma 3.3 holds with arbitrary coefficients and arbitrary unresolved local indices inside the tensors U_s , including indices correlated between different original occurrences. Local support projections and monomial leading-term restrictions have the same property. Finite compositions of these operations may leave all previously produced matrix multiplication factors untouched.*

Proof. Retain an original basis coordinate on every side throughout the construction. In Lemma 3.3, a surviving auxiliary triple is characterized solely by the piece label and the equation just proved. The original coefficient is neither changed nor combined with another coefficient. Appending additional local indices to any original variable therefore gives the same identity for every coefficient separately. The extra indices may be arbitrary; no product distribution or independence between occurrences is used.

A local support projection has the same coefficientwise description. For a monomial restriction, the degree of each term is the sum of three local degrees, and its leading coefficient is either its original coefficient or zero. This too commutes with adjoining unresolved coordinates or earlier local count predicates. The claim for a finite composition follows by induction. At each subsequent stage, tensor the new maps with the identity on all previously produced pairings. $\square$

This preservation statement is a locality statement, not a nonvanishing statement. If an earlier support restriction makes a designated source word absent, its fiber is zero and cannot

be counted as an output. Our finite constructions below explicitly retain every counted word. The later count construction will require a separate guarantee for every complete controlled path.

3.3 Exact types and a finite hierarchy of labels

For a probability vector p on a finite alphabet, write $H(p) = -\sum_i p_i \log p_i$, with $0 \log 0 = 0$. Exact types provide the finite counting form of entropy used here; see Csiszár [13] for the method of types. For rational p and admissible integers N , the number of words having exactly Np_i occurrences of symbol i is

$$\binom{N}{(Np_i)_i} = \exp(NH(p) + O(\log(N+1))). \quad (3.7)$$

The implicit constant depends only on the fixed alphabet. We use exact types so that the number of positions assigned to every later conditional operation is independent of the previously selected direct output.

The following lemma makes precise how multiple separation steps share auxiliary tensors and count each label once. Its accounting is the conditional-entropy chain rule [25], applied to labels that the tensor maps can actually read.

Lemma 3.5 (Accounting for a finite label hierarchy). *Let $b \geq 1$, and let $\mathcal{B} \subseteq \mathcal{S}^b$ be nonempty. Suppose local restrictions and degenerations of $T^{\otimes b}$ retain exactly these original scalar components, with their original coordinates and coefficients. Suppose there are finitely many labels $S_1, \dots, S_t$, each a function of a component in $\mathcal{B}$, with these properties:*

- (i) *the complete record $(S_1, \dots, S_t)$ determines the original component;*
- (ii) *after the earlier labels are fixed, the next label is independently readable by two fixed sides within that conditional pool;*
- (iii) *every component designated by a retained complete record survives all the local restrictions.*

Let α be a rational probability distribution on $\mathcal{B}$. For admissible $N \rightarrow \infty$, there are auxiliary tensors $\mathcal{A}_N$ and degenerations of $T^{\otimes bN} \otimes \mathcal{A}_N$ into L_N independent matrix multiplication tensors of common volume v_N , with $\underline{R}(\mathcal{A}_N) \leq R_N$, such that

$$\log L_N = NH(\alpha) + o(N), \quad \log R_N = NH(\alpha) + o(N), \quad \log v_N = NH(\alpha) + o(N). \quad (3.8)$$

When $H(\alpha) > 0$, this gives $\omega \leq 3b \log 3 / H(\alpha)$.

Proof. Apply the stipulated local restrictions in each of the N blocks. At layer j , write $h = (S_1, \dots, S_{j-1})$ for a preceding record, let p_h be its probability under α , and let $p_{s|h}$ be the conditional law of S_j . The earlier layers have made the full preceding-label word direct on all three sides. In every branch there are exactly Np_h block positions in conditional pool h . Each knowing side can read the next-label word in those positions. It can therefore zero variables whose next-label word does not have counts $Np_h p_{s|h}$. This is a genuine local type projection, not a presumed projection onto an unavailable joint type.

There remain

$$K_{j,h} = \binom{Np_h}{(Np_h p_{s|h})_s}$$

possible next-label words in this pool. Use Lemma 3.3 to separate them. A zero-probability pool is omitted, and a deterministic next label needs no auxiliary tensor. For a positive-entropy pool, (3.7) and (3.6) give identical leading logarithmic contributions

$$Np_h H(S_j | h) + o(N)$$

to its direct count, auxiliary rank, and pairing volume.

The auxiliary tensor for pool (j, h) is appended once. It is reused in every preceding direct branch through a block-diagonal map: on a direct sum, each of the three parties knows the branch, so it may address the appropriate positions using that branch's label. Equivalently, $(\bigoplus_{\ell} V_{\ell}) \otimes \mathcal{A}$ is the direct sum of the $V_{\ell} \otimes \mathcal{A}$, and separate maps may act in its blocks. This does not charge $\text{rank}(\mathcal{A})$ once per ℓ . The size and physical orientation of every conditional supply depend only on (j, h) , whose population is fixed, not on the selected output word.

There are only finitely many layers and conditional pools. Multiplying their counts, ranks, and pairing volumes, and using the entropy chain rule, gives in each case the leading logarithm

$$N \sum_{j=1}^t H(S_j \mid S_1, \dots, S_{j-1}) = NH(\alpha).$$

All error terms sum to $o(N)$. The complete record determines one original scalar component in each block, and every such designated component is nonzero by assumption. Exact conditional counts fix the number of pairings in each physical orientation. Formula (3.1) therefore gives a common matrix multiplication tensor in all final branches. This proves (3.8). Substituting them in (3.3), dividing by N , and taking the limit proves the exponent bound. $\square$

Nothing in this lemma permits arbitrary support deletion. The locally obtainable support and the two-sided readability must be established for each hierarchy. Once they are established, exact conditional types select its desired joint frequencies through accessible labels.

3.4 An explicit four-component construction

First remove the term 101 from T . Scale each of x_1, z_1, y_0, y_2 by t . The five other terms have degree one in t , whereas 101 has degree three. Dividing by t and taking the limit gives

$$T \supseteq T' = A + B + C, \quad A = x_0(y_0 z_2 + y_1 z_1), \quad B = (x_2 y_0 + x_1 y_1) z_0, \quad C = x_0 y_2 z_0. \quad (3.9)$$

A local predicate on each side identifies exactly one group: $Z > 0$ identifies A , $X > 0$ identifies B , and $Y = 2$ identifies C . Exactly one of these predicates holds on every retained term. We call any such three-group partition a *W-partition*; the three predicates, rather than the original numerical indices, are its one-hot flags.

Delete C by setting $y_2 = 0$, and take $(A + B)^{\otimes N}$, with $4 \mid N$. Retain coarse words with $N/2$ copies of A and $N/2$ copies of B . This is local on X , since a zero X index means A and a positive one means B . The Z side also identifies the word, with the roles reversed. There are $\binom{N}{N/2}$ coarse words. Lemma 3.3 separates them by supplying their label to Y , and gives each word a pairing on X, Z of logarithmic length $N \log 2 + o(N)$.

Inside a fixed coarse word, A 's two terms 002, 011 are each identified by Y and by Z . Retain $N/4$ occurrences of each among the $N/2$ A positions and separate their words. This adds a pairing on Y, Z of logarithmic length $(N/2) \log 2 + o(N)$. Similarly, B 's two terms 200, 110 are each identified by X, Y ; resolving their equal-frequency words adds a pairing on X, Y of the same logarithmic length. These restrictions are conditional on the already-direct coarse word and therefore are local.

The three pairings combine by (3.1). The total logarithmic output count, auxiliary rank, and volume are each $N \log 4 + o(N)$. Theorem 3.2 consequently gives

$$\omega \leq \frac{3 \log 3}{\log 4} = 2.377443751 \dots \quad (3.10)$$

The scalar original terms alone would not supply any volume to which ω could be applied. Here the three pairing orientations supply that volume, while all three auxiliary rank costs have been included.

3.5 Completion and the 75-component bound

A W-partition has a useful local description even when its three groups contain unresolved tensors: side i knows precisely the positions at which its own group occurs. This permits a degeneration of a block of several occurrences before any group has been made direct. We call the following operation COMPLETE.

Lemma 3.6 (Completion). *Let a tensor have a W-partition with labels 0, 1, 2. For an integer $b \geq 1$, a local monomial degeneration of its b -th power retains exactly the coarse words*

$$\{0, 1\}^b \cup \{0, 2\}^b. \quad (3.11)$$

This support has a new W-partition: the constant word 0^b , the words containing 1, and the words containing 2. Conditional on either nonzero new label, the original binary word is independently readable by two sides. The construction preserves arbitrary unresolved fibers.

Proof. The three local predicates are

every position has label 0, some position has label 1, some position has label 2.

Each is readable on the correspondingly active side of the original W-partition. Their sum is one on (3.11) and two when both 1 and 2 occur. Multiply a variable by t when its predicate holds, divide the tensor by t , and take the limit. The surviving predicates form a new one-hot partition. In the nonzero group $i \in \{1, 2\}$, the word uses only $0, i$, so either of those two active sides determines the entire word from the positions of its own label. The degeneration is coefficientwise and hence preserves the fibers by Lemma 3.4. $\square$

For the groups A, B, C in (3.9), assign labels 0, 1, 2 in that order. The group sizes are 2, 2, 1. A completed block has three group sizes

$$2^b, \quad 4^b - 2^b, \quad 3^b - 2^b.$$

Delete its all- A group: its active Z predicate is that every original index is positive. The remaining groups will be denoted P_B and P_C . The X side recognizes P_B by the occurrence of a positive index, and the Y side recognizes P_C by the occurrence of index two. Since only these two groups remain, either side independently determines which group occurs.

Theorem 3.7 (Elementary completion bound). *For every integer $b \geq 2$,*

$$\omega \leq \frac{3b \log 3}{\log(4^b + 3^b - 2^{b+1})}. \quad (3.12)$$

In particular,

$$\omega \leq \frac{9 \log 3}{\log 75} = 2.290107196 \dots$$

Proof. Apply (3.9) to each occurrence, perform the completion of Lemma 3.6 in blocks of size b , and delete the all- A group as just described. This locally retains $s_b = 4^b + 3^b - 2^{b+1}$ original scalar components per block. Here is a complete hierarchy that separates them.

First resolve the binary group label P_B, P_C , known on X, Y . After it is direct, resolve the original word inside each group: the A, B word in P_B is known on Z, X , and the A, C word

in P_C is known on Z, Y . After that word is direct, resolve 002 versus 011 in every A position using Y, Z , and 200 versus 110 in every B position using X, Y . There is no internal choice in C . The complete record identifies one original scalar component. Every such component survives the stated monomial restriction and deletion. The pair of knowing sides at each stage is fixed by the preceding direct labels.

Give the s_b components equal frequency and apply Lemma 3.5. Their entropy is $\log s_b$, while the original rank cost per block is at most 3^b . This proves (3.12).

For $b = 3$, the sizes of P_B, P_C are 56, 19. The prescribed outer frequencies are 56/75, 19/75; the internal component law in each group is uniform. The entropy identity

$$H(56/75, 19/75) + \frac{56}{75} \log 56 + \frac{19}{75} \log 19 = \log 75$$

shows explicitly how the successive choices count each of the 75 terms once. Over N such blocks the three leading logarithms are $\log L_N = \log R_N = \log v_N = N \log 75 + o(N)$. Using the input cost $3^{3N} R_N$ in (3.3) and canceling the *logarithmic* multiplicity and auxiliary costs leaves $(\omega/3) \log 75 \leq 3 \log 3$, as claimed. $\square$

The neighboring block lengths illustrate that this particular completion is not monotone in b :

Block length	Retained scalar components	Exponent bound
2	17	2.326571610...
3	75	2.290107196...
4	305	2.304655406...

No optimization claim over all restrictions of the corresponding power is needed. The length-three construction already gives the stated bound.

3.6 Parity and conditional W constructions

Completion gives one way of making a new W-partition. A different local restriction on three occurrences supplies two useful W labels, which can be resolved in either order. We call this operation PARITY.

Lemma 3.8 (Parity restriction). *In three W-partitioned occurrences, a local monomial degeneration removes exactly the six words in which all three labels occur and retains the other 21 coarse words. On the retained support, the majority label and the odd-count label each define a W-partition. Once both labels are direct, the remaining choice is either deterministic or independently readable by two sides. All statements remain true with arbitrary unresolved fibers.*

Proof. Let n_i be the count of label i among the three positions. Side i knows its occurrence set and hence n_i . Scale a variable by $t^{n_i \bmod 2}$ on that side. The sum of the three parity exponents is one for count patterns (3,0,0) and (2,1,0), up to permutation, and three for (1,1,1). Division by t and passage to the limit gives the claimed support.

On this support exactly one label has count at least two, and exactly one has odd count. Both are local unary tests, giving the two W-partitions. Write these labels as c, h . If $c = h$, the word is ccc . Otherwise, it consists of two c 's and one h . The c side knows the two majority positions and the h side knows the singleton position. Either determines which of the three placements occurs, so Lemma 3.3 applies after c, h have been made direct. All restrictions and the final separation preserve fibers by Lemma 3.4. $\square$

If the three occurrences have internal scalar counts $(a_i)_i, (b_i)_i, (c_i)_i$, the fiber with majority i and odd-count label j has size

$$M_{ij} = \begin{cases} a_i b_i c_i, & i = j, \\ a_i b_i c_j + a_i b_j c_i + a_j b_i c_i, & i \neq j. \end{cases} \quad (3.13)$$

Each term for $i \neq j$ describes one singleton placement. For the boundary weights $(2, 2, 1)$ on all three occurrences, this is

$$(M_{ij}) = \begin{pmatrix} 8 & 24 & 12 \\ 24 & 8 & 12 \\ 6 & 6 & 1 \end{pmatrix}.$$

For example, delete the third majority row and resolve the remaining binary majority label. Within the first direct row delete its first odd-count column; within the second direct row delete its second column. These are permitted conditional local deletions because the majority is already direct. Each row then has size 36, giving 72 original scalar components. Lemma 3.5 yields $\omega \leq 9 \log 3 / \log 72 = 2.311966920 \dots$. This illustrative restriction is weaker than Theorem 3.7. Parity is useful because its two conditional groupings enlarge the finite construction family at nonuniform distributions.

3.7 Finite recipes and weighted laws

We now track the source law and score of a finite tree of W constructions. Fix labels 0, 1, 2 on each original W-partitioned occurrence before applying the tree. These are its *canonical labels*: later physical permutations may change which side reads a flag, but the original labels are still counted in this fixed ordering.

For the score calculations below, fix a finite operation tree and its rational conditional laws. Admissible exact-type populations, together with the efficient GROUP tensors of (3.6), give a family of finite realizations. The *limiting score* of this fixed tree and law data is the limit along increasing populations; each member remains one finite recipe with fixed tensor supplies and maps. A later operation may depend on a label only once that label is direct on all three sides. At every stage, previously produced pairings are separate tensor factors; their internal indices are never used to choose subsequent controls.

For a realization on n original occurrences, let L be its direct multiplicity, R its auxiliary rank bound, and v its additional matrix multiplication volume. The rates are

$$D = \frac{\log L}{n}, \quad A = \frac{\log R}{n}, \quad M = \frac{\log v}{n}.$$

Here M excludes any tensor initially present inside a coarse group. For the W recipes the convenient formal score is $u_\gamma = \gamma(D - A) + M$, where $\gamma \geq 1$. For $0 < \beta \leq 3$,

$$u_{3/\beta} = \frac{3}{\beta} \left(D - A + \frac{\beta}{3} M \right).$$

Thus u_γ rescales the normalized score expression in (3.4) so that the coefficient of the added volume is one. Its definition assumes no bound on ω . Definitions 4.1 and 4.2 will specify the banked realizations and the closure of their law–score pairs. For a fixed tree consisting only of pairwise separation, completion, parity, and conditional exact types, its total $D - A$ tends to zero when the group/type supplies grow. Consequently its limiting score is independent of γ .

A binary leaf deletes one W label locally and resolves the remaining binary word by Lemma 3.3. At law q on that pair, its limiting score is $H(q)$. If positive numbers d_i are

assigned as weights for choosing a law, the weighted score is $H(q) + \sum_i q_i \log d_i$. Maximizing on the retained pair i, j gives $\log(d_i + d_j)$. These weights may guide selection of a finite rational law; they do not change the coefficients of a tensor.

For completion of b occurrences with weights $d_{r,i}$, the three fiber weights are

$$w_0 = \prod_{r=1}^b d_{r,0}, \quad w_i = \prod_{r=1}^b (d_{r,0} + d_{r,i}) - \prod_{r=1}^b d_{r,0} \quad (i = 1, 2). \quad (3.14)$$

When the weights are scalar term counts, this is literal counting of the supports in Lemma 3.6. For arbitrary positive weights it is the corresponding generating polynomial. For parity the analogous matrix is (3.13).

3.8 Physical orientations and canonical laws

A physical permutation acts on the full tensor factors of an occurrence or completed block. Each original occurrence keeps both its canonical label and its net physical side permutation. For $\sigma \in \mathfrak{S}_3$, the *orientation bank* σ is the collection of original occurrences with that net permutation relative to their canonical coordinates. We will arrange that all six banks have the same exact canonical label histogram. Their common normalized histogram is the source law; it is not the aggregate law of the physical flags after the bank orientations are forgotten.

The exact types used below fix canonical histograms within each relative orientation. The next lemma shows how the six global physical permutations then distribute these counts equally among the original banks, while keeping one common auxiliary supply.

Lemma 3.9 (Equal physical orientation supplies). *Fix a finite oriented schedule of operations with prescribed exact conditional populations. Suppose its exact types fix, for each relative physical permutation $\tau \in \mathfrak{S}_3$, the canonical input histogram H_τ of all original occurrences with that permutation, independently of the direct history. Include all six global physical side permutations of this schedule, with the same prescribed populations and histograms in corresponding replicas. Then every original orientation bank has the same exact canonical input histogram. Auxiliary supplies and pairing dimensions are independent of the direct history. All regroupings controlled by preceding direct labels are implementable by local block-diagonal maps.*

Proof. In the replica with global permutation σ , an original occurrence of relative orientation τ has absolute orientation $\sigma\tau$. Thus an absolute bank ρ receives the histogram H_τ from the unique replica $\sigma = \rho\tau^{-1}$. Its total histogram is

$$\sum_{\tau \in \mathfrak{S}_3} H_\tau,$$

independently of ρ and of the direct history. The occurrence positions contributing to a histogram may vary with the history, but its counts are fixed by hypothesis. Each original occurrence is counted once, in its relative-orientation class.

All parties know a preceding direct label, so each can reorder or address the same specified positions using its own local coordinate map. Corresponding relative permutations address corresponding physical banks; no party is required to read another party's local flag. Each fixed pool uses a fixed auxiliary tensor and fixed pairing lengths. Tensoring all those supplies once and acting block-diagonally therefore gives uniform rank bounds and dimensions. Previously produced pairings remain untouched. $\square$

In particular, rotating the flags of an intermediate completed block changes its grouping and its readers, but does not recode the labels of the original occurrences inside it. To recover a canonical label from a physically permuted occurrence, undo that occurrence's permutation.

3.9 Resolving and composing nested recipes

An operation in the finite library may complete two occurrences, physically transpose that completed block, and then complete it with a third occurrence. Resolving its original word calls for a hierarchy inside the final fiber. The following statement supplies that hierarchy for any fixed finite composition of completion gates.

Lemma 3.10 (Resolving a tree of completion gates). *Consider a finite rooted tree whose leaves are W -partitioned occurrences and whose internal nodes apply Lemma 3.6 to their children. Before a child is used at a parent, allow a fixed physical permutation of that whole child block. The retained support has a W -partition at its root. Once the root label is direct, the original leaf-label word admits a finite hierarchy of pairwise-readable refinements. All refinements preserve arbitrary unresolved leaf fibers.*

In a fixed root fiber, let P be any rational law on the retained original leaf-label words. Across N copies of that fiber, exact conditional types and Lemma 3.3 give direct multiplicity, auxiliary rank bound, and added pairing volume whose logarithms are each $NH(P) + o(N)$. The auxiliary product and pairing dimensions are common to every direct output. Canonical original labels are unchanged by the intermediate physical permutations.

Proof. Process the nodes from the root toward the leaves. Suppose a node's label is already direct, and express its children's W labels in that node's current physical coordinates. If the node label is zero, completion forces all those child labels to be zero. Otherwise, if its label is $i \in \{1, 2\}$, its child-label word lies in $\{0, i\}^b$, where b is the number of children. Side zero knows the positions with label zero and side i knows the positions with label i ; either side determines the word. Apply Lemma 3.3 to separate that word. Every child label is now direct on all three sides. Undo the known physical permutation on its label to identify the child's own root label, and recurse into that child. Recursion is required even when the parent label was zero: a known child root may still contain unresolved choices.

There are finitely many nodes. Each successive refinement is a function of the original leaf word, and the complete record recovers that word. For the law P , prescribe the induced rational conditional type of each refinement in each preceding direct pool. These types are locally readable by the two sides just identified. They fix all pool populations, so a single prescribed auxiliary product can be used by block-diagonal maps in every branch. They also fix the counts of pairings in every physical orientation. As in the proof of Lemma 3.5, each pool contributes its conditional entropy to all three leading logarithms. The chain rule sums these contributions to $H(P)$, since the complete record and the original leaf word determine one another. The output pairing dimensions are therefore common to all branches.

The retained support is exactly the intersection of the primitive completion conditions, each implemented by its local monomial degeneration. Every word in P 's support satisfies those conditions. Lemma 3.4 preserves any unresolved leaf fibers through every step. A physical permutation changes the sides reading a node's flags; the final histogram still counts each leaf in its original canonical coordinates. To apply Lemma 3.9, let τ_r be the relative physical permutation of original leaf r . The exact full-word law P fixes the count of canonical label i in relative orientation τ to be

$$H_{\tau,i} = N \sum_w P(w) \#\{r : \tau_r = \tau, w_r = i\}.$$

These counts are independent of the direct output. The lemma therefore supplies equal original orientation banks when this schedule is taken in all six global orientations. No individual occurrence is required to carry a fixed letter. $\square$

To spell out the three-position case used in the library, let $c(i, j)$ be the two-child completion label: it is zero for $(i, j) = (0, 0)$, and is $a \in \{1, 2\}$ if both entries belong to $\{0, a\}$ and at

least one equals a . It is undefined on $(1, 2), (2, 1)$. Write

$$a = c(i, j), \quad z = a - k \pmod{3}, \quad s = c(z, r), \quad k \in \{0, 1, 2\}.$$

The subtraction by k represents the physical rotation of the whole first completed block. In a fixed final fiber s , first resolve the top-level operand labels (z, r) . This fixes $a = z + k \pmod{3}$, after which the inner word (i, j) can be resolved using the pair of sides belonging to that inner completion, with its physical rotation included. If C_s is an arbitrary rational law on the retained triples, the two levels supply

$$H(C_s) = H(Z, R \mid S = s) + \sum_{z, r} \mathbb{P}(Z = z, R = r \mid S = s) H(I, J \mid Z = z, R = r, S = s). \quad (3.15)$$

Here Z is a deterministic function of I, J , so no intermediate label adds a second entropy count. The intermediate rotation changes the grouping flag and its physical reader; it does not recode I, J, R when the original canonical word histogram is calculated.

Proposition 3.11 (Composition of finite recipe laws). *Consider a primitive completion, a finite tree of completion gates as in Lemma 3.10, or a parity operation on b original occurrences. Let s be its outer W label, with a rational law a_s , separated by a fixed recipe tree with limiting score u_0 . In the outer group s , let t be a further W label, with conditional rational law r_{st} , separated by a fixed recipe tree with limiting score u_s . For completion this label is deterministic and $u_s = 0$. For parity it is the other of the majority and odd-count labels. For each permitted (s, t) , choose a rational distribution P_{st} on its remaining original coarse words. If $n_i(w)$ counts canonical input label i in the word w , then the resulting original law and limiting score are*

$$q_i = \frac{1}{b} \sum_{s, t} a_s r_{st} \mathbb{E}_{w \sim P_{st}} n_i(w), \quad (3.16)$$

$$u = \frac{1}{b} \left(u_0 + \sum_s a_s u_s + \sum_{s, t} a_s r_{st} H(P_{st}) \right). \quad (3.17)$$

The formula is feasible for every stipulated finite collection of child recipes and rational distributions; no optimality assertion is required.

Proof. Use the outer child recipe while the internal choices remain unresolved fibers. Its coordinate-preserving identity is valid by Lemma 3.4. Once its label is direct, apply the corresponding conditional child in each group. The group populations are fixed by exact types, so the prescribed auxiliary product is appended once and shared by all branches. After these labels are direct, resolve the remaining words by the hierarchy in Lemma 3.10 for completion, or by the single remaining pairwise-readable placement in Lemma 3.8. Prescribe the conditional types induced by P_{st} at every refinement. Their conditional entropies sum to $H(P_{st})$, and Lemma 3.3 supplies exactly this limiting score across the hierarchy. The relative population of that pool is $a_s r_{st}$.

The same types provide the histograms required for banking. The outer and conditional child recipes fix the number of macro occurrences assigned to each (s, t) pool in each of their original banks. Within every such pool, impose the full-word law P_{st} through its readable refinements. For each original leaf position, this law fixes its marginal counts across the pool, just as in the preceding proof. Grouping those contributions by the leaf's relative physical permutation fixes the total histograms H_τ . This uses each child's whole-bank histogram and the new full-word types, not a fixed letter at an individual position of a child recipe. Lemma 3.9 now makes the final canonical law common to all six original banks.

Logarithms of counts, auxiliary rank bounds, and pairing volumes add. Dividing their sum by the number b of original occurrences per new block proves (3.17). Counting the

original labels in the same conditional pools proves (3.16). A finite common multiple of the rational denominators and the fixed child bank sizes makes all pool sizes integral. Increasing these supplies makes the finitely many group and type errors vanish. Every counted original coarse word survives the stipulated support restrictions, so the outputs on the scalar W input are nonzero. Arbitrary unresolved fibers are preserved coefficientwise; a zero attached fiber remains zero. $\square$

The words in (3.16) are counted in their original canonical coordinates. The resulting q is the exact common law of the original orientation banks, providing the finite input to the banked-value construction of the next section.

4 Banked constructions and attainable values

A local separation identity must remain valid when the positions on which it acts are correlated with positions to be processed later. It is also necessary to distinguish a distribution in the coordinates of a tensor from the aggregate distribution after physically permuting its sides. We make both requirements explicit. This gives the finite objects whose attainable values will enter the differential comparison.

4.1 Canonical banks and preservation of unresolved tensors

Let $\mathcal{S} = \{200, 020, 002, 011, 101, 110\}$ and $\mathcal{W} = \{100, 010, 001\}$. The latter is the support of the ordinary three-state tensor

$$W = x_1 y_0 z_0 + x_0 y_1 z_0 + x_0 y_0 z_1.$$

An element of either alphabet is a triple of local grades. We use $\mathcal{B}$ to denote either alphabet in definitions that apply to both. Write $\mathfrak{S}_3$ for the group of permutations of the three tensor sides. A *banked input of size N* has N occurrences in each orientation $\sigma \in \mathfrak{S}_3$, for a total of $n = 6N$ occurrences. Its original component words are

$$w = (w_{\sigma,t})_{\sigma \in \mathfrak{S}_3, 1 \leq t \leq N}, \quad w_{\sigma,t} \in \mathcal{B}.$$

The triple $w_{\sigma,t}$ is expressed in canonical coordinates: its physical grade triple is $\sigma w_{\sigma,t}$. Thus physical side j observes the canonical coordinate $\sigma^{-1}(j)$. Both T and the ordinary W tensor are symmetric under these physical permutations, so this banking does not change the underlying input tensor. Its purpose is to record types and permitted transposes accurately.

A banked word has exact law $\alpha \in \Delta(\mathcal{B})$ if, separately for each σ , every symbol s occurs $N\alpha_s$ times. In particular, α is not the histogram obtained by forgetting the bank orientations. All normalizations below divide by $6N$.

We allow additional, unresolved local indices. To state the preservation requirement precisely, for every original component word w attach an arbitrary three-tensor P_w on these additional indices. The tensors P_w may share their variable spaces, may correlate all the positions, and may be zero. Denote the resulting input by $\mathcal{T}(P)$: its restriction to the original word w is the original scalar component at w tensored with P_w . This description includes preceding local count restrictions by setting the corresponding fibers to zero. It does not assert that all such families arise as independent products.

Definition 4.1 (Strong banked realization). A strong realization consists of a finite auxiliary tensor $\mathcal{A}$, a finite degeneration by three local maps, a set Λ of direct labels, and positive integers a, b, c , with the following properties.

- (i) Every $\lambda \in \Lambda$ designates an original word $w(\lambda)$, and $\lambda \mapsto w(\lambda)$ is injective. All these words have the same exact canonical law in each orientation bank.
- (ii) For every family of unresolved tensors $P = (P_w)$ the same maps give

$$\mathcal{T}(P) \otimes \mathcal{A} \supseteq \bigoplus_{\lambda \in \Lambda} (P_{w(\lambda)} \otimes \langle a, b, c \rangle). \quad (4.1)$$

In this display the original local coordinates of $w(\lambda)$ are retained as well; only their fixed basis factors are suppressed. Original coefficients and unresolved local indices are unchanged.

- (iii) The auxiliary input, its charged border-rank bound, and the output dimensions a, b, c are independent of λ .

The identity is required also for zero P_w . Such an output is a formal zero fiber and contributes no nonzero direct summand. A realization used in an attainable-value calculation must separately establish that all its counted fibers are nonzero on its specified input.

The maps in this paper retain each original local index and act on auxiliary coordinates conditionally on that index and on already direct labels. Consequently they commute with local projections depending only on the original indices. The same statement holds when the maps are expressed as leading monomial degenerations. This observation concerns locality; it does not prove nonvanishing after an earlier projection. Nonvanishing will be checked by support identities for the algebraic primitives and by the universal path condition in Lemma 5.3 for arithmetic-progression (AP) count constructions.

Lemma 3.3 supplies (4.1) term by term. Indeed its sphere and slice argument decides the retained auxiliary terms without referring to any coefficient of P_w . Local singleton projections and direct matchings have the same preservation property. The conditional composition rules of Section 4.2 use only these identities; they never presume independence of unresolved fibers.

For the banking operation of Lemma 3.9, group the original input occurrences of a finite schedule by their relative physical orientation τ . The lemma requires their total canonical histogram H_τ to be fixed independently of the direct history and shared by all six globally permuted replicas. Exact types of complete original words supply these grouped histograms in the completion and parity recipes. When an already strong child is physically permuted as a whole, each of its original input banks supplies a fixed histogram by Definition 4.1. The letter at an individual position may still vary between direct outputs.

In outer replica σ , the group of relative orientation τ has absolute orientation $\sigma\tau$. Each absolute bank therefore receives one copy of H_τ for every τ , and has histogram $\sum_\tau H_\tau$. Equal conditional pool sizes alone would not imply this histogram identity. The pool sizes instead fix how many child calls and auxiliary supplies are used; the grouped types or strong child banks supply the required letter counts. A physical permutation of an intermediate macro acts on every original occurrence inside it.

4.2 Finite recipes and the closure of their values

A *structural recipe* is a finite expression describing local projections and leading degenerations, direct-label-controlled maps, locally imposed exact joint and conditional types, auxiliary GROUP calls, and tensor products on specified collections of positions. Its bank sizes, rational types, and integer group parameters are finite. A conditional choice is permitted only after its controlling label is direct on all three sides. Each byproduct matrix multiplication factor is kept as a separate factor: later decisions use original-word labels and do not read, identify, or spend its indices.

We use the following two classes of such recipes.

- (a) The W class starts with singleton projections, binary deletion followed by GROUP, and the finite COMPLETE and PARITY compositions of Proposition 3.11. It allows rational time-sharing on separate positions, physical permutation banking, and finite AP count compositions of previously constructed recipes.
- (b) The six-state class starts with singleton projections, the direct matching construction of Proposition 4.9, and the five-state face substitutions of Proposition 4.10 using finite W recipes. It has the same time-sharing, banking, and finite AP closure.

Both classes also allow two explicit finite rearrangements. First, one may reindex the orientation banks by $\sigma \mapsto \sigma\tau$ for a fixed $\tau \in \mathfrak{S}_3$, with the corresponding canonical letter relabeling; Lemma 4.4 proves its precise effect. Second, one may *conditionally recompile* an earlier finite recipe: expand its finite direct-label refinement tree, take finitely many replicas, impose a rational exact type of complete terminal transcripts through its induced conditional types, and replace corresponding GROUP calls by GROUP on the longer conditional pools. All other original support predicates are retained on their old macroblocks. This is a finite construction rule using the same local primitives, not an operation on scores. Its instances must preserve Definition 4.1 and nonzero support like every other recipe; Lemma 4.6 proves those requirements for the uniform terminal type used below. Neither rearrangement authorizes an infinite composition.

Here an AP composition is the finite count-window construction of Lemma 5.3. Its children are finite recipes, not calls to an attainable-value function. The full expression can therefore be expanded to a finite tree. Equivalently, take the union of all finite levels of this construction rule. The W class is defined independently of the six-state class.

A recipe belongs to these classes only when its local identities establish Definition 4.1, its counted original words survive all of its restrictions, and its auxiliary profile and output dimensions are fixed across its direct histories. These are algebraic conditions, not inequalities for numerical scores. The primitive support proofs, exact conditional-type composition, and Lemma 5.3 establish these conditions for the listed generators. In particular, introducing the AP closure here does not assume the differential inequality that it will later prove.

The auxiliary tensor in every recipe is a product of finite group tensors. We charge the product of their ranks, which is also their tensor product's rank by Fourier diagonalization and flattening. It is harmless to charge a larger certified bound. No tensor already present in an unresolved fiber is included in the byproduct volume.

For a realization with $n = 6N$ original occurrences, $L = |\Lambda|$ counted nonzero direct summands, auxiliary rank bound R , and common additional volume $v = abc$, put

$$D = \frac{\log L}{n}, \quad A = \frac{\log R}{n}, \quad M = \frac{\log v}{n}. \quad (4.2)$$

For $0 < \beta \leq 3$ and $\gamma \geq 1$, respectively, the two scores are

$$f_\beta = D - A + \frac{\beta}{3}M, \quad u_\gamma = \gamma(D - A) + M. \quad (4.3)$$

These are formal tensor rates. Their definitions impose no assumption on ω . In particular, on the same realization $u_{3/\beta} = (3/\beta)f_\beta$.

Definition 4.2 (Attainable-value functions). For each finite realization let α be its canonical bank law. Take all pairs (α, t) with t no greater than its score, and take their ordinary Euclidean closure in $\Delta(\mathcal{B}) \times \mathbb{R}$. The upper boundaries of these closed sets are denoted by $F_\beta(\alpha)$ for the six-state class and $U_\gamma^W(q)$ for the W class.

Allowing smaller scores makes these sets downward closed. The closure allows irrational laws and limits of finite recipes of increasing size; it does not introduce an infinite composition acting on a finite tensor. In particular, if $F_\beta(\alpha) > c$, then some actual finite realization has score greater than c , possibly at a law arbitrarily close to α .

Proposition 4.3 (Bounds, concavity, and continuity). *For $0 < \beta \leq 3$ and $\gamma \geq 1$,*

$$0 \leq F_\beta(\alpha) \leq H(\alpha) \leq \log 6, \quad 0 \leq U_\gamma^W(q) \leq \gamma H(q) \leq \gamma \log 3. \quad (4.4)$$

Both functions are concave and continuous on their closed probability simplices. They have supporting affine functionals at every relative interior point.

Proof. Every additional pairing has size at most the rank charged for its group tensor. Because byproducts remain separate, their volumes multiply, and $M \leq A$. Source-word injectivity and exact types give

$$L \leq \left(\frac{N!}{\prod_s (N\alpha_s)!} \right)^6 \leq \exp(6NH(\alpha)), \quad D \leq H(\alpha).$$

The same formula holds with the W alphabet. Thus $f_\beta \leq D$ and $u_\gamma \leq \gamma D$, proving the upper bounds also after closure.

A specified original word can be retained by singleton projections. Taking one such word of any prescribed rational type in each bank gives $L = R = v = 1$ and score zero. Density supplies the lower bound zero at every law.

To mix two finite constructions, repeat them enough times that a rational fraction λ of the positions in each orientation is allocated to the first and the remaining fraction to the second. Tensor their maps, auxiliary inputs, and outputs. The new law and score are

$$\lambda\alpha^{(1)} + (1 - \lambda)\alpha^{(2)}, \quad \lambda f^{(1)} + (1 - \lambda)f^{(2)}.$$

Their direct transcripts inject into concatenated original words, all bank histograms and dimensions remain uniform, and all support conditions are preserved. Rational approximation and closure give concavity for real λ and limiting laws.

For upper semicontinuity, suppose $\alpha_k \rightarrow \alpha$. The upper bounds in (4.4) give a bounded subsequence of values approaching the limsup. Approximate each value by a pair in the defining closed set. Closedness then places the limiting pair in that set, so its value is at most the value at α . This also shows that each finite upper boundary is attained in the closed set.

For lower semicontinuity at any α , including at a face, set

$$\varepsilon_k = \max_{s:\alpha_s > 0} \left(1 - \frac{\alpha_{k,s}}{\alpha_s} \right)_+.$$

Then $\varepsilon_k \rightarrow 0$. If $\varepsilon_k > 0$, the vector

$$r_k = \frac{\alpha_k - (1 - \varepsilon_k)\alpha}{\varepsilon_k}$$

is a probability vector. Indeed its coordinates are nonnegative, including those where $\alpha_s = 0$, and its total is one. Concavity and nonnegativity give $F_\beta(\alpha_k) \geq (1 - \varepsilon_k)F_\beta(\alpha)$. If $\varepsilon_k = 0$, conservation of total mass implies $\alpha_k = \alpha$. This proves lower semicontinuity; the proof for U_γ^W is identical. A finite continuous concave function has a supporting affine functional at each relative interior point, by separation of its hypograph. $\square$

Lemma 4.4 (Symmetry of canonical laws). *For $\tau \in \mathfrak{S}_3$, define $(\tau_*\alpha)(s) = \alpha(\tau^{-1}s)$. Then*

$$F_\beta(\tau_*\alpha) = F_\beta(\alpha), \quad U_\gamma^W(\tau_*q) = U_\gamma^W(q).$$

Proof. Start with a strong realization and move old bank $\sigma\tau$ to new bank σ , reinterpreting its canonical symbol s as τs . The physical grade is unchanged, because $\sigma(\tau s) = (\sigma\tau)s$. This is a fixed permutation of original occurrence positions, performed identically on all three sides, together with a change of canonical bookkeeping. Conjugate the old local maps by this position permutation. Their coordinatewise identities continue to hold for arbitrary unresolved tensors, with the same relabeling of their original positions. Every new bank has canonical law $\tau_*\alpha$. Source-word injectivity, nonzero support, the auxiliary tensor, and all byproduct dimensions are unchanged. Thus the same D, A, M are attained at the permuted law. The inverse rearrangement uses τ^{-1} ; taking closures proves both equalities. This canonical law symmetry is distinct from replicating a schedule in all six physical orientations, which preserves its canonical law. $\square$

Remark 4.5 (A useful exact-law correction). If a construction or a convex mixture has law $\hat{\alpha}$, and a desired law α has positive coordinates, put

$$\lambda = \max_{s:\hat{\alpha}_s > 0} \left(1 - \frac{\alpha_s}{\hat{\alpha}_s} \right)_+.$$

For $\lambda > 0$, the vector $r = (\alpha - (1 - \lambda)\hat{\alpha})/\lambda$ is a probability law. Mixing the construction with singleton constructions of average law r therefore gives α exactly, at loss at most $K\lambda$ if the old score is at most K , where $K \geq 0$. Moreover

$$\lambda \leq \max_s \frac{(\hat{\alpha}_s - \alpha_s)_+}{\alpha_s}.$$

When the data are rational, this is a finite rational time-sharing operation. For limiting data it is justified by the defining closure. This correction will be used when converting approximate finite laws to exact target laws.

4.3 Conditional pooling and monotonicity in the formal parameter

A finite GROUP call pays a small excess of auxiliary rank over its number of labels. Tensor-repeating that particular finite map does not remove its normalized excess. The following lemma replaces each GROUP call by a new one acting on a long conditional pool. It is the requisite justification for monotonicity in γ . The face substitution below evaluates the W value at $\gamma = 3/\beta$; monotonicity will let the certificate use a lower estimate obtained at a smaller formal parameter.

Lemma 4.6 (Recompiling a fixed finite recipe). *Fix a finite recipe on n original occurrences with canonical law q , L terminal labels, charged auxiliary rank R , and additional volume v . Fix $\gamma \geq 1$. There is a sequence of recipes on larger exact banks, with the same canonical law q , such that*

$$\liminf u_\gamma^{\text{new}} \geq \frac{\gamma \log L - \gamma \log R + \log v}{n}, \quad \liminf (D^{\text{new}} - A^{\text{new}}) \geq 0. \quad (4.5)$$

All source-word restrictions of the old recipe hold separately in every old macroblock of the new recipes.

Proof. Expand the fixed recipe into its finite sequence of conditional refinements, including the finite recipes used at any earlier AP nodes. Refine a history whenever its next local operation, its informed pair, or its auxiliary parameters differ. Terminal histories can be padded by deterministic steps so that every transcript is a leaf of one finite decision tree. Every intermediate label is determined by the designated original word: GROUP labels encode its source label sequence, matching labels identify the word, coarse flags are functions of it, and AP labels are fixed by its disjoint count windows. Thus this expansion adds no

terminal multiplicity; its leaves are exactly the L original strong labels. A GROUP context refers to a particular such call with its preceding direct history fixed. If several pools are processed, order them and include their labels in the history. A physical orientation is part of the context when needed.

Give the L complete terminal transcripts their uniform probability law. At context c , let w_c be its visitation probability, let r_c be the conditional law of its GROUP label, and write

$$\begin{aligned} h_c &= H(r_c), \\ K_c &= \text{number of available labels}, \\ R_c &= \text{charged group rank}, \\ m_c &= \text{old pairing size}. \end{aligned}$$

Repeated calls are regarded as distinct contexts. All probabilities are rational. Every actually used label is among the K_c available tags, so

$$h_c \leq \log K_c \leq \log R_c, \quad \log m_c \leq \log R_c. \quad (4.6)$$

The last two inequalities follow because there are at most $|G|$ tags and at most $|G|$ retained pairing coordinates in a group tensor of rank $|G|$.

Choose B through sufficiently divisible multiples. Across B replicas of the old complete recipe, retain precisely the exact type with B/L occurrences of each terminal transcript. This condition is implemented in prefix order: at a prefix of probability w , impose the induced exact conditional type of its next label on its Bw replicas. Each next type is readable on the sides that already read that label. For a direct matching label or an AP target label, all three sides read it; for a GROUP label, its two informed sides read it before the GROUP call. Previously direct labels identify the relevant pool on every side. The conditional type restrictions are therefore local.

At GROUP context c replace the old Bw_c copies of its auxiliary call by one application of Lemma 3.3 to the entire conditional label sequence of that pool. Its number of possible labels is

$$K'_c = \frac{(Bw_c)!}{\prod_s (Bw_c r_c(s))!}, \quad \log K'_c = Bw_c h_c + O(\log B).$$

Choose the pairing size to be K'_c and choose the efficient group parameters of Lemma 3.3. Consequently

$$\log R'_c = Bw_c h_c + o(B), \quad \log m'_c = Bw_c h_c + o(B). \quad (4.7)$$

A deterministic pool requires no auxiliary. The number of contexts is fixed before B increases, so all their errors sum to $o(B)$.

Here are the support and resource details of this replacement. The current pool is selected by previously direct histories. Both informed sides can read its conditional label word independently, so the new GROUP call is legitimate. Its payload comprises every remaining original coordinate and every old support predicate. The pointwise identity preserves that payload. The new direct tag identifies the entire conditional word on all three sides, so each old per-macro GROUP label is locally recoverable from it. Every subsequent old label-controlled map can therefore be implemented using the new tag; the old pairing register is replaced and is never consulted. Other direct operations are retained on the old macroblocks, followed by the specified conditional-type projections. Thus each final tuple consists of old admissible transcripts, and every old per-block count window, COMPLETE/PARITY predicate, and singleton or matching restriction still holds. All its original scalar fibers are nonzero. The exact conditional counts fix every pool and every auxiliary multiplicity independently of the new output history. The auxiliaries can be appended once and the

maps used block-diagonally on direct histories. New pairing dimensions may differ from the old ones; this does not affect a later map, since byproduct indices are never used for a later decision.

The product of the conditional multinomial counts is the full multinomial count of terminal transcript tuples. Hence the new direct multiplicity is

$$L' = \frac{B!}{((B/L)!)^L}, \quad \log L' = B \log L + O(\log B). \quad (4.8)$$

No terminal transcript is lost beyond these prescribed types: non-GROUP maps act as before, and each replacement GROUP retains every conditional word of its selected type. Conversely, these types specify exactly the chosen leaf counts. This also proves injectivity into original words. Each old terminal word has the same law in every canonical bank. Concatenate the B already banked replicas within corresponding original orientation banks, retaining every old orientation label and internal relative permutation. No additional outer replicas are introduced at this stage. Each resulting canonical bank therefore has the old law q . The construction respects Definition 4.1, not just a total histogram after forgetting orientations, with exactly Bn original occurrences as used in the displayed counts.

The chain rule for the uniform terminal transcript gives

$$\sum_{c \text{ a GROUP context}} w_c h_c \leq \log L. \quad (4.9)$$

The omitted contributions are conditional entropies of other direct labels and are nonnegative. Equations (4.7) and (4.8) now prove the second assertion in (4.5).

For the first assertion, the original auxiliary profile is fixed across terminal histories. Averaging its logarithmic charges and its pairing volumes over those histories therefore gives

$$\sum_c w_c \log R_c = \log R, \quad \sum_c w_c \log m_c = \log v.$$

If the old rank bound charges unused resources or is larger than the product rank, the first equality can instead be replaced by $\leq$, which only strengthens the conclusion below. Non-GROUP operations have no additional auxiliary charge or byproduct. From (4.7) and (4.8), the limiting improvement of the unnormalized γ -score is at least

$$\begin{aligned} & \sum_c w_c [\gamma \log R_c - \log m_c - (\gamma - 1)h_c] \\ &= \sum_c w_c [(\log R_c - \log m_c) + (\gamma - 1)(\log R_c - h_c)] \geq 0, \end{aligned}$$

by (4.6). Dividing by n proves the first assertion. The finite common multiples needed above exist because the expanded tree, its rational laws, and its old block sizes were fixed first. $\square$

Proposition 4.7 (Monotonicity). *For $1 \leq \gamma_0 \leq \gamma_1$ and every W law q ,*

$$U_{\gamma_1}^W(q) \geq U_{\gamma_0}^W(q).$$

Proof. Apply Lemma 4.6 to any finite recipe scored at γ_0 . On each of the recompiled realizations,

$$u_{\gamma_1} - u_{\gamma_0} = (\gamma_1 - \gamma_0)(D - A).$$

The second assertion of that lemma makes the limiting difference nonnegative, while its first assertion preserves the old γ_0 -score. Thus the closure at γ_1 contains a value at least as large as the old law. Approximate an arbitrary pair in the γ_0 closed attainable set by finite recipes and take a diagonal sequence of their recompilations. Their laws converge to the same q ; closedness proves the claim. $\square$

4.4 A direct matching lower bound

The first boundary estimate separates original scalar components without an auxiliary tensor. The matching argument uses progression-free hashing and collision removal as in Coppersmith and Winograd [10, Section 6]; we give the details with the bank normalization needed here.

Lemma 4.8 (Progression-free labels). *For all sufficiently large integers L there is a set $J \subseteq \{1, \dots, L\}$ containing no nonconstant three-term arithmetic progression and satisfying*

$$\log |J| = \log L - O(\sqrt{\log L}).$$

For primes $P \rightarrow \infty$, a subset of $\mathbb{F}_P$ of size $P \exp(-O(\sqrt{\log P}))$ has the same property for progressions modulo P .

Proof. This is the sphere construction of Behrend [4]. Put $d = \lfloor \sqrt{\log L} \rfloor$ and $m = \lfloor L^{1/d}/2 \rfloor$. Encode vectors in $\{0, \dots, m-1\}^d$ as integers in base $2m$. Their encodings are nonnegative and smaller than $(2m)^d \leq L$. There are at most $d(m-1)^2 + 1$ possible squared norms, so one norm shell contains at least $m^d / (d(m-1)^2 + 1)$ vectors. Its logarithmic size is $\log L - O(\sqrt{\log L})$: both the base overhead $d \log 2$ and the norm-shell loss $O(\log d + \log m)$ have that order.

If three encodings satisfy $a + c = 2b$, digitwise addition on either side produces no carries, so their vectors satisfy the same midpoint relation. If all three have the same squared norm, writing the first and third as $b \pm u$ gives $\|b + u\|^2 + \|b - u\|^2 = 2\|b\|^2 + 2\|u\|^2$ and forces $u = 0$. The three vectors coincide. Translating all encodings by one puts them in $\{1, \dots, L\}$ without changing this property.

For a prime P , apply the integer construction inside $\{1, \dots, \lfloor P/4 \rfloor\}$. An integer of the form $a + c - 2b$ from this interval has absolute value less than P . If it vanishes modulo P , it is zero over the integers. This gives the asserted modular set. $\square$

Proposition 4.9 (Banked marginal matching). *Let $\alpha \in \Delta(\mathcal{S})$, and let $\alpha_X, \alpha_Y, \alpha_Z$ be its three canonical local-grade marginals. For every $0 < \beta \leq 3$,*

$$F_\beta(\alpha) \geq H_{\text{marg}}(\alpha) := \frac{H(\alpha_X) + H(\alpha_Y) + H(\alpha_Z)}{3}. \quad (4.10)$$

The constructions proving this bound are strong realizations with no auxiliary input and no additional matrix multiplication volume.

Proof. First take α rational and N through its admissible denominators. Write $p_i = \alpha(2e_i)$ and $E_i = \alpha(\mathbf{1} - e_i)$. On canonical side i , the frequency of grade two is exactly p_i . If b_i is the frequency of grade one, then

$$b_i = E_{i+1} + E_{i+2}, \quad E_i = \frac{1}{2}(b_{i+1} + b_{i+2} - b_i),$$

with subscripts modulo three. Thus the three marginal types determine the whole joint type. Local restrictions to those types, separately in each orientation bank, retain precisely the original words of canonical law α in every bank. No nonlocal joint-type filter is used.

View the retained support as a three-partite hypergraph. A vertex is a complete retained local index word on one physical side; an edge is a retained triple of local words. There are

$$B_N = \left(\frac{N!}{\prod_{s \in \mathcal{S}} (N\alpha_s)!} \right)^6$$

edges. Each physical side sees each canonical marginal in exactly two banks, so all three vertex parts have the same size

$$V_N = \prod_{i=0}^2 \left(\frac{N!}{\prod_{a=0}^2 (N\alpha_i(a))!} \right)^2.$$

Consequently, with $n = 6N$,

$$\log B_N = nH(\alpha) + O(\log N), \quad \log V_N = nH_{\text{marg}}(\alpha) + O(\log N). \quad (4.11)$$

Permuting positions within each bank acts transitively on each local exact-type vertex set and preserves the edge set. Every vertex therefore has degree $\Delta_N = B_N/V_N$. Two vertices on different sides determine at most one edge, since the original grades sum to two coordinatewise.

Choose a prime $P > 3$ between X_N and $2X_N$, where $X_N = \exp(\sqrt{N}) \max(3, \Delta_N)$; integer rounding or enlarging X_N by a fixed amount is immaterial. Such a prime exists for all sufficiently large N by Bertrand's postulate. Let $J \subset \mathbb{F}_P$ be the set of Lemma 4.8. Choose independent uniform field elements d_X, d_Y and $w_1, \dots, w_n$, and for the three physical index words I, J', K define local hashes

$$\begin{aligned} h_X(I) &= d_X + \sum_t w_t I_t, \\ h_Y(J') &= d_Y + \sum_t w_t J'_t, \\ h_Z(K) &= \frac{1}{2} \left(d_X + d_Y + \sum_t w_t (2 - K_t) \right). \end{aligned}$$

The symbol J' here is an index word; the progression-free set is J . On every edge $h_X + h_Y = 2h_Z$. Retaining variables whose hashes lie in J therefore forces all three hashes on a surviving edge to coincide. A fixed edge survives with probability $|J|/P^2$.

For clarity, condition on this edge surviving with common hash $a \in J$. The equations fix d_X and d_Y once the vector (w_t) is chosen; they leave that vector uniformly distributed. For a different edge sharing its X vertex, survival imposes the nonzero linear equation $\sum_t w_t (J''_t - J'_t) = 0$. Its conditional probability is $1/P$. The same reasoning holds for a shared Y vertex. For a shared Z vertex, tightness gives $I' + J'' = I + J'$; survival is equivalent to $\sum_t w_t (I'_t - I_t) = 0$, again of probability $1/P$. The difference vectors are nonzero because two shared vertices would force the edges to agree. Their entries belong to $\{-2, -1, 0, 1, 2\}$, so they remain nonzero modulo P .

There are fewer than $3\Delta_N$ neighboring edges. A union bound shows that the expected number of isolated surviving edges is at least

$$\frac{B_N |J|}{P^2} \left(1 - \frac{3\Delta_N}{P} \right) = V_N \exp(-o(N)). \quad (4.12)$$

Here $\log(P/\Delta_N) = O(\sqrt{N})$, $\log(P/|J|) = O(\sqrt{\log P}) = o(N)$, and $3\Delta_N/P \rightarrow 0$. Thus some hash choice attains this many isolated edges. Delete every vertex not belonging to an isolated surviving edge. No extra edge can remain: an extra edge touching any retained vertex would contradict that vertex's edge being isolated in the filtered hypergraph. The output is therefore a direct sum of scalar original components.

The restriction preserves each chosen original component, with any unresolved tensor attached to it, unchanged. All chosen words have the same exact canonical type in every bank. Thus it is a strong realization with $A = M = 0$ and $D \geq H_{\text{marg}}(\alpha) - o(1)$ by (4.11)–(4.12). Passing to the attainable closure proves (4.10) for rational laws. Approximation by rational laws, including on faces, and continuity prove the general statement. $\square$

4.5 The five-component face

On one face the six-state tensor has a genuine three-state W partition. The remaining distinctions within its coarse groups must be separated only after that partition has been resolved. This gives the second boundary estimate used in the certificate.

Proposition 4.10 (*W constructions on a face*). *Suppose $\alpha = (p_0, p_1, p_2, E_0, E_1, E_2) \in \Delta(\mathcal{S})$ satisfies $E_1 = 0$. Put*

$$Q = (p_2 + E_0, p_0 + E_2, p_1), \quad (4.13)$$

$$J(\alpha) = \text{en}(p_2) + \text{en}(E_0) + \text{en}(p_0) + \text{en}(E_2) - \text{en}(Q_0) - \text{en}(Q_1), \quad (4.14)$$

where $\text{en}(s) = -s \log s$ and $\text{en}(0) = 0$. Then for $0 < \beta \leq 3$,

$$F_\beta(\alpha) \geq \frac{\beta}{3} (U_{3/\beta}^W(Q) + J(\alpha)). \quad (4.15)$$

In particular, if $1 \leq \gamma_0 \leq 3/\beta$, the same lower bound holds with $U_{\gamma_0}^W$ in place of $U_{3/\beta}^W$.

Proof. Scale x_1, z_1, y_0, y_2 by a degeneration parameter. The five terms other than 101 have degree one, whereas 101 has degree three. Dividing by the parameter and taking its leading coefficient leaves

$$A = x_0(y_0 z_2 + y_1 z_1), \quad B = (x_2 y_0 + x_1 y_1) z_0, \quad C = x_0 y_2 z_0.$$

The local predicates $z > 0$, $x > 0$, and $y = 2$ identify, respectively, A, B, C , and exactly one holds on each retained term. These are W flags, with the canonical coarse law Q in (4.13). The fixed permutation identifying the three active parties with the three coarse labels is applied to the physical banks; it bijects their orientations.

Apply a strong W recipe to these coarse occurrences. Its unresolved fibers contain the internal binary choices in A and B . The strong identity retains those choices without restricting them. Once a coarse word is direct, all sides know its A and B positions. Within A , the choice between 002 and 011 is independently readable on Y and Z ; within B , the choice between 200 and 110 is independently readable on X and Y . Apply exact conditional types and GROUP to the A and B pools separately in every original orientation bank. Their total entropy per original occurrence is

$$Q_0 H\left(\frac{p_2}{Q_0}, \frac{E_0}{Q_0}\right) + Q_1 H\left(\frac{p_0}{Q_1}, \frac{E_2}{Q_1}\right) = J(\alpha),$$

with an empty pool contributing zero. Lemma 3.3 gives this additional logarithmic volume and matching direct multiplicity, with equal auxiliary rank to leading order. Exact types fix pool sizes across all coarse histories, so the internal auxiliary product is charged once. The final labels are distinct original scalar words of the prescribed canonical bank law, and all their fibers are nonzero.

For a coarse realization scored at $\gamma = 3/\beta$, its contribution to the six-state score is $(\beta/3)u_\gamma$ by (4.3). The two internal pools contribute $(\beta/3)J(\alpha)$ in the limit. To obtain the supremum at an interior face law, approximate its W value by finite recipes, correct their small coarse-law discrepancies by Remark 4.5, and choose rational conditional types. Every step is finite, with errors made arbitrarily small before taking the attainable closure. This proves (4.15) when the five present coordinates are positive. Both sides are continuous on the closed face by Proposition 4.3 and continuity of conditional entropy, so limits give the remaining cases. The last assertion is Proposition 4.7. $\square$

5 Arithmetic-progression restrictions and differential comparison

An attainable value contains information about nearby source laws, not merely about a single tensor restriction. This section makes that information quantitative. If a supporting plane loses too little value when the mean grade changes, one can steer long grade sequences into many disjoint count windows at a total cost smaller than the entropy of the windows. Their labels then give an improved tensor construction, contradicting the supporting plane. The steering argument must control every path permitted by its controls: an estimate that only holds with high probability would not justify counting all the output tensors. We first construct the shared count labels and prove their finite tensor gain. We then build controls whose cost is strictly smaller than that gain.

All source laws in this section are the canonical laws in each of the six physical orientation banks of Definition 4.1. In particular, they are not the aggregate physical law obtained by forgetting the banks.

Let

$$\mathcal{P} = \{x \in \mathbb{R}^3 : x_0 + x_1 + x_2 = 0\}.$$

We use orthonormal coordinates on this plane when forming covariances, gradients, and Hessians. For the six-state alphabet, the grade of a source letter is its vector in $S = \{2e_i, \mathbf{1} - e_i : 0 \leq i \leq 2\}$. For a law α , write

$$\mu(\alpha) = \mathbb{E}_\alpha g, \quad C(\alpha) = \mathbb{E}_\alpha [(g - \mu)(g - \mu)^\top] \big|_{\mathcal{P}}.$$

The covariance is positive definite when every letter has positive probability. Indeed, the three grades $2e_i$ alone affinely span the mean plane. For a twice continuously differentiable chart $\alpha(\mu)$ whose grade mean is μ , set

$$\mathcal{L}u = C(\alpha(\mu)) : D^2u = \text{Tr}(C(\alpha(\mu))D^2u).$$

This is the full covariance contraction; the corresponding probabilistic generator has an additional factor $1/2$.

Smooth touching tests are a standard formulation in viscosity-solution theory [11]; see in particular [12, Section 2]. The differential inequality below is derived directly from finite tensor constructions.

Theorem 5.1 (Lower tests for the six-state value). *Fix $0 < \beta \leq 3$, and let $\alpha(\mu)$ be a C^2 chart of strictly positive six-state laws, parametrized by their means on an open subset of $\{\mu : \sum_i \mu_i = 2\}$. Suppose that a C^2 function φ touches $F_\beta \circ \alpha$ from below at an interior point μ_* . If g_* is any supporting supergradient of F_β at $\alpha(\mu_*)$, then*

$$\mathcal{L}\varphi(\mu_*) \leq g_* \cdot \mathcal{L}\alpha(\mu_*) - 1. \quad (5.1)$$

Here “touches from below” means equality at the indicated point and the inequality $\varphi \leq F_\beta \circ \alpha$ in a neighborhood. No derivatives of F_β are assumed. A supporting supergradient is a vector such that

$$F_\beta(q) \leq F_\beta(\alpha(\mu_*)) + g_* \cdot (q - \alpha(\mu_*)) \quad \text{for every source law } q.$$

Such vectors exist at positive laws by Proposition 4.3.

Write $h = F_\beta$, $\mu_0 = \mu_*$, $\alpha_0 = \alpha(\mu_0)$, and

$$\ell(q) = h(\alpha_0) + g_* \cdot (q - \alpha_0).$$

For a finite construction with law q and utility u , its *supporting-plane cost* is $\ell(q) - u$. It is nonnegative because $u \leq h(q) \leq \ell(q)$. This cost measures lost utility; it is not an additional auxiliary rank charge.

5.1 Locally readable target windows

We require many target indices with no nonconstant three-term arithmetic progression. We use the integer form of the progression-free set construction of Behrend [4], proved in Lemma 4.8. Using such labels to separate tensor products follows the progression-free extraction method of Coppersmith–Winograd [10, Sections 4–7]; the count-window restriction needed here is given explicitly below.

Lemma 5.2. *For all sufficiently large integers L , there is a set $J \subseteq \{1, \dots, L\}$ with no nonconstant three-term arithmetic progression and*

$$\log |J| \geq \log L - O(\sqrt{\log L}).$$

Proof. This is the integer assertion of Lemma 4.8. $\square$

For a length- n word with grades $g_1, \dots, g_n$, its centered grade sum is $\sum_{t=1}^n (g_t - \mu_0)$. Use the targets

$$y_j^{\text{grade}} = (j, j, -2j)\sqrt{n}/L, \quad j \in J, \quad (5.2)$$

and coordinate windows with radius $\rho = \sqrt{n}/(10L)$. Each side reads its own centered sum. Because the three centered sums add to zero, a retained triple whose three local window labels are j_0, j_1, j_2 satisfies

$$|j_0 + j_1 - 2j_2| \leq 3/10.$$

The left side is an integer, and hence is zero. The progression-free property gives $j_0 = j_1 = j_2$. Windows are disjoint on every side, including the side whose target spacing is twice as large. Thus a nonzero retained tensor has a shared direct target label.

In a physical orientation bank $\sigma \in \mathfrak{S}_3$, the mean μ_0 , the target vector, and the local windows are all permuted by σ . The coordinate-sum test is expressed in canonical coordinates. A physical side reads the corresponding permuted coordinate. The coordinate-sum argument is unchanged, so the target remains shared. There is no averaging of the canonical grade covariance across orientations in this operation.

5.2 Exact finite compilation

At each target and prefix, a control specifies a probability law on the source alphabet and a cost. A *permitted path* chooses at every step a letter in the prescribed law's support. In the next lemma, every node law and utility must be supplied by an earlier finite strong realization. The lemma turns these realizations into tensor maps when every permitted path ends inside its assigned target windows. It is useful to state it for either alphabet, and with a general coefficient $\lambda > 0$ on the direct-label logarithm. For the six-state score $D - A + (\beta/3)M$ this coefficient is $\lambda = 1$; for the W score $\gamma(D - A) + M$ it is $\lambda = \gamma$.

The exact populations and multinomial entropy estimates use the method of types [13, Section II]. The proof also establishes the local tensor maps, their uniform auxiliary supply, and nonvanishing of every counted output.

Lemma 5.3 (Compiling a finite tree). *Fix a finite horizon n , a finite set of disjoint shared terminal windows J , and a rational target law a . At every target and prefix node $v = (j, h)$, prescribe an earlier finite strong realization with exact canonical law q_v , utility u_v , and uniform auxiliary and byproduct dimensions. Assume that every path permitted by these laws belongs to its target windows. Put*

$$w_{j,h} = a_j \prod_{t=1}^{|h|} q_{j,h_{<t}}(h_t), \quad \bar{q} = \frac{1}{n} \sum_{|h| < n, j \in J} w_{j,h} q_{j,h}.$$

Then finite strong realizations with exact law $\bar{q}$ attain, in the limit of their outer bank sizes,

$$u_{\text{out}} \geq \frac{\lambda H(a)}{n} + \frac{1}{n} \sum_{|h| < n, j \in J} w_{j,h} u_{j,h}. \quad (5.3)$$

The limiting notation only removes a multinomial type-count loss; every member of the family is a finite tensor construction using one fixed auxiliary tensor and nonzero counted outputs.

Proof. Use B length- n macroblocks in each of the six physical orientations. Thus the total number of source occurrences is $6Bn$. First apply the local terminal-window projections. On any surviving term, the preceding argument makes the target of each macroblock a shared label. On every orientation restrict the target-label word to the exact type a . The number of such words is

$$\binom{B}{(Ba_j)_{j \in J}},$$

with logarithm $BH(a) + O(\log B)$, since J is fixed. The six orientations have independently chosen target words of this type.

For each shared target assignment, process sites in order. At the node $v = (j, h)$, there are exactly

$$b_v = Bw_v$$

macroblocks in every orientation having that target and canonical prefix. Apply its prescribed strong realization to their next occurrences, as an equal collection across the six orientations. If that realization consumes N_v occurrences in each orientation, tile b_v / N_v copies of it. Its exact per-bank histogram guarantees that the number of these macroblocks receiving next canonical letter i is

$$b_{j,hi} = b_{j,h} q_{j,h}(i). \quad (5.4)$$

These numbers do not depend on which of its direct output labels occurs. That label does identify, on every side, exactly which macroblocks enter each child node, so the next maps may address the corresponding original occurrence positions conditionally on the shared direct label.

All the numbers w_v and q_v are rational. There are finitely many nodes and finitely many integers N_v . Taking B through common multiples of their denominators and the necessary N_v 's makes every nonzero b_v an exact tiling. Zero populations require no call. This proves the induction (5.4) throughout the tree. It also shows that every branch uses the same number of every raw auxiliary tensor. Append their product once. Apply the appropriate maps inside each already direct branch; there is no multiplication of auxiliary rank by the number of branch labels. Uniformity of the raw byproduct dimensions makes the final byproduct dimensions uniform as well. The byproducts remain separate tensor factors.

We justify carefully why the earlier count restrictions do not invalidate the raw calls. Definition 4.1 and the pointwise construction in Lemma 3.4 give a coordinatewise identity for every correlated unresolved payload, preserving the original source coordinates. Conditional maps can address the listed original positions without changing the source word. Their identities therefore commute with the previously applied local count projectors. The same statement holds for a degeneration after taking its leading coefficient. Equivalently, compute the unfiltered raw identity first and then apply the original count projections to its designated fibers.

This identity alone would allow a formal zero fiber, which must not be counted. Here every complete designated word has, in each macroblock, a path permitted by the chosen controls. It consequently satisfies that macroblock's target windows. Its local raw words satisfy the support conditions of the corresponding earlier strong realizations. Starting from

the independent source tensor, every such complete source word is present, and it survives all of these predicates. Thus every designated output counted here is nonzero. This also proves that every target assignment of the retained exact type supplies all the claimed raw outputs, even though the initial projection may correlate the unprocessed suffixes.

Bank labels are fixed physical orientations throughout. At a node the child realization supplies law q_v in canonical coordinates in each bank; its physical law in bank σ is σq_v . Both the prefix classification and the window centering use that same conversion. This AP compilation introduces no additional relative permutation of a site's source orientation: all internal physical permutations of a raw recipe are already included in its strong realization identity and its exact per-bank type. The population assigned to an original input bank is not redefined after observing an output label. Summing the node contributions therefore gives the exact final canonical law $\bar{q}$ in every bank.

For clarity about the probability notation, choose one macroblock uniformly from any fixed final output branch, within a fixed canonical orientation. The exact counts give

$$\mathbb{P}(j, h) = w_{j,h}, \quad \mathbb{P}(\text{next letter} = i \mid j, h) = q_{j,h}(i).$$

This is a counting identity. Different macroblocks need not be independent. The law of one sampled block is exactly the law used in the tree's expected-cost calculation, so its single-step canonical covariance is the covariance of $q_{j,h}$, not a symmetrization across banks.

It remains to count multiplicities and charges. If the child realization at v has direct multiplicity L_v , rank bound R_v , and volume V_v , its rates are normalized by $6N_v$. Its b_v/N_v copies therefore contribute precisely w_v/n times each raw rate to the rate per $6Bn$ source occurrences. The target types contribute

$$\frac{6 \log \binom{B}{(Ba_j)_j}}{6Bn} = \frac{H(a)}{n} + O\left(\frac{\log B}{B}\right)$$

to the direct rate and nothing to the auxiliary rate. This proves (5.3) in the limit.

No extra transcript entropy has been counted: the target is recovered uniquely from the final original word's disjoint count windows. Once the target and earlier prefixes are fixed, injectivity of each child realization recovers its direct label from its designated original subword. Induction therefore makes the complete output label inject into the complete source word. The coordinatewise identity, exact bank law, uniform dimensions, and nonvanishing just proved are precisely the strong realization invariants. Only finite compositions and tensor products have been used for every admissible finite B . $\square$

For any affine functional ℓ , define the node cost by $c_v = \ell(q_v) - u_v$. The prefix weights sum to one at each time, so the averaged law in the lemma satisfies

$$\frac{1}{n} \sum_v w_v \ell(q_v) = \ell(\bar{q}).$$

Thus (5.3) has the equivalent form

$$u_{\text{out}} \geq \ell(\bar{q}) + \frac{\lambda H(a) - \mathbb{E} \sum_{t=1}^n c_t}{n}. \quad (5.5)$$

The expectation denotes the finite sum $\sum_v w_v c_v$. For the six-state supporting plane fixed above, $\lambda = 1$ and every node cost is nonnegative. To contradict that plane, it is therefore enough to send every permitted path into its window while keeping the expected cost strictly below $H(a)$. The remaining argument constructs such controls from a failed lower test.

5.3 A lower test supplies inexpensive small drifts

At a lower contact, $\ell \circ \alpha - \varphi$ has a local minimum zero. Thus its first derivative vanishes and

$$Q_0 = g_* \cdot D^2 \alpha(\mu_0) - D^2 \varphi(\mu_0) \succeq 0. \quad (5.6)$$

If (5.1) fails, $\text{Tr}(Q_0 C(\alpha_0)) < 1$. Choose $\varepsilon > 0$ so small that

$$Q = Q_0 + \varepsilon I \succ 0, \quad \tau = \text{Tr}(QC(\alpha_0)) < 1. \quad (5.7)$$

Taylor's theorem and $\varphi \leq h \circ \alpha$ give, after reducing the neighborhood if necessary,

$$0 \leq c(v) := \ell(\alpha(\mu_0 + v)) - h(\alpha(\mu_0 + v)) \leq \frac{1}{2} v^\top Q v. \quad (5.8)$$

By the definition of the closed attainable set, the law and utility in this display can be approximated simultaneously, to any prescribed positive accuracy, by a finite earlier construction. We postpone these finitely many approximations until the entire control tree has been fixed.

Transform centered grades by $Q^{1/2}$. A grade g then gives the increment $X = Q^{1/2}(g - \mu_0)$. A small prescribed mean m is obtained by taking the law $\alpha(\mu_0 + Q^{-1/2}m)$. We call these chart laws soft controls; their ideal cost satisfies

$$c(m) \leq |m|^2/2. \quad (5.9)$$

There is a fixed bound $B > 0$ on the magnitude of every increment, and

$$\mathbb{E}|X|^2 \longrightarrow \tau \quad \text{as } m \longrightarrow 0. \quad (5.10)$$

The limit concerns the second moment, which equals the covariance trace at mean zero.

We also allow exact singleton controls. The point μ_0 is in the relative interior of the grade convex hull, so there is a $\kappa > 0$ such that for every unit vector $u \in \mathcal{P}$, one singleton increment satisfies $u \cdot X \leq -\kappa$. Its utility is zero. The costs of the finitely many singleton choices have a finite common upper bound C_* .

5.4 A control that retains every path

We now construct controls for the bounded increments just described whose cost is smaller than the target-label gain in (5.5). The terminal guarantee must hold for every permitted path. Soft controls will be made safe for every letter in the entire source alphabet, so later rational approximation cannot create an unsafe new outcome.

Lemma 5.4 (Shrinking-corridor control). *Assume that every $|m| \leq m_0$ is available as a soft mean, with cost at most $|m|^2/2$, bounded increments $|X| \leq B$, and the limit (5.10). Assume also the singleton controls above. Fix $c > 0$ and a target bound $C_y > 0$. For every $\eta > 0$ there is a constant K_η , independent of L , with the following property. For each integer $L \geq 2$, all sufficiently large finite horizons n , and every target y with $|y| \leq C_y \sqrt{n}$, there is a finite control tree such that every permitted path satisfies*

$$\left| \sum_{t=1}^n X_t - y \right| < w, \quad w = c\sqrt{n}/L,$$

and its expected total cost is at most

$$\mathbb{E} \sum_{t=1}^n c_t \leq (\tau + 2\eta) \log L + K_\eta. \quad (5.11)$$

The expectation is only an accounting device for the tree's transition laws; the terminal inclusion holds for every path.

Proof. Reduce m_0 if necessary so that all its means lie in the available local chart. Choose a wall cutoff d_0 such that

$$d_0 \geq 16B, \quad d_0 m_0 \geq 1024B^2.$$

Choose a sufficiently large that

$$a \geq 1024B^2, \quad a \log \left(1 + \frac{\kappa}{8d_0} \right) \geq 2C_* + 2.$$

Finally choose D_0 so large that

$$D_0 > 2C_y + 1, \quad D_0 \geq B, \quad \frac{aB^2}{D_0^2} < \eta/4. \quad (5.12)$$

The numerical factors are convenient slack, not optimized constants. All these choices are independent of L and n .

If r steps remain, put

$$s = r + w^2/D_0^2, \quad R_s = D_0\sqrt{s}, \quad J_s(e) = \frac{|e|^2}{2s} - a \log \left(1 - \frac{|e|^2}{D_0^2 s} \right),$$

where e is the current displacement from the target. The permitted region is $|e| < R_s$. Initially $e = -y$, so this region contains the starting point with a fixed relative margin. At the terminal time its radius is w .

Write $\delta = R_s - |e|$. If $\delta \geq d_0$, choose the soft mean

$$m = -\text{clip}_{m_0} \left(\frac{e}{s} + \frac{2ae}{R_s^2 - |e|^2} \right), \quad (5.13)$$

where clipping truncates the vector's length to at most m_0 . If $\delta < d_0$, choose an exact singleton with inward radial component at least κ .

First verify support. In the soft region every increment in the entire alphabet has magnitude at most $B < d_0$, while $R_s - R_{s-1} = O(s^{-1/2})$. Thus every outcome stays in the next region once the minimum s is large. In the singleton region, $|e| \rightarrow \infty$ uniformly as that minimum grows, and the inward radial component gives

$$|e + X| \leq |e| - \kappa/2$$

for all sufficiently large s . It too stays in the next region, whose shrinkage is eventually at most $\kappa/4$. Induction proves the support assertion at every time. At fixed L , the minimum $s = w^2/D_0^2 = c^2 n/(D_0^2 L^2)$ tends to infinity with n , so these statements are uniform over the finite horizon.

We next prove the uniform one-step estimate

$$c_{\text{step}} + \mathbb{E}J_{s-1}(e + X) - J_s(e) \leq \frac{\tau/2 + \eta}{s}. \quad (5.14)$$

Here c_{step} is the cost of the chosen control at the current state; the constant c in $w = c\sqrt{n}/L$ remains fixed. There are three regimes. We give their estimates explicitly, and then explain uniformity.

Interior regime. Suppose along a sequence with $s \rightarrow \infty$ that $t = |e|^2/(D_0^2 s)$ stays bounded away from one. The drift in (5.13) is then unclipped and tends to zero. Taylor expansion in space and in the time variable, with remainder $o(s^{-1})$, is valid uniformly on any such smaller region. Indeed the third spatial derivatives are $O(s^{-3/2})$, and the needed time and mixed derivatives have their corresponding smaller orders, while $|X| \leq B$. Using $c_{\text{step}} \leq |m|^2/2$ and $m = -\nabla J_s$, the leading temporal and drift contributions, after multiplication by s , sum to

$$-\frac{at}{1-t} - \frac{2a^2 t}{D_0^2(1-t)^2}.$$

For completeness, the temporal contribution is $D_0^2 t/2 + at/(1-t)$; the combined drift and cost contribution is $-D_0^2 t/2 - 2at/(1-t) - 2a^2 t/[D_0^2(1-t)^2]$. The quadratic noise gives $\tau/2 + o(1)$ by (5.10). The remaining barrier noise is at most

$$\frac{aB^2}{D_0^2(1-t)} + \frac{2aB^2 t}{D_0^2(1-t)^2}.$$

The second of these terms is absorbed by the negative term quadratic in a , since $a \geq B^2$. The first, together with $-at/(1-t)$, is at most aB^2/D_0^2 , since $D_0^2 \geq B^2$. Consequently the limsup of the left side of (5.14) multiplied by s is at most $\tau/2 + \eta/4$.

Soft control near the wall. Now suppose $\delta \geq d_0$ and $\delta/R_s \rightarrow 0$. The drift is radial and inward, and

$$|m| = (1 + o(1)) \min\{m_0, a/\delta\}. \quad (5.15)$$

This follows because the radial length before clipping is $|e|/s + 2a|e|/[\delta(R_s + |e|)]$, whose second term is $(1 + o(1))a/\delta$ and whose first is smaller by $O(\delta/\sqrt{s})$. Define

$$z_* = \frac{D_0^2 + 2e \cdot X + |X|^2}{R_s^2 - |e|^2}.$$

The logarithmic part of the potential changes by

$$a\{-\log(1 - z_*) + \log(1 - 1/s)\}.$$

For large s , $|z_*| \leq 3B/\delta \leq 1/2$. The inequality $-\log(1 - z) \leq z + 2z^2$ for $|z| \leq 1/2$ therefore gives an upper bound

$$-(1 - o(1))\frac{a|m|}{\delta} + \frac{20aB^2}{\delta^2} + o\left(\frac{a|m|}{\delta}\right). \quad (5.16)$$

To see the orders here, the mean of the linear part is

$$\mathbb{E}z_* = -\frac{2|e|}{R_s + |e|}\frac{|m|}{\delta} + O\left(\frac{D_0^2 + B^2}{R_s \delta}\right),$$

and $\mathbb{E}z_*^2 \leq 9B^2/\delta^2$ for large s . The displayed error in the mean is $o(|m|/\delta)$ by (5.15) and $\delta/R_s \rightarrow 0$. We used the harmless constant 20 to absorb the second-moment bound.

The cost is at most $(1 + o(1))a|m|/(2\delta)$. The quadratic part of the potential has increment

$$\frac{|e|^2}{2s(s-1)} + \frac{e \cdot m}{s-1} + \frac{\mathbb{E}|X|^2}{2(s-1)}.$$

Its middle term is nonpositive, and the other terms are $O(s^{-1})$, which is $o(a|m|/\delta)$ in this regime. Finally,

$$\delta|m| \geq (1 - o(1)) \min\{d_0 m_0, a\} \geq (1 - o(1))1024B^2.$$

Thus the noise term in (5.16) is strictly smaller than the remaining negative drift term. The left side of (5.14) is negative for all sufficiently large s along any sequence in this regime.

Singleton control near the wall. If $\delta < d_0$, the support estimate already proved gives the new wall distance $\delta' \geq \delta + \kappa/4$. Using $R_s^2 - |e|^2 = \delta(R_s + |e|)$, and noting that the other radius factors tend to one uniformly, the logarithmic potential decreases by at least

$$a \log \left(1 + \frac{\kappa}{8d_0} \right)$$

for sufficiently large s . This pays the cost $c_{\text{step}} \leq C_*$ with fixed slack by the choice of a . The quadratic-potential increment is $O(s^{-1/2})$ and hence cannot consume that slack. The left side of (5.14) is again negative.

These estimates imply a uniform threshold for (5.14). Otherwise a sequence of violating states with $s \rightarrow \infty$ would have a subsequence on which t stays away from one, or tends to one. The former contradicts the interior estimate with its strict $3\eta/4$ margin. The latter has a further subsequence in one of the two wall regimes, where the left side is negative. This also proves that the threshold works for every time in a sufficiently large horizon.

The initial value of $J_s(-y)$ is bounded by a constant depending only on the chosen control constants and C_y , not on L . The final value is nonnegative. Summing (5.14) and telescoping therefore bounds the expected cost by

$$(\tau/2 + \eta) \sum_{r=1}^n \frac{1}{r + w^2/D_0^2} + O_\eta(1) \leq (\tau/2 + \eta) \log \left(1 + \frac{D_0^2 L^2}{c^2} \right) + O_\eta(1).$$

Since the logarithm is at most $2 \log L + O(1)$, this is (5.11). The factor two in this logarithmic time ratio cancels the factor $1/2$ in the one-step covariance contribution. $\square$

5.5 The contradiction and the order of limits

Proof of Theorem 5.1. Suppose the conclusion fails and choose Q and $\tau < 1$ as in (5.7). Choose $\eta > 0$ with $\tau + 2\eta < 1$. Transform the grade-sum targets in (5.2) by $Q^{1/2}$. Their norms are at most $C_y \sqrt{n}$ for a fixed C_y , independent of L and n . Choose $c > 0$ so that $c \|Q^{-1/2}\| \leq 1/10$. A transformed ball of radius $c\sqrt{n}/L$ then lies inside all three original coordinate windows. Fix the resulting constants of Lemma 5.4. For each L , Lemma 5.2 gives a target set J . Give its targets equal probabilities, so $a_j = 1/|J|$ and $H(a) = \log |J|$. The average ideal steering cost is bounded by

$$\mathbb{E} \sum c_t \leq (\tau + 2\eta) \log L + K_\eta.$$

Choose L so large that

$$G := \log |J| - \{(\tau + 2\eta) \log L + K_\eta\} > 0, \quad (5.17)$$

using the uniform upper bound on the expectation. Then choose a sufficiently large finite n for all the control estimates and fix the complete finite control tree supplied by Lemma 5.4 for every target.

For each soft node the law is the appropriate chart law and the ideal utility is h at that law. Exact singleton nodes stay exact. Include all source letters as possible children of every soft node, whether or not an ideal law assigns them positive probability, and give every such child the safe continuation supplied by the controller. Thus the pathwise window property is independent of small perturbations of the soft transition probabilities.

There are now only finitely many soft nodes. Approximate each one's ideal law and utility by an earlier finite strong realization from the closed attainable class. The finite realization has a rational exact law, which need not lie exactly on the smooth chart. This causes no difficulty. Its actual supporting-plane cost is $\ell(q_v) - u_v \geq 0$, and approaches the

ideal cost as the law and utility errors tend to zero. In a finite tree, all prefix probabilities, the expected total cost, and the averaged source law depend continuously on these finitely many data. Choose the approximation errors so small that the actual expectation is at most the ideal expectation plus $G/2$. No accuracy bound uniform in n is needed, since this finite n has already been fixed.

Apply Lemma 5.3 to these finite realizations. Let $\bar{\alpha}$ be its exact averaged law. The supporting affine terms sum linearly, because at every time the prefix weights sum to one. Thus its limiting utility satisfies

$$\begin{aligned} u_{\text{out}} &\geq \frac{\log |J|}{n} + \frac{1}{n} \sum_v w_v \{ \ell(q_v) - c_v \} \\ &= \ell(\bar{\alpha}) + \frac{\log |J| - \mathbb{E} \sum_t c_t}{n} \geq \ell(\bar{\alpha}) + \frac{G}{2n}. \end{aligned}$$

Taking the finite outer bank size sufficiently large makes its remaining type-count loss less than $G/(4n)$. We have therefore constructed an actual finite admissible realization whose utility exceeds $\ell(\bar{\alpha})$, contradicting the supporting-plane bound $h \leq \ell$.

The order of choices was: local control constants; a finite target set; a sufficiently large finite horizon and its full tree; finitely many finite approximating raw recipes; and finally a sufficiently large outer common multiple. Each approximant has finite inductive depth, so their maximum depth is finite and the new macro uses only earlier constructions. The potentially very large common multiple affects existence neither of the maps nor of the strict excess. No infinite-depth realization has been applied to a finite tensor. The contradiction proves $\text{Tr}(Q_0 C) \geq 1$, which is exactly (5.1). $\square$

Write $U_\gamma = U_\gamma^W$ for the three-state attainable value.

Corollary 5.5 (Lower tests for the W value). *Fix $\gamma \geq 1$. On the relative interior of the three-letter simplex, put*

$$C_W(q) = \text{diag}(q) - qq^\top, \quad \mathcal{L}_W = C_W(q) : D^2.$$

Every C^2 lower test ψ for U_γ satisfies

$$\mathcal{L}_W \psi \leq -\gamma \tag{5.18}$$

at its contact point.

Proof. Use W grades e_0, e_1, e_2 , whose means equal their law q . The chart is affine, its covariance is $C_W(q)$, and the concavity, supporting planes, and boundedness needed above hold by Proposition 4.3. The argument from a failed lower test again gives a quadratic drift cost and a trace τ . All control and target-window proofs apply because the centered grades sum to zero and span the same plane. The sole change in the utility accounting is that a direct target contributes $\gamma \log |J|$. Lemma 5.3 uses $\lambda = \gamma$, while the hard-wall cost is still $(\tau + 2\eta) \log L + O_\eta(1)$. Thus a contradiction occurs if $\tau < \gamma$, proving (5.18). No pre-existing payload volume enters this utility. $\square$

The following interior-maximum argument is the strict-comparison strategy of viscosity-solution theory [12, Section 5.C]. Its short proof uses the lower-test inequalities just established.

Corollary 5.6 (Strict comparison). *Let Ω be a bounded domain in the mean plane with compact closure, and let a chart α as in Theorem 5.1 extend continuously to $\bar{\Omega}$. Suppose $f \in C(\bar{\Omega}) \cap C^2(\Omega)$, $f \leq F_\beta \circ \alpha$ on $\partial\Omega$, and, for every $\mu \in \Omega$ and every supporting supergradient g there,*

$$\mathcal{L}f(\mu) > g \cdot \mathcal{L}\alpha(\mu) - 1.$$

Then $f \leq F_\beta \circ \alpha$ throughout $\overline{\Omega}$. For the affine W chart, the analogous conclusion holds for U_γ when $\mathcal{L}_W f > -\gamma$.

Proof. The value functions are continuous by Proposition 4.3. If $f - F_\beta \circ \alpha$ has a positive maximum c , compactness and the boundary inequality place a maximizing point in the interior. The function $f - c$ touches the value from below there. Its derivatives equal those of f , contradicting Theorem 5.1. The W proof uses Corollary 5.5 in the same way. $\square$

5.6 Controlling the supporting-plane term

The lower-test inequality retains $g \cdot \mathcal{L}\alpha$ when the chart is not exactly harmonic. Although the supporting supergradient is unknown, concavity and a uniform value bound control this term. The certificate will apply the following estimate where every chart coordinate has a positive lower bound.

Lemma 5.7 (Supporting-plane chart error). *Suppose $0 \leq h \leq K$ is a concave value on the full source simplex, α is an interior law, and g is a supporting supergradient there. For a probability-law chart through α ,*

$$|g \cdot \mathcal{L}\alpha| \leq K \max_i \frac{|\mathcal{L}\alpha_i|}{\alpha_i}.$$

In particular, if $\alpha_i \geq m > 0$ and $\|\mathcal{L}\alpha\|_\infty \leq \varepsilon$, the right side is at most $K\varepsilon/m$.

Proof. Add a multiple of the all-ones vector to g so that $g \cdot \alpha = h(\alpha)$. This leaves its supergradient inequality on the simplex unchanged, and does not change $g \cdot \mathcal{L}\alpha$ because $\sum_i \alpha_i = 1$. The resulting supporting plane is $q \mapsto g \cdot q$. Evaluating it at simplex vertices and using $h \geq 0$ gives $g_i \geq 0$. Therefore

$$|g \cdot \mathcal{L}\alpha| \leq \sum_i g_i |\mathcal{L}\alpha_i| \leq \left(\max_i \frac{|\mathcal{L}\alpha_i|}{\alpha_i} \right) \sum_i \alpha_i g_i \leq K \max_i \frac{|\mathcal{L}\alpha_i|}{\alpha_i}.$$

$\square$

6 The three-state boundary seed

Proposition 4.10 bounds the six-state value on a five-component face by a three-state value and an explicit conditional entropy. We now construct the three-state lower bound needed for that estimate. Binary separation gives an entropy baseline throughout the simplex. A finite family of completion and parity constructions improves this baseline on a closed curve in its interior. The lower-test inequality then carries that improvement inside the curve.

Throughout this section put

$$\gamma_0 = \frac{300000}{225925} = \frac{3}{2.25925}, \quad U(q) = U_{\gamma_0}(q), \quad \text{en}(t) = -t \log t,$$

where $\text{en}(0) = 0$. As in the definition of the W value, its score is $\gamma_0(D - A) + M$, and M excludes the volume of any payload already inside a coarse component. All decimals specifying coefficients or thresholds below denote exact rational numbers.

6.1 The affine chart and the baseline

For $q = (q_0, q_1, q_2)$ in the probability simplex, set

$$z = \frac{3q_0 - 1}{2} + \frac{i\sqrt{3}}{2}(q_2 - q_1), \quad q_i = \frac{1 + 2\operatorname{Re}(\zeta^i z)}{3}, \quad \zeta = e^{2\pi i/3}. \quad (6.1)$$

The three deterministic laws correspond to $1, \zeta^2, \zeta$. If Z takes these values with probabilities q_0, q_1, q_2 , respectively, then $\mathbb{E}Z = z$, $\mathbb{E}Z^2 = \bar{z}$, and $\mathbb{E}|Z|^2 = 1$. Thus the full covariance contraction, with $\partial_z = (\partial_x - i\partial_y)/2$, is

$$\mathcal{L}_W = (\bar{z} - z^2)\partial_z^2 + 2(1 - z\bar{z})\partial_z\partial_{\bar{z}} + (z - \bar{z}^2)\partial_{\bar{z}}^2. \quad (6.2)$$

There is no factor $1/2$ in this formula. Equivalently, in affine simplex coordinates it is $(\operatorname{diag}(q) - qq^\top) : D^2$.

Define

$$F_0(q) = (\gamma_0 - 1) \sum_{i=0}^2 \operatorname{en}(q_i) + (2 - \gamma_0) \sum_{i=0}^2 \operatorname{en}(1 - q_i). \quad (6.3)$$

Since $\operatorname{en}''(t) = -1/t$, the diagonal covariance entries give

$$\mathcal{L}_W \sum_i \operatorname{en}(q_i) = -\sum_i (1 - q_i) = -2, \quad \mathcal{L}_W \sum_i \operatorname{en}(1 - q_i) = -\sum_i q_i = -1.$$

Consequently $\mathcal{L}_W F_0 = -\gamma_0$. On each edge the two entropy sums coincide with the binary entropy. Lemma 3.3 achieves that entropy by binary separation, with $D - A$ tending to zero. Both coefficients in (6.3) are positive. Hence, for $0 < \delta < 1$, $(1 - \delta)F_0 \leq U$ on the boundary and $\mathcal{L}_W((1 - \delta)F_0) > -\gamma_0$ in the interior. The comparison consequence of Corollary 5.5 and continuity of the values prove, upon letting δ decrease to zero, that

$$U(q) \geq F_0(q) \quad \text{on the whole simplex.} \quad (6.4)$$

6.2 A polynomial improvement and its domain

Let

$$\rho(\theta) = \sum_{k=0}^4 c_k \cos(3k\theta), \quad \Omega_W = \{re^{i\theta} : 0 \leq r \leq \rho(\theta)\}, \quad (6.5)$$

$$(c_0, c_1, c_2, c_3, c_4) = (.28309034, -.05388682, .00841589, .00450821, -.00680398).$$

Here and below inequalities on this compact set include its boundary; comparison is applied to its open interior. The proposed improvement is a real polynomial H_W . Its exact specification is short despite its degree. Put

$$(s_0, s_1, s_2, s_3) = (.0152567903, .0002597033, -.0014351728, -.0017937434), \\ (s_4, s_5, s_6) = (-.0008593989, -.0002090472, -.0000227468).$$

Set $a_{3k,0} = s_k / (.32)^{3k}$ for $0 \leq k \leq 6$. All other pure coefficients $a_{j,0}, a_{0,j}$ are zero, except the already assigned $a_{0,0}$. For $d = 0, \dots, 110$, and $i + j = d$, define

$$a_{i+1,j+1} = \frac{d(d-1)a_{ij} - (i+2)(i+1)a_{i+2,j-1} - (j+2)(j+1)a_{i-1,j+2}}{2(i+1)(j+1)}. \quad (6.6)$$

A negative index denotes a zero coefficient. Every coefficient on the right has degree at most $d + 1$, so this is a finite rational recursion. Finally set

$$H_W(z) = \operatorname{Re} \sum_{i+j \leq 112} a_{ij} z^i \bar{z}^j. \quad (6.7)$$

Here are explicit bounds on this polynomial; their verification uses only rational arithmetic. Put $R_W = .357$, and, for $k = 0, 1, 2$, put

$$S_k = \sum_{i+j \geq k} |a_{ij}| (i+j)(i+j-1) \cdots (i+j-k+1) R_W^{i+j-k},$$

where the empty product is one. The recursion gives

$$S_0 < .058178, \quad S_1 < 1.490164, \quad S_2 < 53.686778. \quad (6.8)$$

For clarity, a finite procedure for checking the residual is as follows. For each a_{ij} , add its four contributions

monomial	coefficient
$z^i \bar{z}^j$	$-(i+j)(i+j-1)a_{ij}$
$z^{i-2} \bar{z}^{j+1}$	$i(i-1)a_{ij}$
$z^{i+1} \bar{z}^{j-2}$	$j(j-1)a_{ij}$
$z^{i-1} \bar{z}^{j-1}$	$2ija_{ij}$

and omit negative exponents. Call the resulting coefficients b_{ij} . Equation (6.6) makes $b_{ij} = 0$ for $i+j \leq 110$, exactly. The remaining rational sum satisfies

$$\sum_{i,j} |b_{ij}| R_W^{i+j} < 6.739223 \cdot 10^{-13} < 7 \cdot 10^{-13}. \quad (6.9)$$

This proves $|\mathcal{L}_W H_W| < 7 \cdot 10^{-13}$ on $|z| \leq R_W$.

The geometry of the shell has similarly direct bounds. Write

$$r_* = \sum_k |c_k| = .35670524, \quad r_1 = \sum_k 3k|c_k| = .33437745, \quad r_2 = \sum_k (3k)^2 |c_k| = 2.13289155.$$

The lower triangle inequality gives $\rho \geq .20947544 > .2$, whereas $\rho \leq r_* < .357$. Formula (6.1) therefore implies $q_i > .095$ throughout Ω_W . For $z(\theta) = \rho(\theta)e^{i\theta}$,

$$|z'| \leq r_* + r_1 = .69108269, \quad |z''| \leq r_* + 2r_1 + r_2 = 3.15835169 < 3.16, \quad |q_i''| < 2.11. \quad (6.10)$$

Differentiating each monomial in (6.7) gives $|(H_W \circ z)''| \leq S_2 |z'|^2 + S_1 |z''| < 30.348$. One can also verify the needed baseline bound without cancellation. For a single summand $f(t) = (\gamma_0 - 1) \operatorname{en}(t) + (2 - \gamma_0) \operatorname{en}(1 - t)$, on $.095 < t < .572$ we have $|f'(t)| < 2.5$ and $|f''(t)| < 5.1$. These follow, for example, from $1.32 < \gamma_0 < 1.33$, $-\log(.095) < 2.4$, and $-\log(.428) < .9$. Using $|q_i'| \leq 2|z'|/3$ and $|q_i''| \leq 2|z''|/3$ gives

$$\left| \frac{d^2}{d\theta^2} (F_0(q(\theta)) + H_W(z(\theta))) \right| < 49.387 < 68. \quad (6.11)$$

The slightly larger bound 68 will be used in interpolation.

6.3 What is certified by the finite library

We next describe the mathematical objects in the library. A record fixes a finite tree of completion, parity, binary-separation, and side-permutation operations, with rational conditional distributions on its branches. Proposition 3.11 compiles this hierarchy into a family of finite realizations using growing exact-type populations and efficient GROUP parameters, preserving arbitrary payloads and the canonical bank convention. The *utility of a record* is the limiting attainable score of this family. Every positive accuracy is achieved with finite supplies, so the record gives a lower bound on the closed value U . The arithmetic below evaluates these laws and limiting utilities; the final strict excess over $\log 3$ will later select an actual finite realization.

Here is an explicit description of its word groups, which is also useful for checking the library independently. Number the W labels 0, 1, 2 and define the partial completion map

$$c(0,0) = 0, \quad c(0,1) = c(1,0) = c(1,1) = 1, \quad c(0,2) = c(2,0) = c(2,2) = 2;$$

the pairs (1, 2) and (2, 1) are discarded. A two-position completion has outer label $c(i, j)$. A three-position completion has outer label $c(c(i, j) - k \pmod{3}, r)$, where $k \in \{0, 1, 2\}$; an undefined intermediate label discards the word. Its inner label is deterministic. Independent permutations of the three labels are allowed in the two or three positions. These are precisely finite compositions of the completion restriction. For parity on three positions discard the six words with all labels different. A retained word has a majority label and an odd-count label; on a constant word both equal its sole label. Either label can be the outer label and the other the inner label. A nonconstant outer/inner group has three words and a constant group has one. This is the parity restriction proved earlier, followed by two W separations.

Suppose the block length is $L \in \{2, 3\}$. Let t_i be the outer law, v_{ij} the inner law conditional on i , and C_{ij} the law of the retained words in that group. Let $n(w)$ count the original canonical source labels, obtained by undoing each positional reindexing. The intermediate cyclic rotation changes only the completed-block flag and fiber membership, not these source labels. If the child constructions have limiting utilities u_t, u_{v_i} , the original law and limiting utility are given exactly by

$$q_{\text{out}} = \frac{1}{L} \sum_{i,j} t_i v_{ij} \mathbb{E}_{C_{ij}} n(w), \quad (6.12)$$

$$u_{\text{out}} = \frac{1}{L} \left(u_t + \sum_i t_i u_{v_i} + \sum_{i,j} t_i v_{ij} H(C_{ij}) \right). \quad (6.13)$$

For completion the inner utility is zero. The last term is supplied by Proposition 3.11: once the outer label is direct, resolve the gate refinements in the order of Lemma 3.10, using each gate's own pair of knowing sides. The flattened final word need not be readable by one fixed pair. Exact conditional types fix the populations and canonical histograms of every pool, including under the positional permutations, and the entropy chain rule counts the remaining word law once. For parity, after its two labels are direct, the remaining choice is the primitive three-word group of Lemma 3.8. These are precisely the hypotheses behind the last term of (6.13). Each auxiliary separation has $D - A \rightarrow 0$; hence the limiting utilities of these pure completion/parity records satisfy $u = M \leq H(q) \leq \log 3$.

The following finite menu protocol specifies the candidates used in the arithmetic check. Its deterministic integer implementation is `menu.hpp` in the accompanying certificate sources.

1. For weights $d = (i/64, j/64, 1)$ with $0 \leq i \leq j \leq 64$, the terminal operation deletes a least-weight label and separates the remaining two, with probabilities proportional to their weights, rounded as specified below. The zero pair uses the deterministic record.

2. The initial record repeats two-position completion twelve times, placing a maximum-weight label first at each step, and ends with this terminal operation. Positive integer weights are renormalized at each recursive call, with a lower bound of one integer unit.
3. Fourteen improvement layers may retain an earlier record or assemble earlier records using two-position completion, three-position completion, or either order of the parity labels above. All positional permutations and the three intermediate cyclic offsets are available. A child call also permits terminal binary separation. Candidate comparison uses $u + \sum_i q_i \log d_i$; lookup examines the nearby indices within two of the nearest $64d_i / \max d$ after sorting the weights, and all compatible side permutations. The outer weights in parity may themselves be chosen from the rounded exponentials of such child scores.
4. Each conditional word receives a positive integer weight equal to the rounded product of its positional weights, clamped below by one. The cumulative normalization in (6.14) fixes its *exact* conditional law. Append all six side permutations of the selected final records and the three deterministic records.

The preliminary scores that select operation types, the integer roots or exponentials used as search weights, and ties in selection have no mathematical accuracy requirement: they only choose among the explicitly admissible finite trees just described. In particular, a computed score is never used as the utility of a record. Utilities and laws are evaluated anew by (6.12)–(6.13). The supplied finite program fixes all these selection choices, including ties; the certificate verifies feasible mixtures of its resulting records. Neither global optimization of the records nor optimality of a mixture is needed for the lower bound.

6.4 Arithmetic errors and exact distributions

Let $B = 2^{52}$. For positive integer word weights $w_1, \dots, w_m$ set

$$p_j = \frac{1}{B} \left(\left\lfloor \frac{B \sum_{k \leq j} w_k}{\sum_k w_k} \right\rfloor - \left\lfloor \frac{B \sum_{k < j} w_k}{\sum_k w_k} \right\rfloor \right). \quad (6.14)$$

These probabilities are nonnegative and telescope to one. Zero-probability words are omitted. This formula, not an unrounded idealized weight law, defines every conditional distribution used by a record.

The evaluator represents a real number by an integer divided by B . Multiplication and division truncate toward zero, and an input constant x is stored as $\lfloor Bx + 1/2 \rfloor$. Logs are evaluated by reducing the argument to $x \in [1, 2]$ with powers of two and using

$$\log x = 2 \sum_{j=0}^{39} \frac{t^{2j+1}}{2j+1} + \varepsilon(x), \quad t = \frac{x-1}{x+1}, \quad 0 \leq \varepsilon(x) \leq \frac{2 \cdot 3^{-81}}{81(1-1/9)}. \quad (6.15)$$

Since $|t| \leq 1/3$, summing the geometric propagation of rounding errors in Horner evaluation gives an error below $7/B$ on $[1, 2]$. For a probability represented at scale B , reduction uses at most 52 doublings. Keeping the factor p_j in $-p_j \log p_j$, rather than bounding all reduced-log errors by their worst case separately, gives entropy error less than $400/B$ for any conditional word distribution here (which has at most 27 words). For example, $\sum_j p_j |h_j| \leq 1 + H(p) / \log 2 \leq 1 + \log_2 27$, where h_j is the reduction exponent; the log errors and the at most 27 final product roundings are already less than $100/B$. The allowance $400/B$ includes the other normalization and evaluation roundings.

Let e_r^{init} be the ℓ^1 error between the exact law of a record and its stored law after r initial completion steps, and let f_r^{init} be its utility error. The terminal binary law is exact at scale

B , with utility error at most $400/B$. In a completion, averaging normalized count vectors contracts the child-law error. There are three groups; rounding the count averages and restoring the missing mass to the first coordinate contributes less than $15/B$. The outer child utility is divided by two. Perturbing its weights in the conditional entropy term contributes at most $2e_r^{\text{init}}$, and the local entropy and arithmetic errors contribute less than $250/B$. Thus

$$e_r^{\text{init}} \leq \frac{15r}{B}, \quad f_r^{\text{init}} \leq \frac{1}{2}f_{r-1}^{\text{init}} + 2e_r^{\text{init}} + \frac{250}{B}. \quad (6.16)$$

At $r = 12$ these give $e_0 \leq 180/B$ and $f_0 \leq 1700/B$ for the improvement layers. (The displayed initial recurrence actually gives $f_{12}^{\text{init}} \leq 1159.991/B$.)

For an improvement layer let e_h, f_h be uniform errors over its records. A parity assembly combines at most an outer law and an inner law, so their errors total at most $2e_{h-1}$. There are at most nine groups. The sum of product and count roundings leaves fewer than 21 missing probability units before restoration; adding that mass to one coordinate costs less than $42/B$ in ℓ^1 . The two child utility contributions are divided by three, giving coefficient $2/3$; completion has the smaller coefficient $1/2$. Inner utilities are at most $\log 3$, and the conditional word entropy is at most $\log 27$. Applying these bounds to (6.13), with the just obtained law-error allowance, gives the conservative recurrence

$$e_h \leq 2e_{h-1} + \frac{42}{B}, \quad f_h \leq \frac{2}{3}f_{h-1} + 3e_h + \frac{250}{B}. \quad (6.17)$$

To see that the last fixed allowance is sufficient, the entropy errors contribute at most $400/(LB) \leq 200/B$; the at most nine group-weight products, their utility products, the three inner-utility products, and the final division contribute less than $50/B$ after normalization. For parity, each conditional group actually has at most three words, so the weight-perturbation estimate in (6.17) has additional slack. Retaining a record from an earlier layer preserves these bounds. At $h = 14$ they give

$$e_{14} \leq \frac{3637206}{B} < 8.5 \cdot 10^{-10}, \quad f_{14} < 3.635 \cdot 10^{-9} < 10^{-8}. \quad (6.18)$$

Lemma 6.1 (Correction to an exact target law). *Let q be a strictly positive probability vector, let $K \geq 0$, and suppose a construction has law $\hat{q}$ and attainable utility $u \leq K$. Then the closed attainable value satisfies*

$$U(q) \geq u - K \max_i (\hat{q}_i - q_i)_+ / q_i.$$

For rational laws, the correction is realized by finite rational time-sharing; for arbitrary real q , the assertion concerns the defining closure.

Proof. Apply Remark 4.5 to the W value with target law q , old law $\hat{q}$, and score bound K . Its singleton mixture gives the stated loss. Rational approximation and closure permit real mixture proportions. $\square$

For mixture evaluation, project the stored first two coordinates and utility down to scale 2^{30} . A feasible triple of projected library points determines nonnegative barycentric weights by exact determinant ratios. Applying those same weights to the true records gives a genuine attainable law $\hat{q}$. The singleton points are in the library, so every projected simplex target admits a feasible triple. The implementation's pivot search is used only to locate a triple with a sufficient utility; feasibility is verified by its nonnegative determinant ratios.

The difference between two floor remainders in a first or second coordinate is less than 2^{-30} ; the third coordinate is their complement. Combining this with (6.18) gives $|\hat{q}_i - q_i| < 2.8 \cdot 10^{-9}$, including the much smaller error in a shell target's trigonometric

evaluation. We may cap the utility claimed for a mixture at $K = 1.5$. Lemma 6.1 then bounds its correction loss on the shell by $1.5(2.8 \cdot 10^{-9})/.095$. Adding the library utility and projection errors proves the following bounds, also recording the grid bound needed later:

Quantity	Absolute error allowance
True library law versus stored law, ℓ^1	$8.5 \cdot 10^{-10}$
True library utility versus stored utility	10^{-8}
Mixture versus target, each coordinate	$2.8 \cdot 10^{-9}$
Mixture evaluation and correction, $q_i > .095$	$.07 \cdot 10^{-6}$
Mixture evaluation and correction, $q_i > .03$	$.16 \cdot 10^{-6}$
H_W at a shell sample	10^{-9}
F_0 at an exact stored law	10^{-11}

The last two bounds can be checked directly as follows. Store the scaled polynomial coefficients $a_{ij}R_W^{i+j}$, evaluate in polar coordinates with $r/R_W < 1$, and sum the finitely many roundings. There are 6441 monomials, degree at most 112, and (6.8) bounds the effects of coordinate error. For cosine, reduce the angle to $[0, \pi]$ and use its degree-48 Taylor polynomial in Horner form. Its Taylor remainder and propagated fixed-point errors total less than $300/B$. Coefficient rounding, at most 112 radial multiplications, and the two products per monomial give an error well below 10^{-9} , including this trigonometric error. Applying (6.15) to the six entropy terms in (6.3) gives the stated 10^{-11} allowance.

6.5 The finite checks and the continuum conclusion

The finite shell check uses

$$\theta_l = \frac{l\pi}{6480}, \quad 0 \leq l \leq 2160. \quad (6.19)$$

At each angle it forms $r = \rho(\theta_l)$ and the target law by (6.1), evaluates a feasible library mixture as above, and subtracts the stored values of F_0 and H_W . Thus the finite statement to check is a collection of 2161 integer inequalities: if g_l is the resulting gap in units of B^{-1} , then

$$\min_l g_l = -4180063084, \quad 10^6 g_l > -5B \quad (0 \leq l \leq 2160). \quad (6.20)$$

In particular the recorded minimum is exactly $-4180063084/2^{52}$ in utility units, approximately $-0.928160457 \cdot 10^{-6}$. This finite arithmetic result is reproducible from the coefficient and menu specifications above and the supplied integer checker. Its mathematical meaning is a lower bound from feasible constructions, with the proved error allowances; it is not a claim that the numerical candidate is an optimizer. Using the conservative threshold $-5 \cdot 10^{-6}$ in (6.20), subtracting the preceding evaluation and exact-law correction allowances, and transporting from rounded targets to exact sample targets gives

$$U(q(\theta_l)) - F_0(q(\theta_l)) - H_W(z(\theta_l)) > -5.2 \cdot 10^{-6}. \quad (6.21)$$

It remains to fill the intervals between these samples; no sampling assumption is made here. If a twice differentiable scalar function has $|f''| \leq M$ on an interval of length h , its difference from its linear interpolant is at most $Mh^2/8$. Mix the two sample constructions with their linear interpolation weights. Their law is the chord between the two sample laws. By (6.10), each coordinate differs from the desired curved law by at most $2.11h^2/8$. Correct this discrepancy using Lemma 6.1; since $q_i > .095$, the utility loss is at most $1.5(2.11/.095)h^2/8$. The interpolation error of $F_0 + H_W$ is at most $68h^2/8$ by (6.11). With $h = \pi/6480$ their total is

$$\left(68 + \frac{1.5 \cdot 2.11}{.095}\right) \frac{h^2}{8} < 2.977 \cdot 10^{-6} < 3.1 \cdot 10^{-6}. \quad (6.22)$$

Both the shell and the polynomial are invariant under rotation by $2\pi/3$ and conjugation: the recurrence preserves $i - j \equiv 0 \pmod{3}$ and has real coefficients. The value is invariant under label permutations. These symmetries extend the interval $[0, \pi/3]$ to the entire boundary.

Proposition 6.2 (Certified W seed). *With the preceding exact coefficients and domain,*

$$U(q) \geq F_0(q) + H_W(z) - 9 \cdot 10^{-6} \quad (z \in \partial\Omega_W),$$

and

$$U(q) \geq F_0(q) + H_W(z) - 22 \cdot 10^{-6} \quad (z \in \Omega_W). \quad (6.23)$$

Proof. Equations (6.21) and (6.22) give the boundary assertion, with room to spare. Put

$$f = (1 - 10^{-10})F_0 + H_W - 21 \cdot 10^{-6}.$$

Since $F_0 \geq 0$, this is below the boundary lower bound. Furthermore

$$\mathcal{L}_W f \geq -\gamma_0 + 10^{-10}\gamma_0 - 7 \cdot 10^{-13} > -\gamma_0.$$

If $f - U$ had a positive maximum in the interior, subtracting that maximum from f would produce a smooth lower test for U , contrary to Corollary 5.5. Therefore $U \geq f$ inside. Finally $F_0 < 2$, so the loss $10^{-10}F_0$ is absorbed by the remaining 10^{-6} in (6.23). $\square$

At the target exponent $\beta = 1129/500$, we have $3/\beta > \gamma_0$. Proposition 4.7 therefore transfers both the baseline and the seed to $U_{3/\beta}$. Together with Proposition 4.10, this is the promised attainable lower bound on the five-component face.

7 A rational certificate for the comparison bound

We now construct a smooth family of six-component laws and a polynomial that lies below the attainable value on its boundary. The polynomial has slightly more curvature than the lower-test inequality permits at an interior contact point. This forces it below the attainable value throughout the family. Its value at the center is strictly greater than $\log 3$.

There are two numerical difficulties. First, the chart is only approximately harmonic, so we must retain the supporting-covector term in Theorem 5.1. Second, a finite list of point evaluations does not prove a differential or boundary inequality on a continuous region. We address the first by moving the boundary slightly into the positive simplex, and the second by coefficient majorants, polynomial interval bounds, and interpolation of actual attainable constructions. All decimal constants in this section that define polynomials or arithmetic thresholds denote exact rational numbers.

Theorem 7.1 (The certificate bound). *Let $\beta = 1129/500$. The construction class defining F_β satisfies*

$$F_\beta(\alpha_*) \geq 1.098619383270680538 > \log 3,$$

where $\alpha_* = (p, p, p, 1/3 - p, 1/3 - p, 1/3 - p)$ and

$$p = 2^{-500} \left\lfloor 2^{500} \frac{18623}{100000} + \frac{1}{2} \right\rfloor.$$

The proof occupies this section. The finite checks are specified together with analytic bounds that turn their integer conclusions into uniform inequalities. Section B gives the implementation and reproduction interface. In particular, the computations used below are finite certificate verifications; they do not presume convergence of a numerical optimizer or of an infinite harmonic series.

7.1 The chart and its exact coefficients

Put $\zeta = \exp(2\pi i/3)$ and $R = 31/25$. For a real polynomial P on the complex plane define, with subscripts modulo three,

$$r_i(z) = \frac{1 + \Re(\zeta^i z)}{3}, \quad p_i(z) = P(\zeta^i z), \quad E_i(z) = r_i(z) + p_i(z) - p_{i+1}(z) - p_{i+2}(z). \quad (7.1)$$

Their six-coordinate sum is one. In the six-state support, their mean satisfies $\mu_i = 1 - r_i$. Thus

$$z = 3r_0 - 1 + i\sqrt{3}(r_2 - r_1)$$

is an invertible affine coordinate on the mean plane. Whenever all six coordinates are positive, (7.1) is a chart to which Theorem 5.1 applies.

Lemma 7.2 (Complex covariance normalization). *For $\alpha = (p_0, p_1, p_2, E_0, E_1, E_2)$, set*

$$r_i = 1 - \mu_i, \quad z = x + iy = 3r_0 - 1 + i\sqrt{3}(r_2 - r_1), \quad D_i = r_i(1 - r_i) + 2p_i.$$

Then $D_i = \text{Var}(g_i)$. In the (x, y) coordinates the covariance contraction has matrix

$$C = \begin{pmatrix} 9D_0 & 3\sqrt{3}(D_1 - D_2) \\ 3\sqrt{3}(D_1 - D_2) & -3D_0 + 6D_1 + 6D_2 \end{pmatrix}. \quad (7.2a)$$

Writing

$$d = \frac{D_0 + D_1 + D_2}{3}, \quad f = \frac{D_0 + \zeta D_1 + \zeta^2 D_2}{3},$$

the operator is

$$\mathcal{L}u = 36(d u_{z\bar{z}} + f u_{zz} + \bar{f} u_{\bar{z}\bar{z}}). \quad (7.2b)$$

Proof. Since g_i takes values zero, one, and two, with probability p_i of the last, $\mathbb{E}g_i^2 = \mu_i + 2p_i$. Hence $\text{Var}(g_i) = \mu_i(1 - \mu_i) + 2p_i = D_i$. For the centered grade $\xi = g - \mu$, the sum of the three coordinates is zero, so

$$\mathbb{E}\xi_i \xi_j = \frac{D_k - D_i - D_j}{2} \quad (\{i, j, k\} = \{0, 1, 2\}).$$

The induced coordinate increment is $\Delta z = -3\xi_0 + i\sqrt{3}(\xi_1 - \xi_2)$. Substitution gives (7.2a), and consequently

$$\mathbb{E}|\Delta z|^2 = 18d, \quad \mathbb{E}(\Delta z)^2 = 36f.$$

For Wirtinger derivatives $\partial_z = (\partial_x - i\partial_y)/2$ and $\partial_{\bar{z}} = (\partial_x + i\partial_y)/2$, a directional derivative is $\Delta z \partial_z + \Delta \bar{z} \partial_{\bar{z}}$. Squaring it and taking expectations proves (7.2b). In particular the coefficient 36 belongs to the full contraction used in Theorem 5.1; it is not the generator divided by two. $\square$

Here is a finite rational definition of P . For integers $N \geq 8$ and $b \geq 1$, put $Q = 2^b$ and $\text{rnd}(t) = \lfloor t + 1/2 \rfloor$. Let

$$(Z_0, \dots, Z_7) = (18623, -2696, -2677, 194, 393, 17, 1, 8), \quad Z_k = 0 \quad (k \geq 8).$$

We define symmetric integers $A_{ij} = A_{ji}$ for $i + j \leq N$. The pure coefficients are

$$A_{0k} = A_{k0} = \text{rnd}(QZ_k/100000).$$

Let $c_{00} = 8$, $c_{10} = c_{01} = 2$, $c_{20} = c_{02} = -1$, and $c_{11} = -2$, with every other c_{ij} zero; after computing A_{ij} set $C_{ij} = 72A_{ij} + Qc_{ij}$. For $1 \leq i \leq j$, in increasing total degree, form the following sum. For $0 \leq x < i$, $0 \leq y < j$, $(x, y) \neq (0, 0)$, put

$$h = 1 + y - x \pmod{3}, \quad h \in \{0, 1, 2\}, \quad a = i - x - 1 + h, \quad c = j - y + 1 - h,$$

and set

$$w_0(a, c) = c(c - 1), \quad w_1(a, c) = ac, \quad w_2(a, c) = a(a - 1).$$

Then define

$$S_{ij} = \sum_{\substack{0 \leq x < i, 0 \leq y < j \\ (x, y) \neq (0, 0)}} C_{xy} A_{a, c} w_h(a, c), \quad A_{ij} = \text{rnd} \left(-\frac{S_{ij}}{C_{00}ij} \right), \quad A_{ji} = A_{ij}. \quad (7.3)$$

The indices a, c are nonnegative, and both $a + c$ and $x + y$ are smaller than $i + j$. Thus (7.3) is a definition by finite recursion. For both choices used here, $21 < C_{00}/Q < 22$, so its denominator is positive. Finally put

$$P_{N, b}(z) = 2^{-b} \sum_{i+j \leq N} A_{ij} z^i \bar{z}^j, \quad P = P_{780, 500}, \quad P^{\text{ref}} = P_{420, 224}. \quad (7.4)$$

The chart used in the proof is defined by P . The smaller polynomial P^{ref} makes the finite evaluations less expensive; uniform bounds on its difference from P will transfer those evaluations to the chart. The symmetry of the coefficients makes these polynomials real-valued and invariant under conjugation. Rotation permutes the three functions p_i ; it need not fix P itself.

7.2 Uniform residual and polynomial comparisons

The reason for the recursion is most transparent at the coefficient level. For the chart $P_{N, b}$ form

$$\mathcal{C}(z, \bar{z}) = \sum_{i, j} (C_{ij}/Q) z^i \bar{z}^j = 36[r_0(1 - r_0) + 2P_{N, b}].$$

Write $\mathcal{C}^{[s]}$ for the terms whose exponent difference is s modulo three. Then

$$\mathcal{L} = \mathcal{C}^{[0]} \partial_z \partial_{\bar{z}} + \mathcal{C}^{[2]} \partial_z^2 + \mathcal{C}^{[1]} \partial_{\bar{z}}^2.$$

The coefficient of $z^{i-1} \bar{z}^{j-1}$ in $\mathcal{L}P_{N, b}$ is

$$\frac{C_{00}ijA_{ij} + S_{ij}}{Q^2}.$$

Indeed, the three second derivatives give exactly the shifts and factors in (7.3). A term outside $x < i$, $y < j$ has zero derivative factor. Every other coefficient on the right has already been computed. Nearest rounding therefore leaves a defect of absolute value at most $C_{00}ij/(2Q^2) < 36ij/Q$. Reflection handles $i > j$.

For any polynomial $F = \sum f_{ij} z^i \bar{z}^j$, the elementary inequality

$$\sup_{|z| \leq R} |F(z)| \leq \sum_{i, j} |f_{ij}| R^{i+j} \quad (7.5)$$

provides a uniform, rather than sampled, bound. A convenient majorant for all low-degree rounding defects is

$$\rho = \sum_{k=2}^N \frac{36(k+1)k^2 R^{k-2}}{Q}.$$

For the remaining products put, for $s = 0, 1, 2$,

$$D_{s,k} = \frac{R^k}{Q} \sum_{\substack{a+c=k \\ a-c \equiv s \pmod{3}}} |C_{ac}|, \quad V_{s,m} = \frac{R^{m-2}}{Q} \sum_{a+c=m} |A_{ac}| v_s(a, c),$$

where $(v_0, v_1, v_2) = (ac, c(c-1), a(a-1))$. A product with covariance degree k and differentiated chart degree m has degree $k + m - 2$. Every coefficient outside the recursion is consequently included in

$$\sup_{|z| \leq R} |\mathcal{L}P_{N,b}(z)| \leq \rho + \sum_{s=0}^2 \sum_{k+m > N} D_{s,k} V_{s,m}. \quad (7.6)$$

All terms on the right are nonnegative rational numbers. They can be bounded using integer arithmetic: set $U = 2^{160}$, replace each $D_{s,k}$ and $V_{s,m}$ by its upper integer multiple at scale U , round each summand of ρ upwards at the same scale, and divide the two sums by U^2 and U , respectively. For $N = 780, b = 500$ this gives

$$\epsilon := \sup_{|z| \leq R} |\mathcal{L}P(z)| < 3.328667 \cdot 10^{-22} < 3.329 \cdot 10^{-22}. \quad (7.7)$$

The exact upper numerator and denominator, as well as the complete coefficients regenerated by (7.3), form part of the finite certificate. Thus the small low-degree defect is attached to the defined polynomial, not assumed for an arbitrary supplied coefficient list.

For two polynomials, with missing coefficients interpreted as zero, define

$$\Delta_a = \sum_{i,j} |P_{ij} - P'_{ij}| R^{i+j} (i+j)^a, \quad a = 0, 1, 2,$$

using $(i+j)^0 = 1$ also at the constant term. A unit real directional derivative of a monomial costs at most $i+j$ and a second one at most $(i+j)(i+j-1)$. Hence

$$|P - P'| \leq \Delta_0, \quad \|\nabla(P - P')\| \leq \Delta_1/R, \quad \|D^2(P - P')\|_{\text{op}} \leq \Delta_2/R^2. \quad (7.8)$$

Computing the coefficient differences exactly before taking absolute values, and rounding the resulting positive sums upwards, gives the following bounds on the entire closed disk:

Quantity	Upper bound
$ P - P^{\text{ref}} $	$3.517 \cdot 10^{-17}$
$\ \nabla(P - P^{\text{ref}})\ $	$1.233 \cdot 10^{-14}$
$\ D^2(P - P^{\text{ref}})\ _{\text{op}}$	$4.327 \cdot 10^{-12}$
$ P - P^{\text{ref}}_{\leq 236} $	$1.902 \cdot 10^{-11}$
$ P - P^{\text{ref}}_{\leq 220} $	$7 \cdot 10^{-11}$

Here the subscript denotes truncation by total degree. The same sums with factors k and $k(k-1)$ prove, for either full or reference chart,

$$\|\nabla p_i\| < .592, \quad \|D^2 p_i\|_{\text{op}} < 3.05. \quad (7.9)$$

Truncating either chart at degree 40 changes its value by less than .0002 and its gradient by less than .006. These derivative estimates will control interpolation and root locations; value closeness alone would not suffice.

7.3 Geometry of the domain

Work first in the sector $0 \leq \theta \leq \pi/3$, writing $z = re^{i\theta}$. Define its probability domain by $r \leq R$ and $E_1 \geq 0$. We shall prove that the other five coordinates are uniformly positive there. Rotations and conjugation then provide the full domain Ω .

From (7.9),

$$\|\nabla E_i\| < G := \frac{1}{3} + 3(.592) < 2.11, \quad \|D^2 E_i\|_{\text{op}} < 9.15. \quad (7.10)$$

Divide the sector into 480 radial and 480 angular cells, with centers

$$r_j = \frac{R(2j+1)}{960}, \quad \theta_l = \frac{(2l+1)\pi}{2880}, \quad 0 \leq j, l < 480.$$

Evaluate the degree-40 reference chart at these centers, including its radial derivative and unit-tangential derivative $r^{-1}\partial_\theta$. The following are the exact integer center tests, after division by the arithmetic scale:

$$\begin{aligned} p_i &\geq .0516 \quad (i = 0, 1, 2), & E_0, E_2 &\geq .055, \quad d \geq .488, \quad (7.11) \\ E_1 < .018 &\implies -\partial_r E_1 \geq .183 \quad \text{and} \quad \|\nabla E_1\| \geq .3208, \\ E_1 \geq -.008 &\implies (9d - 2)^2 \geq 18 \sum_{i=0}^2 (D_i - D_{i+1})^2. \end{aligned}$$

For completeness, their arithmetic specification uses $T = 2^{45}$, the rational approximation $3141592653589793238/10^{18}$ to π , and sine and cosine series through degree 64 on arguments reduced to $[0, 2\pi)$. Every division in these recurrences is rounded down. The Taylor remainder is below 10^{-30} ; propagation of rounding errors is bounded by $65e^{2\pi}/T$. Including the argument approximation gives error below $2 \cdot 10^{-9}$ in either trigonometric value. There are 861 possible chart coefficients through degree 40; their absolute majorant is below .921, and the first-derivative majorant is below .592. Product and summation errors consequently leave a common allowance 10^{-7} for the displayed scalar fields and gradient vectors. These rules specify finite integer evaluations at all 230400 centers, and all tests (7.11) hold.

Let $b = \pi/2880$. Every point of a cell is within distance

$$\delta \leq \sqrt{(R/960)^2 + R^2 b^2} < .001870298$$

of its center. Adding the truncation tails, arithmetic allowance, and Lipschitz variation gives

$$\begin{aligned} p_i &> .0516 - .0002 - .592\delta - 10^{-7} > .0502926, \\ E_0, E_2 &> .055 - .0006 - G\delta - 10^{-7} > .0504548. \end{aligned} \quad (7.12)$$

If the true E_1 at a point is at most .01, its truncated center value is below $.01 + .0006 + G\delta + 10^{-7} < .018$. Thus the derivative tests apply. Using the Hessian variation and the degree-40 gradient tail yields

$$\begin{aligned} -\partial_r E_1 &> .183 - .018 - 9.15\delta - Gb - 10^{-7} > .1455857, \\ \|\nabla E_1\| &> .3208 - .018 - 9.15\delta - 10^{-7} > .2856866. \end{aligned} \quad (7.13)$$

Only the first line needs the term Gb , accounting for rotation of the radial unit vector between the point and the center.

In particular $-\partial_r E_1 > .144$ wherever $E_1 \leq .01$. At the origin, $E_1 = 1/3 - P(0) > .1471$. Once a ray reaches .01, its value decreases strictly while below that level and cannot cross

upwards through it. Consequently each ray has at most one zero. Let $r_*(\theta)$ be this zero if it lies below R , and R otherwise. Implicit differentiation on the smooth face and clipping at R give

$$|r'_*| \leq GR/.144 < 18.2, \quad |r''_*| \leq \frac{9.15((r'_*)^2 + R^2) + 2.11(2|r'_*| + R)}{.144} < 22000, \quad (7.14)$$

where the second inequality is used only on intervals wholly in a smooth face. The clipped radius is Lipschitz; its second derivative is not used across a face-cap junction.

The eigenvalues of the matrix in (7.2) are

$$9d \pm 3\sqrt{2\sum_i (D_i - D_{i+1})^2}.$$

The center tests give a smaller eigenvalue at least 2; $d \geq .488$ ensures the required sign before taking square roots. A cell containing a point with $E_1 \geq 0$ has computed center $E_1 > -.004546$, so the test is active. Throughout the sector $r_i \in [-.08, .746667]$, and hence

$$\|\nabla D_i\| \leq 1.16/3 + 2(.592) < 1.571.$$

The change from the truncated center to the full value is less than $2(.0002) + 1.571(.001871) + 10^{-7} < .00334$. If all D_i change by at most t , then $\|\Delta C\|_{\text{op}} \leq 15t$: the three coefficients of D_i in a unit-direction quadratic form are $3 + 6\cos(2a + 2\pi i/3)$; they sum to 9, at most one is negative, and a negative coefficient has magnitude at most 3. Their absolute sum is at most 15. We conclude

$$\lambda_{\min}(C) > 2 - 15(.00334) > 1.949 \quad \text{on } \overline{\Omega}. \quad (7.15)$$

All full/reference differences in (7.8) fit within the remaining strict slacks, so these conclusions hold for $P_{780,500}$.

7.4 The barrier and the interior inequality

Finite Taylor polynomials, explicit remainder bounds, and subdivision into certified boxes are standard techniques of validated numerics; see [21, Section 2 and Algorithm 1] and [26, Section 2]. We specify the integer arithmetic and all error allowances for the present certificate below.

Let

$$\mathcal{I} = \{(i, j) : 0 \leq j \leq i, i + j \leq 22, i - j \equiv 0 \pmod{3}\}.$$

The 52 exact rational coefficients a_{ij} in Table 1 define

$$\varphi(z) = \Re \sum_{(i,j) \in \mathcal{I}} a_{ij} z^i \bar{z}^j. \quad (7.16)$$

For $i > j$, the real part gives $a_{ij}(z^i \bar{z}^j + z^j \bar{z}^i)/2$; a diagonal term occurs once. This polynomial is invariant under $z \mapsto \zeta z$ and conjugation. Its constant term is 1.098619383270680538.

Lemma 7.3 (Interior certificate). *On $\overline{\Omega}$,*

$$\mathcal{L}\varphi > -1 + 9.8 \cdot 10^{-8}.$$

Proof. We specify a finite interval verification and its error accounting. Let $q = P_{\leq 236}^{\text{ref}}$ and $w = z/R$. By (7.8), $|P - q| < 1.902 \cdot 10^{-11}$ on the disk; we reserve the looser allowance 10^{-10} below. Define

$$A(w) = 36\varphi_{z\bar{z}}(Rw), \quad B(w) = 36\varphi_{zz}(Rw).$$

Differentiating the real-part expression (7.16) assigns a coefficient factor $18R^{i+j-2}$ to each conjugate contribution. If $i = j$, the identical contributions combine, correctly counting the original diagonal term once.

The five real fields to be verified are

$$\mathcal{L}_q\varphi, \quad E_{1,q}, \quad A, \quad \Re B, \quad \Im B. \quad (7.17)$$

The first supplies the differential lower bound. The second identifies boxes lying outside Ω , and the last three bound the error in the operator when q is replaced by P . The subscript q means that the chart probabilities and covariance are formed from q . Explicitly, put $r = (1 + \Re z)/3$, $D = r(1 - r) + 2q$, and let Π_s retain exponent differences congruent to s modulo three. Then

$$d_q = \Pi_0 D, \quad f_q = \Pi_2 D, \quad \mathcal{L}_q\varphi = d_q A + 2\Re(f_q B), \quad E_{1,q} = (r + 2q)(\zeta z) - 3\Pi_0 q(z).$$

All derivative polynomials are evaluated at $w = z/R$.

Convert rational coefficients to integers at scale $U = 2^{115}$ by truncation toward zero. Divisions in polynomial products are rounded down. Every constituent field has degree below 320 and fewer than 52000 possible monomials; the checked coefficient absolute sums, divided by U , are below 2^{36} . Before quantization, $2^{36} + 1$ is a valid bound. The sums of coefficient-quantization errors and product-division errors are therefore below $6 \cdot 10^{-19}$. These errors concern the functions on $|w| \leq 1$, by (7.5).

Put $N = 128$, $J = 14$, $h = \pi/(6N)$. The boxes

$$z = R \frac{2u + 1 + v}{2N} \exp(i(2s + 1 + t)h), \quad |t|, |v| \leq 1, \quad 0 \leq s, u < N, \quad (7.18)$$

cover the whole closed sector. For each field, expand in t and v to degree J in each variable. At scale $T = 2^{69}$ the resulting integer array $a = (a_{ij})_{0 \leq i, j \leq 14}$ represents $T^{-1} \sum a_{ij} t^i v^j$.

Here are explicit arithmetic rules for generating these arrays. Evaluate $\arctan(1/m)$ by the first 36 terms of its alternating power series, flooring each unsigned term magnitude at scale U before applying its alternating sign, and form

$$h_U = \left\lfloor \frac{16 \arctan_U(1/5) - 4 \arctan_U(1/239)}{6N} \right\rfloor.$$

For a monomial $w^i \bar{w}^j$, let $m = i - j$. Reduce the phase index $m(2s + 1)$ modulo $12N$, multiply by h_U , and evaluate sine and cosine with terms of orders $0, \dots, 83$. Generate the order- k angular Taylor coefficient by differentiating and multiplying by $mh_U/(kU)$, rounding each division down. An extra phase index $4Nm$ gives the rotated constituent in $E_{1,q}$. Sum the monomial contributions by total degree and convert once to scale T with floor division. Finally substitute $(2u + 1 + v)/(2N)$ by polynomial Horner evaluation, discarding powers above J in v and rounding each division down. These rules produce precisely the arrays for (7.17).

We next bound what these finite Taylor arrays omit. For each constituent with integer coefficients v_{ij} at scale U (so its polynomial coefficients are v_{ij}/U), the following rational inequality is checked:

$$\sum_{i,j} \frac{|v_{ij}|}{U} \left[\frac{((i+j)\pi_+/(6N))^{15}}{15!} + 4 \binom{i+j}{15} (2N)^{-15} \right] < 10^{-10}, \quad \pi_+ = 22/7. \quad (7.19)$$

The first term is a real sine/cosine Taylor remainder. The second bounds the radial remainder; the angular Taylor polynomial has coefficient sum at most $\exp(319\pi/(6N)) < 4$. Taylor's theorem for the radial monomial on $[0, 1]$ supplies its binomial factor. The edge field has two constituents, so its tail bound is doubled.

For clarity, the remaining arithmetic estimates are collected here:

Source of error	Uniform allowance
Coefficient assembly	$6 \cdot 10^{-19}$
Angular recurrence and conversion to scale T	$4 \cdot 10^{-16}$
Taylor truncation, edge/other fields	$2 \cdot 10^{-10} / 10^{-10}$
Radial Horner floor divisions	$320 \cdot 15^2 / T$
At most 28 subdivisions	$28 \cdot 15^3 / T$
Total, for either sign of any field	$3 \cdot 10^{-10}$

To justify the angular line, the Machin computation errs by less than $2/U$ in h , and reduced phases by less than $3100/U$. Taylor recurrence errors in sine and cosine are below $50000/U$, and each of the 15 angular derivative coefficients errs by less than $300000/U$. Multiplying these bounds by the checked coefficient absolute sums gives the stated conversion allowance. The radial substitutions contract the coefficient absolute norm: their affine coefficients are nonnegative and have sum at most one. A child substitution has the same contraction property after absolute values. Consequently the floor errors add over Horner steps and subdivision levels rather than amplifying. There are at most 15^2 entries per Horner step and fewer than 15^3 elementary contributions per subdivision, giving the last two lines.

For an integer array define

$$\ell_\sigma(a) = \sigma a_{00} - \sum_{(i,j) \neq (0,0)} |a_{ij}|, \quad \sigma \in \{-1, 1\}.$$

The represented polynomial is bounded below by $\ell_\sigma(a)/T$ after multiplication by σ . Therefore the exact field satisfies

$$\sigma F \geq \ell_\sigma(a_F)/T - 3 \cdot 10^{-10} \quad (7.20)$$

everywhere in the box. If a bound is insufficient, bisect by $(t, v) = (y, (\varepsilon + x)/2)$, $\varepsilon = \pm 1$, thereby alternating the two subdivision directions. The child coefficient of $y^i x^k$ is

$$\sum_{j \geq k} \left[a_{ij} \binom{j}{k} \varepsilon^{j-k} / 2^j \right].$$

A box is discarded only when

$$\ell_{-1}(a_{E_{1,q}}) > \lfloor T/10^8 \rfloor + 1. \quad (7.21)$$

Its field error and the chart-change allowance $5 \cdot 10^{-10}$ total less than 10^{-8} , so every such box has full-chart $E_1 < 0$.

To certify the interior inequality, observe that

$$|\mathcal{L}_P \varphi - \mathcal{L}_q \varphi| \leq 2 \cdot 10^{-10} |A| + 4 \cdot 10^{-10} (|\Re B| + |\Im B|). \quad (7.22)$$

For each derivative field $F \in \{A, \Re B, \Im B\}$ and either sign put $x_{F,\sigma} = -\ell_\sigma(a_F)$. Use the integer allowance

$$e = 2 \sum_{F \in \{A, \Re B, \Im B\}} \sum_{\substack{\sigma = \pm 1 \\ x_{F,\sigma} > 0}} \left(\left\lfloor \frac{2x_{F,\sigma}}{10^{10}} \right\rfloor + 4 \right).$$

The sum of the two positive negative-part bounds dominates the absolute value of a represented field. Hence e/T bounds the right-hand side of (7.22), up to less than $2 \cdot 10^{-18}$ from the field-approximation errors. The coefficient used for A is deliberately larger than necessary. Each integer cushion exceeds the floor loss in its summand.

The acceptance condition is

$$\ell_1(a_{\mathcal{L}_q\varphi}) > -T + \lfloor T/10^7 \rfloor + e. \quad (7.23)$$

Together with (7.20) and (7.22), reserving $2 \cdot 10^{-9}$ for all evaluation, chart-change, and threshold-rounding errors, it proves $\mathcal{L}_P\varphi > -1 + 9.8 \cdot 10^{-8}$ on the box.

Start with every pair (s, u) in (7.18), apply the two certification conditions, and recursively subdivide any unresolved box. All 128 angular executions terminate with every descendant accepted or discarded by these conditions at depth at most 28. The finite certificate records the complete sector inputs and successful termination for all sectors, not a collection of floating-point sample values. Arithmetic can be interpreted as exact integers with the specified floor divisions. For the fixed-width implementation, array norms are below 2^{109} , Horner products below 2^{117} , and subdivision products below 2^{121} , within signed 128-bit range. Section B records the machine-semantics checks and reproduced finite coverage. Rotations and conjugation cover the other sectors. This proves the lemma. $\square$

7.5 The six-component boundary inequality

We apply the three-state seed to a face of the six-component domain, and the matching construction to its circular cap. It is important to keep these two sources of attainable values distinct.

Put $U = U_{\gamma_0}$, where $\gamma_0 = 300000/225925$. Since $3/\beta > \gamma_0$, Proposition 4.7 permits us to use this lower parameter in the face estimate of Proposition 4.10. At $E_1 = 0$ define

$$Q = (p_2 + E_0, p_0 + E_2, p_1), \quad J = \text{en}(p_2) + \text{en}(p_0) + \text{en}(E_0) + \text{en}(E_2) - \text{en}(Q_0) - \text{en}(Q_1). \quad (7.24)$$

The face lower bound is

$$B_e = \frac{1129}{1500}(U(Q) + J). \quad (7.25)$$

At the cap put $l_h = E_{h+1} + E_{h+2}$. The side- h marginal is $(1 - l_h - p_h, l_h, p_h)$, so Proposition 4.9 gives

$$B_c = \frac{1}{3} \sum_{h=0}^2 [\text{en}(l_h) + \text{en}(p_h) + \text{en}(1 - l_h - p_h)]. \quad (7.26)$$

A finite lower estimate for U . For evaluation in (7.25), use the entropy baseline, the improved shell bound, and finite-library mixtures from Section 6. The library bounds are tabulated on the three-state simplex grid of spacing $1/500$. Within a grid square whose four vertices are in the simplex, bilinear interpolation is a convex combination of the four vertex laws and reproduces the target law exactly. Concavity of U therefore makes the interpolant of lower utilities a valid lower bound. Every grid vertex used here has all coordinates greater than .03.

At each grid vertex subtract $2 \cdot 10^{-6}$ from the computed library utility. Subtract a further $2 \cdot 10^{-8}$ from the maximum of this interpolant and F_0 . Inside the shell also consider $F_0 + H_W - 24 \cdot 10^{-6}$, using a radial guard 10^{-8} to ensure that the point really lies in the shell. Take the maximum of the available bounds and cap it at 1.45. The resulting evaluated estimate is nonnegative. The error recurrences, entropy evaluation, and distribution correction proved above allow at most 10^{-7} additional overstatement; we include it in the boundary arithmetic allowance. Neither selection of the maximum nor downward capping requires the library to be optimal.

Locating boundary points. Set

$$K = 65536, \quad h = \frac{\pi}{3K}.$$

Evaluate the boundary at the $K + 1$ endpoint angles. Use the degree-220 reference chart and arithmetic scale $B = 2^{52}$. The value error relative to P is below $7 \cdot 10^{-11}$, and each E -coordinate error is below $2.5 \cdot 10^{-10}$. The chart coefficient tail was already bounded in Section 7.2. For evaluation, combine conjugate coefficients as real cosines and evaluate radial powers by Horner's rule. The fundamental cosine is computed by the degree-48 Taylor polynomial on $[0, \pi]$ after symmetry reduction, with error at most $300/B$. Higher angular harmonics are formed by the recurrence $c_{s+1} = 2c_1c_s - c_{s-1}$; their error amplification is controlled by the k^2 -weighted coefficient sum. The total absolute coefficient sum of the reference chart on the disk is below .414, and its k^2 -weighted sum is below 6. These majorants bound the additional coefficient, trigonometric, and Horner errors within the stated allowances.

If the computed $E_1(Re^{i\theta})$ is nonnegative, take the clipped radius R ; otherwise perform 38 bisections on $[0, R]$. By (7.13), the combined evaluation and bisection error in $r_*(\theta)$ is below $2 \cdot 10^{-9}$: the dominant term is $(2.5 \cdot 10^{-10})/.144$, and $R2^{-38}$ is smaller than the remaining slack. The same bound holds when a sign uncertainty changes whether the root is clipped.

Use the band

$$b_K = 2.7h + 5 \cdot 10^{-9} \quad (7.27)$$

to classify a complete angular cell from its left endpoint. This is valid because $|\partial_\theta E_1(Re^{i\theta})| < GR < 2.62$. A value below $-b_K$ identifies a full face cell; one above b_K identifies a full cap cell. In the other cells, subdivide into 8192 equal subcells and use the analogous band at that scale. Where the sign remains uncertain, evaluate both bounds.

At an endpoint the face bound is computed whenever the cap value is below $2.1b_K$, and the cap bound whenever it is above $-2.1b_K$. Such a face evaluation may occur at a clipped cap point with $E_1 \neq 0$. Its three-state law is normalized by setting

$$\hat{Q}_0 = p_2 + E_0, \quad \hat{Q}_1 = p_0 + E_2, \quad \hat{Q}_2 = 1 - \hat{Q}_0 - \hat{Q}_1 = p_1 + E_1.$$

On the face, $\hat{Q} = Q$. At a sampled point off the face, use $\hat{Q}$ in the right side of (7.25), with the same internal coordinates and first two coarse coordinates in (7.24). This is an auxiliary comparison expression; the interpolation and displacement estimates below transport it to actual face laws, where Proposition 4.10 supplies attainability.

The band rule ensures that both endpoints required for a full-cell test have been evaluated: for example, a left value below $-b_K$ forces the next value below $-b_K + 2.62h$ plus evaluation error, which is below $2.1b_K$. This also guarantees the face samples near cap points at which the inset will become active. There are six transition cells in the finite check.

Barrier evaluation and derivative bounds. For (7.16), store the scaled radial coefficients at $B_\varphi = 2^{44}$, rounded to nearest integers. Evaluate powers of r/R and the appropriate cosine or sine for angular differentiation. For a term of radial degree $d = i + j$ and angular frequency $s = i - j$, an a -th radial and b -th angular derivative has multiplier $d(d-1) \cdots (d-a+1)s^b$ with the appropriate phase shift. Exact coefficient sums give the following value and derivative error bounds:

$$\text{value error} < 1.2 \cdot 10^{-7}, \quad \text{error in each derivative through order three} < .002. \quad (7.28)$$

One may obtain these by summing, for each monomial, its coefficient rounding error, the at most d radial-power division errors, and its trigonometric error, multiplied by the displayed

derivative factor. The absolute radial coefficient sums weighted by d^k , $k = 0, \dots, 4$, are respectively below

$$35000, \quad 750000, \quad 15000000, \quad 300000000, \quad 6000000000.$$

These conservative rational bounds make the calculation independent of cancellation in the evaluated polynomial.

At every sampled point where a face bound is evaluated, the integer checks in physical polar coordinates give

$$|\partial_r^a \partial_\theta^b \varphi| < \begin{cases} 2.9, & a + b = 1, \\ 79, & a + b = 2, \\ 1999, & a + b = 3. \end{cases} \quad (7.29)$$

When evaluated using normalized radius r/R , the threshold contains R^a , converting it to precisely this physical-coordinate statement. Cap samples satisfy $|\varphi_\theta| < 999$ and $|\varphi_{\theta\theta}| < 4999$ before the errors (7.28) are restored.

A useful uniform fourth-derivative bound is

$$\sup_{r \leq R} |\partial_r^a \partial_\theta^b \varphi| \leq \sum_{\substack{(i,j) \in \mathcal{I} \\ i+j \geq a}} |a_{ij}| \frac{(i+j)!}{(i+j-a)!} R^{i+j-a} |i-j|^b < 435 \cdot 10^6 \quad (a+b=4). \quad (7.30)$$

These are five finite rational sums. Their largest value is less than 434439204, so the stated integer bound has spare margin. Taylor's theorem with (7.30) extends the sampled derivative estimates between boundary samples.

Interpolation on a smooth face. On a full face cell the additional integer secant test is

$$\frac{|\hat{r}_*(\theta+h) - \hat{r}_*(\theta)|}{h} < 3.05,$$

where hats denote the computed physical radii. The coarse bound $|r_*''| < 22000$ and the root-location errors show $|r_*'| < 3.05 + 22000h + 4 \cdot 10^{-9}/h < 3.41$. Substitute this in (7.14) to get

$$|r_*'| < 3.41, \quad |r_*''| < 1000. \quad (7.31)$$

Write $z(\theta) = r_*(\theta)e^{i\theta}$. Then

$$|z'|^2 \leq 3.41^2 + R^2, \quad |z''| \leq 1000 + 2(3.41) + R.$$

Equations (7.9) and (7.10) now give

$$|Q_0''|, |Q_1''| < 2900, \quad |J''| < 30000, \quad |(\varphi \circ z)''| < 5100. \quad (7.32)$$

Here are details for the last two estimates. In differentiating $\text{en}(s)$, use $\text{en}'(s) = -1 - \log s$ and $\text{en}''(s) = -1/s$. The four internal arguments in J exceed .05, and $Q_0, Q_1 > .10$. The first-derivative bounds for an internal p and E are $.592|z'|$ and $2.11|z'|$; their second derivatives are bounded by $3.05|z'|^2 + .592|z''|$ and $9.15|z'|^2 + 2.11|z''|$. The six entropy chain-rule terms are bounded using $|\text{en}'(s)| < 2$ on $[\text{.05}, 1]$ and $|\text{en}'(s)| < 1.31$ on $[\text{.10}, 1]$. Their sum is below 30000. For the barrier, expand its first and second polar derivatives about a sampled endpoint, using (7.29), (7.28), and (7.30). The polar displacement is at most $(3.41 + 1)h + 4 \cdot 10^{-9}$. Throughout the cell the first derivatives are below 2.91 and the second derivatives below 80.3. Therefore

$$|(\varphi \circ z)''| \leq 80.3(3.41 + 1)^2 + 2.91(1000) < 5100.$$

For a function with $|f''| \leq M$, its difference from linear interpolation over an interval of length h is at most $Mh^2/8$. Concavity of U lets us interpolate the closed attainable law–utility pairs at neighboring endpoints, giving a lower bound at the chord law. The first two Q coordinates differ from the curved target by at most $2900h^2/8$. On the actual face the third coordinate is $Q_2 = p_1$; using it directly gives

$$|Q_2''| \leq 3.05(3.41 + 1.24)^2 + .592(1000 + 2(3.41) + 1.24) < 663.$$

Since $Q_0, Q_1 > .10$ and $Q_2 > .05$, the curvature divided by a coordinate lower bound is at most 29000. The exact-law correction of Lemma 6.1 costs at most $1.5 \cdot 29000h^2/8$ in U . Adding the conditional-entropy and barrier interpolation costs gives

$$[5100 + .754(30000 + 1.5(29000))] \frac{h^2}{8} < 2 \cdot 10^{-6}, \quad (7.33)$$

where $.754 > 1129/1500$.

Interpolation on a cap. On an actual cap all entropy arguments in (7.26) exceed .049. The chart derivative bounds and $|z'| = |z''| = R$ give $|B_c''| < 1600$. For example, the gradient bounds for the three marginal arguments are 4.22, .592, and 4.812, and the Hessian bounds are 18.3, 3.05, and 21.35; substituting these in the entropy chain rule already gives the stated bound. The cap sample derivative checks extend to $|\varphi_{\theta\theta}| < 10000$. A direct coefficient bound gives $|\varphi_{\theta\theta\theta}| < 3875770$, so the second derivative can move by less than 62 in one full cell. It follows that the full-cap interpolation loss is less than $(1600 + 10000)h^2/8 < .4 \cdot 10^{-6}$.

Transition subcells. No second derivative across the clipped corner is needed. In a subcell the actual boundary point and its sampled clipped point have complex displacement less than

$$(18.2 + R)h/8192 + 2 \cdot 10^{-9} < 4.1 \cdot 10^{-8}.$$

This changes Q_0, Q_1 by less than $1.12 \cdot 10^{-7}$ each, and their reconstructed third coordinate by less than $2.24 \cdot 10^{-7}$. The singleton correction costs at most

$$\frac{1129}{1500} 1.5 \max \left\{ \frac{1.12 \cdot 10^{-7}}{.10}, \frac{2.24 \cdot 10^{-7}}{.05} \right\} < 5.06 \cdot 10^{-6}.$$

The conditional-entropy differential for one split is $\log((a+b)/a) da + \log((a+b)/b) db$. All internal coordinates remain above .049 on this displacement, so each logarithmic factor is less than 3.1. Summing the four internal-coordinate changes and multiplying by 1129/1500 gives less than $.52 \cdot 10^{-6}$. The local barrier first derivatives, extended from (7.29) with (7.30), are below 3; its change is less than $.13 \cdot 10^{-6}$. Thus the total face *displacement* loss is less than $6.5 \cdot 10^{-6}$. This estimate remains valid if the sampled clipped point is just on the cap: reconstructing Q_2 absorbs its small active-coordinate mass into the five-state face law, and the same correction reaches the actual face law.

For a cap transition, evaluate the smooth entropy expression in (7.26) even if the sampled cap point has $E_1 < 0$. Attainability is asserted only at the actual cap target, where Proposition 4.9 applies; the displacement estimate compares the two entropy expressions. All their arguments stay above .049, and $|B_c'| < 26$. The cap derivative checks, followed by the local second bound just established, give $|\varphi_\theta| < 1001$. The subcell width is below $2 \cdot 10^{-9}$, so displacement together with the cap endpoint arithmetic allowance costs less than $2.4 \cdot 10^{-6}$.

Arithmetic checks and retained margins. The finite evaluations compare the face and cap lower estimates against φ . The smallest recorded integer gaps at scale $B = 2^{52}$ are

$$g_e = 128315602072/B > 28.49 \cdot 10^{-6}, \quad g_c = 152568202355/B > 33.87 \cdot 10^{-6}. \quad (7.34)$$

All 65536 cells, and all 8192 subcells of each of the six transition cells, satisfy the corresponding tests. On full face cells the secant tests used above also hold. The conditional laws and mixture checks used to produce these values are the finite rational constructions of Section 6.

A conservative endpoint arithmetic allowance is $10 \cdot 10^{-6}$ for a face evaluation and $.2 \cdot 10^{-6}$ for a cap evaluation. To see that the former suffices, account separately for the $2 \cdot 10^{-9}$ root-location error with the chart gradient bounds, correct the resulting coarse-law error by singleton dilution with coordinates above .03, include the 10^{-7} possible error in the computed lower estimate for U , and add the entropy and barrier errors. Each coordinate moves by less than $6 \cdot 10^{-9}$, with less than $12 \cdot 10^{-9}$ for the reconstructed coordinate; this dilution costs less than $6 \cdot 10^{-7}$ before the prefactor. The entropy and barrier evaluation estimates above fit well within the remaining allowance. A cap needs no root correction; the chart error in its entropy arguments and the barrier value error give the stated $.2 \cdot 10^{-6}$ bound. All trigonometric, logarithmic, coefficient, and integer-division errors have been included.

We use the recorded face minimum in (7.34) and charge the full face arithmetic allowance in transition cells separately from displacement. The resulting bounds are

Boundary case	Gap used	Arithmetic loss	Interpolation/displacement loss
Full face	$28.49 \cdot 10^{-6}$	$10 \cdot 10^{-6}$	$2 \cdot 10^{-6}$
Transition face	$28.49 \cdot 10^{-6}$	$10 \cdot 10^{-6}$	$6.5 \cdot 10^{-6}$
Full cap	$8 \cdot 10^{-6}$	$.2 \cdot 10^{-6}$	$.4 \cdot 10^{-6}$
Transition cap	$8 \cdot 10^{-6}$	$2.4 \cdot 10^{-6}$ combined	

Every entry leaves more than $5.5 \cdot 10^{-6}$.

All arithmetic has an exact-integer interpretation with specified truncating or floor divisions. In the finite-width implementation, probabilities and counts are below $3B$, and scores and accumulated utilities below $100B$. Projected-law determinants and barycentric numerators have magnitude at most 2^{60} ; utility-gap products are below 2^{94} , ratio-test products below 2^{122} , and barrier products below $10^{36} < 2^{127}$. Narrowing casts are checked. These bounds justify the integer interpretation independently of the additional runtime checks described in Section B. We have proved the following uniform statement.

Lemma 7.4 (Continuous boundary margin). *On the boundary of the full six-state probability domain,*

$$F_\beta(\alpha(z)) \geq \varphi(z) + 5.5 \cdot 10^{-6}.$$

7.6 The positive inset and the comparison argument

The supporting-covector estimate requires all six probabilities to be bounded away from zero. Set $\eta = 2 \cdot 10^{-9}$ and replace each active face $E_i = 0$ by $E_i = \eta$, retaining the cap where it is reached first. Let the resulting domain be Ω_η . It contains the origin, has compact closure, and all six coordinates are at least η there. We now transfer Lemma 7.4 to its boundary.

On a removed terminal segment of a ray, $0 \leq E_1 \leq \eta$ and $-\partial_r E_1 > .144$. Its length is at most

$$\Delta r \leq \eta / .144 < 1.389 \cdot 10^{-8}. \quad (7.35)$$

This also handles an original cap endpoint with $0 \leq E_1 < \eta$. The five other coordinates remain greater than .05 and each changes by less than $2.11\Delta r < 2.94 \cdot 10^{-8}$. Write α_b and α_η for the original and new endpoint laws. With

$$\lambda = \max_{i: (\alpha_b)_i > 0} \left(1 - \frac{(\alpha_\eta)_i}{(\alpha_b)_i} \right)_+,$$

the active coordinate contributes nothing, and $\lambda < 2.94 \cdot 10^{-8} / .05 = 5.88 \cdot 10^{-7}$. If $\lambda > 0$, the vector

$$s = \frac{\alpha_\eta - (1 - \lambda)\alpha_b}{\lambda}$$

is a probability distribution. Mix a construction at α_b with singleton constructions of average law s . Singleton utility is nonnegative, and the original utility is at most $\log 6 < 2$. The loss is below $2\lambda < 1.18 \cdot 10^{-6}$. When $\lambda = 0$ no correction is required. The closure of rational mixtures gives the same conclusion for limiting endpoint laws.

To bound the change of the barrier, every point of the inset segment has a face-check sample within angular distance $h = \pi / (3 \cdot 65536)$. The band test covers even cap endpoints where the inset becomes active. The radial Lipschitz bound, root error, and inset displacement give total polar-coordinate distance at most

$$d_* = 19.2h + 2 \cdot 10^{-9} + 1.389 \cdot 10^{-8} < .000307.$$

Taylor's theorem for $\partial_r \varphi$, using (7.29), (7.28), and (7.30), gives

$$|\partial_r \varphi| < 2.902 + 79.002d_* + \frac{1999.002}{2}d_*^2 + \frac{435 \cdot 10^6}{6}d_*^3 < 3.$$

Thus the barrier changes by less than $3\Delta r < 4.2 \cdot 10^{-8}$. The two transfer losses total less than $1.3 \cdot 10^{-6}$, leaving more than $4.2 \cdot 10^{-6}$ slack on the inset boundary. At unchanged cap points there is no loss. Symmetry treats all faces.

Proof of Theorem 7.1. Let g be a supporting covector of F_β at a point of the inset, normalized by $g \cdot \alpha = F_\beta(\alpha)$. Nonnegativity at simplex vertices implies $g_i \geq 0$. Proposition 4.3 gives $F_\beta \leq \log 6 < 2$. Rotation covariance in (7.2b) and the residual bound (7.7) imply $|\mathcal{L}p_i| \leq \epsilon$ and $|\mathcal{L}E_i| \leq 3\epsilon$. Lemma 5.7 consequently gives

$$|g \cdot \mathcal{L}\alpha| \leq 2 \max_i \frac{|\mathcal{L}\alpha_i|}{\alpha_i} \leq \frac{6\epsilon}{\eta} < 9.987 \cdot 10^{-13} < 10^{-12}. \quad (7.36)$$

The chart and value function are continuous on this compact inset. If $\varphi - F_\beta \circ \alpha$ had a positive maximum, the boundary slack would place it at an interior point. Subtracting the maximum from φ gives a smooth lower test there. Theorem 5.1, Lemma 7.3, and (7.36) would imply

$$1 \leq g \cdot \mathcal{L}\alpha - \mathcal{L}\varphi < 10^{-12} + 1 - 9.8 \cdot 10^{-8} < 1,$$

a contradiction. Therefore $F_\beta(\alpha(z)) \geq \varphi(z)$ throughout the inset. At the origin $\alpha(z) = \alpha_*$, and

$$\varphi(0) - \log 3 = 1.098619383270680538 - \log 3 > 7.094 \cdot 10^{-6} > 0.$$

For example, the logarithm inequality follows from its positive $2 \sum_{j \geq 0} t^{2j+1} / (2j+1)$ series with $t = (3-1)/(3+1) = 1/2$ and a geometric bound on the tail; no floating-point comparison is needed. This proves the theorem. $\square$

8 From the certificate to the exponent bound

We now pass from the strict comparison to the exponent bound. This last step uses the finite construction class defining the attainable value, rather than a limiting tensor.

Proof of the square assertion in Theorem 1.1. Set $\beta = 1129/500$. By Theorem 7.1, at the center α_* of the certified chart,

$$F_\beta(\alpha_*) \geq 1.098619383270680538 > \log 3 + 7.094 \times 10^{-6}.$$

By the definition of F_β as the closure of scores of admissible finite constructions, there is a finite construction whose score is strictly greater than $\log 3$. Its law may approximate α_* ; exact equality of the law is not needed in the rank inequality.

Let its six banks contain N original occurrences each and put $n = 6N$. Let L be its number of nonzero direct labels, R its charged ancillary border-rank bound, and $v = abc$ its common matrix multiplication volume. The construction gives a finite degeneration

$$T^{\otimes n} \otimes \mathcal{A} \supseteq \bigoplus_{\ell=1}^L \langle a, b, c \rangle, \quad \underline{R}(\mathcal{A}) \leq R.$$

All parameters here—including group sizes, control horizons, rational approximants and exact bank sizes—have already been chosen. The score inequality reads

$$\frac{\log L - \log R + (\beta/3) \log v}{n} > \log 3,$$

or $Lv^{\beta/3} > 3^n R$. On the other hand, Theorem 3.2 and $\underline{R}(T) = 3$ give $Lv^{\omega/3} \leq 3^n R$. Since $v \geq 1$, the inequality $\omega \geq \beta$ is impossible. Therefore $\omega < 1129/500$. $\square$

For the rational exponent in this paper, the sufficient finite inequality has the equivalent integer form

$$L^{1500}v^{1129} > (3^n R)^{1500}.$$

Thus the strict comparison does not depend on taking an asymptotic sum inequality at an infinite tensor or discarding an ancillary charge. The argument establishes an arithmetic exponent. It does not estimate the size of a useful finite implementation or its numerical stability.

9 Conditional labels and entropy

Lemma 3.5 counted the total pairing volume supplied by successive labels. We now keep track of which pair of sides supplies each contribution. The input is a finite tree describing which label each pair can read; the output is a matrix-multiplication bound in terms of the three accumulated conditional entropies. Preserving previously read labels on every side is essential when the next reader pair changes.

Definition 9.1. A *readable label tree* for the support of a tensor is a finite rooted partition tree of the support whose leaves are in bijection with the supported monomials. Each monomial follows exactly one path. At each nonterminal node, conditional on the path to that node, the next child is a function of either of two specified side coordinates separately. The reader pair may vary with the node.

For a probability law π on the leaves, write $\mathbb{P}(v)$ for the mass passing through a node v . Assign to each pair UV the rate

$$r_{UV}(\pi) = \sum_{v: \text{reader pair } UV} \mathbb{P}(v) H(\text{child} \mid v), \quad H(p_1, \dots, p_q) = - \sum_i p_i \log p_i.$$

Zero-mass contributions are zero. The sum of the three rates is $H(\pi)$.

This identity is the entropy chain rule [25, Section 6]. The exact conditional counts below use the method of types [13, Section II]; the tensor restrictions and their full auxiliary costs are proved explicitly.

Theorem 9.2 (Conditional-label entropy bound). *If a tensor of cost e^C has a readable label tree, then*

$$W(r_{XY}(\pi), r_{XZ}(\pi), r_{YZ}(\pi)) \leq C \quad (9.1)$$

for every leaf law π .

Proof. First suppose π is rational and take n with all $n\pi_s$ integral. Prescribe the resulting counts at every node. Process nonterminal nodes in any order in which a parent precedes its children.

The induction invariant is that the current tensor is a direct sum over all processed word choices, and that each side identifies the entire processed prefix. Within a summand, the tensor is the original word-support fiber times the dot factors already introduced. At the root this invariant is trivial.

Consider a node v occurring in s positions, with child counts $s_1, \dots, s_q$. Within a prefix summand, all sides know these positions. The two readers can therefore project to child words with those counts and apply Theorem 3.3 with

$$K_v = \binom{s}{s_1, \dots, s_q}.$$

The new tag makes the chosen child word visible on the third side. All previous dot indices are carried unchanged, so the invariant continues.

Exactly one auxiliary tensor is used for this node, even if its positions vary between prefix summands. Indeed,

$$\left(\bigoplus_p T_p \right) \otimes A_{K_v} = \bigoplus_p (T_p \otimes A_{K_v}).$$

The prefix blocks are disjoint on all three sides, so the restrictions within them may depend on p . The number K_v depends only on the prescribed counts, not on the positions of a particular prefix. If $s = 0$, take $K_v = 1$ and do nothing.

At the end, every surviving original word is a specified monomial word with leaf histogram $n\pi$. Its scalar coefficient is nonzero and can be absorbed by a local rescaling in that summand. The remaining tensor is a matrix product of dot factors, with pair sizes

$$l_{UV}(n) = \prod_{v: \text{reader pair } UV} K_v.$$

These sizes do not depend on the particular surviving word. The number of summands is

$$M(n) = \frac{n!}{\prod_s (n\pi_s)!} = \prod_v K_v. \quad (9.2)$$

The second equality is the cancellation of child factorials with parent factorials; it also counts the successive choices of words. Every such word survives all the restrictions.

The full cost of this construction is

$$e^{nC} \prod_v |G_{K_v}|.$$

For the fixed finite tree and law, a node with positive conditional entropy has $\log K_v$ linear in n up to $O(\log n)$; a deterministic node has $K_v = 1$. The logarithmic factorial estimate and [Theorem 3.3](#) therefore give

$$\frac{\log l_{UV}(n)}{n} \rightarrow r_{UV}(\pi), \quad \log \left(\prod_v |G_{K_v}| \right) - \log M(n) = o(n).$$

Apply [Theorem 2.3](#) at each fixed n , divide by n , and use homogeneity and continuity from [Theorem 2.1](#). This proves (9.1). Finally approximate an arbitrary law by rational laws. All the finite entropies, and W , are continuous. $\square$

Corollary 9.3 (Balancing two axes). *Fix a side L . Let S be the sum of the two rates incident to L , and let T be the remaining rate. Then*

$$W(S/2, T, S/2) \leq C. \tag{9.3}$$

For $C > 0$, the normalized pairs $(S/C, T/C)$ from different constructions may be combined convexly and decreased coordinatewise while retaining the normalized bound.

Proof. Average (9.1) with the estimate exchanging the two incident axes, using permutation invariance, subadditivity and homogeneity. The same properties justify convex combinations after normalizing each bound by its cost. Monotonicity permits coordinatewise decrease, and continuity permits closure. $\square$

The auxiliary cost in [Theorem 9.2](#) is not suppressed. Its exponential contribution is precisely what the multiplicity (9.2) cancels after [Theorem 2.3](#). This exact accounting is what permits many successive conditional separations.

10 Recursive completion trees

This section constructs the finite trees to which the entropy bound will be applied. Two small tensors supply the starting points; a local degeneration preserves three readable color flags through repeated completions, and a final deletion makes the entire tree readable.

We use colors B, A, C owned by X, Z, Y , respectively. A *three-flag tensor* is a W -partition, as in [Section 3.4](#), together with a conditional readable tree on the support of each color. Thus exactly one color is assigned to each supported monomial, and its owning side reads whether that color's flag is on. The additional trees will let us continue separating labels after a completion has identified a color word.

10.1 Two starting tensors

The second starting tensor is the five-term tensor T' from (3.9), obtained from the $q = 1$ Coppersmith–Winograd tensor [10, Section 7]. We retain its color flags and make the conditional trees explicit. The formulas below give short polynomial witnesses for both starting budgets.

The first base has budget $R = 2$ and support

$$B = \{100\}, \quad A = \{001\}, \quad C = \{010\}.$$

It is the coefficient of order one in

$$(x_0 + tx_1)(y_0 + ty_1)(z_0 + tz_1) - x_0y_0z_0,$$

whose constant coefficient is zero. Its conditional trees are trivial.

The second base has budget $R = 3$ and support

$$A = \{002, 011\}, \quad B = \{200, 110\}, \quad C = \{020\}. \quad (10.1)$$

The triples give indices on X, Y, Z , with unit coefficients. The flag tests are $z > 0$, $x > 0$, and $y = 2$. Within A , the choice is read by Y, Z ; within B , it is read by X, Y .

To verify the budget, put

$$P(t) = (x_0 + tx_1 + t^2x_2)(y_0 + ty_1 + t^2y_2)(z_0 + tz_1 + t^2z_2).$$

The first coefficient of $P(t) + P(-t) - 2P(0)$ is twice the sum of the six monomials with index sum two; the family has rank at most three. Now give weight one to x_1, z_1, y_0, y_2 and weight zero to all other variables. The term 101 has total weight three and the other five terms have weight one. The leading part is exactly (10.1).

10.2 Completion and termination

Definition 10.1. A completion of size $m \geq 2$ with center i takes m copies of a three-flag tensor and retains the color words in which at most one distinct noncenter color occurs. A pure-center word has new color i ; otherwise its new color is the unique noncenter color j that occurs.

Lemma 10.2. *Completion is a local leading-term degeneration. It produces another three-flag tensor with conditional readable trees. If the old tensor contains b base factors of budget R , the new one has mb such factors and budget R^{mb} .*

Proof. Use the local tests from Lemma 3.6, with color i as center. On the center's owning side, give weight one to coordinates for which all m old center flags are on. On each noncenter's owning side, give weight one when any old flag of that color is on. All other weights are zero. For a supported color word, the total weight is

$$\mathbf{1}_{\{\text{all center}\}} + \sum_{j \neq i} \mathbf{1}_{\{\text{some } j\}}.$$

It equals one exactly on the desired words, and two if both noncenter colors occur. The same tests supply the new solo flags.

Conditional on new color $j \neq i$, the word uses only i, j and contains at least one j . The owning side of i reads its positions, and the owning side of j reads the complementary positions. Thus either side reads the complete color word. After that word is known, apply the old conditional trees successively in the slots. For new color i , the word is already determined and only the old trees are needed. All m copies are charged, including slots with deterministic color. $\square$

To terminate, delete one color using its owning flag. The choice between the two surviving colors is read by both their owning sides. Follow it with their conditional trees. This is a readable tree to which Theorem 9.2 applies.

A *script* consists of a base, a finite list of completions, and a termination pair. Its base-factor count is

$$b = \prod_{\text{completions}} m$$

(the empty product is one), and its cost before the auxiliary tensors of the readable tree are introduced is R^b . No conditional branch is exempt from this cost.

11 An exact finite-family optimization

For each completion script, the entropy bound gives many possible rate pairs, one for each leaf law. We compute their exact support function by optimizing a weighted sum of rates. A convex-geometric step then selects the best point at a prescribed rectangular shape.

Fix a distinguished side L and a parameter $t \geq 0$. For two different colors i, j , let

$$\eta_{ij} = \begin{cases} 1, & \text{if their owning sides include } L, \\ t, & \text{otherwise.} \end{cases}$$

A label read by those sides contributes with coefficient η_{ij} to the score $S + tT$.

For a conditional tree of color i , denote its maximum score by V_i . For base two, all three values start at zero. For base three they start at

$$V_A = \eta_{AC} \log 2, \quad V_B = \eta_{BC} \log 2, \quad V_C = 0. \quad (11.1)$$

A completion with center i and size m updates them simultaneously:

$$V_i^{\text{new}} = mV_i, \quad (11.2)$$

$$V_j^{\text{new}} = \eta_{ij} \log \left[\left(e^{V_i/\eta_{ij}} + e^{V_j/\eta_{ij}} \right)^m - e^{mV_i/\eta_{ij}} \right], \quad j \neq i. \quad (11.3)$$

For a termination keeping i, j , the score is

$$D_{ij} = \eta_{ij} \log \left(e^{V_i/\eta_{ij}} + e^{V_j/\eta_{ij}} \right). \quad (11.4)$$

At weight zero, these expressions mean their continuous limits. Equivalently, replace a weighted log-sum by the maximum score of the allowed words. In particular, in (11.3) the all- i word remains excluded; its zero-weight value is

$$\max_{1 \leq q \leq m} ((m - q)V_i + qV_j).$$

Definition 11.1. Let $\mathcal{F}_7$ contain both bases, every list of at most seven completions with sizes 2 or 3 and arbitrary centers, every termination pair, and every distinguished side L . Define

$$F(t) = \max_{\mathcal{F}_7} \frac{D_{ij}(t)}{b \log R}. \quad (11.5)$$

For an available square exponent estimate $p \geq w(1)$, define

$$F_p(t) = \max\{F(t), (2 + t)/p\}.$$

All ranges in this definition are finite. The recurrences use only three conditional scores per script, even though the underlying readable tree can be much larger.

The entropy maximization identity used below is the standard conjugacy between log-sum-exp and negative entropy [6, Example 3.25]. We include its elementary proof and the conditional-tree argument needed for these recurrences.

Lemma 11.2. *The values in (11.1)–(11.4) are the exact maxima of $S + tT$ over their conditional or complete leaf laws. Thus F is the support function, in direction $(1, t)$, of the downward convex hull of the normalized rate pairs $(S/(b \log R), T/(b \log R))$ from $\mathcal{F}_7$.*

Proof. For $\lambda > 0$, entropy optimization gives

$$\lambda \log \sum_d e^{a_d/\lambda} = \max_{(p_d)} \left\{ \lambda H((p_d)) + \sum_d p_d a_d \right\}. \quad (11.6)$$

Jensen's inequality applied to the logarithm proves the upper bound; the law proportional to $e^{a_d/\lambda}$ attains it.

For a noncenter completion branch, sum over all length- m words in $\{i, j\}$ other than the all- i word. For each such word the old optimal scores add across its slots. Condition on the color word and, as each slot is processed, on the histories already read in preceding slots. The remaining law in that slot is an admissible law for its old conditional tree, so its score is at most the corresponding old optimum. Summing gives the upper bound even for dependent slot laws. Independent conditional maximizing laws attain it. Equation (11.6) now gives (11.3). The center and termination formulas follow in the same way. The zero-weight cases follow by continuity on the compact leaf-law simplex.

Taking the maximum over scripts gives the claimed support values; convexification and downward closure do not change them for nonnegative weights. $\square$

The geometric step uses the supporting-halfspace description of a closed convex set [22, Theorem 8.24]. The reduction to a ray and the restriction of the parameter interval are proved below for this family. Let $\mathcal{H}_F$ be the closed downward convex hull of the normalized rate pairs in Lemma 11.2, and let $\mathcal{H}_{F_p}$ also include the square rate point $(2/p, 1/p)$ before taking that hull. These hulls are taken in the nonnegative quadrant. Their intersection with the ray $(S/C, T/C) = (2d, kd)$ determines the best balanced bound supplied by these rates.

Theorem 11.3. *For $U = F$ or $U = F_p$, and $0 \leq k \leq 1$, the largest feasible scale on this ray is*

$$d_U(k) := \max\{d \geq 0 : (2d, kd) \in \mathcal{H}_U\} = \min_{0 \leq t \leq 1} \frac{U(t)}{2 + kt}. \quad (11.7)$$

Consequently,

$$w(k) \leq \frac{1}{d_U(k)} = \max_{0 \leq t \leq 1} \frac{2 + kt}{U(t)}. \quad (11.8)$$

A maximizing parameter is the leftmost $t \in [0, 1]$ for which

$$(2 + kt)U'_+(t) \geq kU(t), \quad (11.9)$$

or $t = 1$ if no such point exists.

Proof. By Theorem 9.3, a point $(2d, kd) \in \mathcal{H}_U$ with $d > 0$ yields $w(k) \leq 1/d$. Supporting hyperplanes in nonnegative directions give

$$d_U(k) = \inf_{t \geq 0} \frac{U(t)}{2 + kt}.$$

The pure second-coordinate constraint is the limit as $t \rightarrow \infty$. One may equivalently extend the hull downward to all of $\mathbb{R}^2$; every finite supporting normal is then nonnegative. Continuity of W justifies the closed hull and boundary points.

At $t = 1$, the score is total leaf entropy and is independent of which side is distinguished. Average a maximizing construction over the three choices of that side. This gives the feasible point

$$(2U(1)/3, U(1)/3).$$

The square point already has that ratio if it is a maximizer. Consequently $U(t) \geq (2 + t)U(1)/3$, and for $t \geq 1, k \leq 1$,

$$\frac{2 + kt}{U(t)} \leq \frac{3(2 + kt)}{(2 + t)U(1)} \leq \frac{2 + k}{U(1)}.$$

The last value occurs at $t = 1$, so larger t can be omitted. The averaged point is strictly positive in both coordinates, and U is positive and continuous on $[0, 1]$. Thus the infimum is the minimum in (11.7); taking reciprocals proves (11.8).

Finally U is convex as a support function. Put $G(t) = (2 + kt)U'_+(t) - kU(t)$. For $0 \leq x < y$, the convex secant inequality gives $U(y) - U(x) \leq (y - x)U'_+(y)$, and hence

$$G(y) - G(x) \geq (2 + kx)(U'_+(y) - U'_+(x)) \geq 0.$$

Thus the right derivative of $U(t)/(2 + kt)$, whose sign is the sign of $G(t)$, changes from negative to nonnegative at the first contact in (11.9). This identifies its minimum, including endpoints, upward slope jumps and contact intervals. $\square$

The theorem supplies a finite mathematical prescription for every aspect ratio. For an active script, (11.6) gives exposed leaf laws. At a nonsmooth contact, convex combinations of points on the exposed face meet the desired ray. Coordinatewise decrease is also allowed. These laws and combinations can be approximated as in Theorem 9.2. No assertion that one center sequence is numerically optimal is needed.

Smaller script lists give valid subfamilies, while allowing more completions gives further valid bounds. The truncation to $t \leq 1$ uses all three distinguished-side choices; they should be retained when such subfamilies are formed.

12 Three parameter certificates

We now select three scripts in $\mathcal{F}_7$ and specify their leaf laws. The first two establish the dual-exponent and rectangular conclusions of Theorem 1.1. The third records the square bound $\omega < 2.267$ achieved by this finite family, independently of the stronger square construction in Theorem 1.1. The inequalities in this section use rational enclosures, not numerical feasibility assumptions.

12.1 A dual-exponent witness

Use base two, distinguished side X , and its color B . After b base factors, regard the X -coordinates as binary words of length b ; the logarithmic base cost is $b \log 2$. We seek a final leaf law for which X is uniform on all these words and $S = H(X) = b \log 2$. By Theorem 9.3 and the output-size lower bound, such a law gives

$$w(\min\{1, 2T/(b \log 2)\}) = 2.$$

The construction will preserve the exact incident-entropy equality while increasing the nonincident entropy T .

To identify what preserves this equality, let X also denote the random coordinate induced by a leaf law on a readable tree, including a tree conditional on one color. At a node v , conditioning means that the path reaches that node, and the conditional mutual information is

$$I(X; \text{child} \mid v) = H(X \mid v) - H(X \mid \text{child}, v).$$

Since a leaf determines X , successive conditioning at the nonterminal nodes telescopes to

$$H(X) = \sum_{\nu} \mathbb{P}(\nu) I(X; \text{child} \mid \nu).$$

At an incident node the child is a function of X and the prefix, so its mutual information is $H(\text{child} \mid \nu)$. Subtracting these incident contributions gives

$$H(X) - S = \sum_{\nu \text{ nonincident}} \mathbb{P}(\nu) I(X; \text{child} \mid \nu). \quad (12.1)$$

The terms on the right are nonnegative. Thus $S = H(X)$ precisely when every nonincident choice at a positive-mass node is independent of X , conditional on the preceding choices.

We maintain this equality separately within each color. Partition the binary words into an active set P and its complement Q , with the following properties:

1. color B permits precisely P , while each other color permits precisely Q ;
2. the chosen conditional law of each color makes X uniform on its permitted set;
3. its conditional incident entropy is exactly the logarithm of that set's size.

Write $\rho = |P|/2^b$, and let h_i be color i 's nonincident entropy; initially $(\rho, b, h_i) = (1/2, 1, 0)$.

Lemma 12.1. *The maintained properties are preserved by the following laws. At a completion centered on B of size m ,*

$$\rho' = \rho^m, \quad h'_B = mh_B, \quad (12.2)$$

$$h'_i = m \left(\frac{\rho - \rho'}{1 - \rho'} h_B + \frac{1 - \rho}{1 - \rho'} h_i \right) \quad (i = A, C). \quad (12.3)$$

At a completion centered on $i \in \{A, C\}$, with j the other inactive color,

$$\rho' = 1 - (1 - \rho)^m, \quad h'_B = m \left(\frac{\rho}{\rho'} h_B + \frac{\rho' - \rho}{\rho'} h_i \right), \quad (12.4)$$

$$h'_i = mh_i, \quad h'_j = \log \left[(e^{h_i} + e^{h_j})^m - e^{mh_i} \right]. \quad (12.5)$$

In both cases $b' = mb$.

Proof. At a B-centered completion, the new active set is P^m . A uniform word in its complement determines the B/ i pattern from its P/Q membership in each slot. This pattern is an incident label. Conditioning on not all slots being active, the expected proportions of old B and old i slots are

$$\frac{\rho - \rho^m}{1 - \rho^m}, \quad \frac{1 - \rho}{1 - \rho^m}.$$

Use independent old conditional laws in those slots. They give the required uniform X -law and (12.3). Their old nonincident choices retain independence of the relevant X -coordinates, so (12.1) gives the incident entropy equality.

At a completion centered on inactive i , the new inactive set is Q^m . For new color B, the B/ i pattern is again determined by X ; conditioning on at least one active slot gives the expected proportion ρ/ρ' of old B slots. This proves (12.4).

For new color j , every slot has an old color in $\{i, j\}$, with at least one j . Both old colors have exactly the same uniform X -marginal on Q . Consequently every such pattern has the same uniform marginal on Q^m , independently of the law chosen for the pattern. The pattern label is nonincident, and is independent of the entire X -word. Optimize its entropy plus the old nonincident scores by (11.6), with weight one. This gives (12.5). The old independent slot laws preserve the zero terms on the right of (12.1). Pure-center words use only the old i -law, proving the remaining formula. $\square$

Terminate by keeping B and the inactive color with larger h , placing mass ρ on B. The incident termination label and the uniform conditional laws make X uniform on all 2^b binary words. Hence

$$S = b \log 2, \quad T = bD, \quad D = \frac{\rho h_B + (1 - \rho) \max(h_A, h_C)}{b}. \quad (12.6)$$

By [Theorem 9.3](#) and the size lower bound,

$$\alpha \geq \min\{1, 2D / \log 2\}. \quad (12.7)$$

Use the following seven completions:

$$-2, \quad +2, \quad +2, \quad -3, \quad +2, \quad -2, \quad +3. \quad (12.8)$$

A minus sign means center B. A plus sign means the inactive color with larger h , with an arbitrary fixed tie choice. The absolute value is the completion size. Resolving the first tie as A, the centers are B, A, C, B, C, B, A.

For a short certificate, set $\delta = |h_A - h_C|$. A minus step leaves D unchanged and gives

$$\delta' = \frac{m(1 - \rho)}{1 - \rho^m} \delta.$$

At a plus step put

$$z = \log((1 + e^{-\delta})^m - 1).$$

Substitution in [Theorem 12.1](#) yields

$$\delta' = |z|, \quad D' = D + \frac{1 - \rho'}{mb} \max\{0, z\}. \quad (12.9)$$

Indeed h_i is the larger inactive entropy, and $h'_i = mh_i + z$, so the new maximum is $mh_i + \max\{0, z\}$.

After the first three steps, at $b = 8$,

$$\rho_8 = 1 - (3/4)^4, \quad \delta_8 = \log(9/7), \quad D_8 = \frac{0.5625}{4} \log 3.$$

The two plus steps here have $z = \log 3$ and $z = \log(7/9)$. The subsequent bounds are

b	certified information
24	$\rho_{24} = \rho_8^3, \quad 0.3194 < \rho_{24} < 0.3195, \quad 0.348 < \delta_{24} < 0.354$
48	$1 - \rho_{48} = (1 - \rho_{24})^2 > 0.463, \quad 0.536 < \rho_{48} < 0.537, \quad 0.635 < z_{48} < 0.651$
96	$\rho_{96} = \rho_{48}^2, \quad \delta_{96} = 2\delta_{48} / (1 + \rho_{48}) < 0.86$
288	$1 - \rho_{288} = (1 - \rho_{48}^2)^3 > 0.360, \quad z_{288} > 0.628.$

Their elementary verification is given in [Section 12.4](#). Since $\log 2 < 0.6932$ and $\log 3 > 1.098$,

$$\begin{aligned} \frac{2D_{288}}{\log 2} &> \frac{2}{0.6932} \left[\frac{0.5625 \cdot 1.098}{4} + \frac{0.463 \cdot 0.635}{48} + \frac{0.360 \cdot 0.628}{288} \right] \\ &= \frac{1548637}{3327360} > 0.465. \end{aligned} \quad (12.10)$$

This proves the dual-exponent conclusion without any approximate equality being used to certify exponent two.

12.2 A one-parameter rectangular curve

Use base three, one size-three completion centered on A, keep B, C, and distinguish X. This is the 75-component support of [Theorem 3.7](#); the leaf law below tunes its two balanced outer dimensions and its inner dimension. Every word in $\{0, 1, 2\}^3$ occurs on X. For a nonzero word x , its outer color is B, and the number of supported leaves in its fiber is $d(x) = 2^{\#\{\text{zero entries of } x\}}$. For the zero word, its outer color is C, and $d(0) = 3^3 - 2^3 = 19$. The fiber histogram is therefore

d	1	2	4	19
$\#\{x : d(x) = d\}$	8	12	6	1.

The total number of leaves is 75.

Give x probability proportional to $d(x)^t$, and choose a leaf uniformly within its fiber. Put

$$\begin{aligned} Z(t) &= 8 + 12 \cdot 2^t + 6 \cdot 4^t + 19^t, \\ g(t) &= (\log Z)'(t), \quad h(t) = \log Z(t) - t g(t). \end{aligned} \tag{12.11}$$

Then $H(X) = h(t)$ and $H(\text{leaf} \mid X) = g(t)$. In the nonzero branch, the A/B pattern and the choices within B are incident to X; the choices within A are hidden on Y, Z. In the zero branch all choices after branching are on Y, Z. The outer B/C label is incident to X. To check that the hidden choices do not leak information about X, fix a nonzero outer branch and its A/B pattern. Then $d(x) = 2^{\#A}$ is constant over that pattern. Each leaf has probability $d(x)^{t-1}/Z(t)$, so the hidden choices in its A-slots are independent of the remaining X-coordinates, including after any preceding slot histories. The zero branch has fixed X. The information identity (12.1) therefore gives precisely $S = h(t)$, $T = g(t)$. Its base cost is 3^3 , and [Theorem 9.3](#) gives

$$w\left(\frac{2g(t)}{h(t)}\right) \leq \frac{6 \log 3}{h(t)}. \tag{12.12}$$

Here $w(k) = W(1, k, 1)$ can also be read for $k > 1$; we use only aspect ratios in $[0, 1]$.

At $t = 2/3$, write $u = 2^{2/3}$, $v = 19^{2/3}$. Then

$$g = \frac{12u(1+u) \log 2 + v \log 19}{8 + 12u + 6u^2 + v}.$$

Cubing verifies

$$1.58739 < u < 1.58742, \quad 7.1203 < v < 7.1205, \quad 49.2878 < Z < 49.2890.$$

Using the logarithm bounds below gives

$$1.11844 < g < 1.11856, \quad 3.1519 < h < 3.1522.$$

In particular

$$\frac{2g}{h} > \frac{2 \cdot 1.11844}{3.1522} > 0.709, \quad \frac{6 \log 3}{h} < \frac{6 \cdot 1.0987}{3.1519} < 2.092.$$

Monotonicity proves $w(0.709) < 2.092$.

12.3 A finite-family square witness

Use base three and the completions

$$(B, 2), \quad (A, 3), \quad (C, 2), \quad (B, 2), \quad (A, 3),$$

then keep B, C. The number of original factors is $b = 72$. Let d_i be the number of conditional monomials of color i . Initially $(d_B, d_A, d_C) = (2, 2, 1)$. A size- m completion at i updates counts to

$$d_i^m \quad \text{at } i, \quad (d_i + d_j)^m - d_i^m \quad \text{at } j \neq i.$$

After three steps the vector is $(20691584, 13993344, 10144225)$. After four it is

$$(428141648429056, 774902581936128, 522705468255425) \geq 10^{12}(428, 774, 522)$$

coordinatewise. The final sum of the kept counts is

$$N = 2995461856366867388223311649574852466276480577,$$

and in particular

$$N \geq 10^{36}(1202^3 + 1296^3 - 2 \cdot 774^3) > 2.95 \cdot 10^{45}. \quad (12.13)$$

The substitution of lower counts is legitimate: the polynomial $(a + b)^3 + (a + c)^3 - 2a^3$ has nonnegative coefficients.

Under the uniform leaf law, the sum of the three pair rates is $\log N$. Averaging over permutations balances all three at $\log N/3$. Thus [Theorem 9.2](#) gives

$$w(1) \leq \frac{3 \cdot 72 \log 3}{\log N} < \frac{216 \cdot 1.0987}{1.0817 + 45 \cdot 2.3025} < 2.267.$$

The three constructions above yield their stated bounds once the following elementary enclosures are verified.

12.4 Elementary interval verification

For $1 \leq y \leq 2$, put $s = (y - 1)/(y + 1)$. The logarithm series has the rational enclosure

$$2 \sum_{j=0}^7 \frac{s^{2j+1}}{2j+1} \leq \log y \leq 2 \sum_{j=0}^7 \frac{s^{2j+1}}{2j+1} + \frac{2s^{17}}{17(1-s^2)}. \quad (12.14)$$

It follows by summing the positive series for $2 \operatorname{arctanh} s$ and bounding its tail geometrically. For a positive rational x , write $x = 2^a y$ with $1 \leq y \leq 2$ and use $\log x = a \log 2 + \log y$. Rational substitution in [\(12.14\)](#) gives

x	bounds for $\log x$
2	(0.69314, 0.69316)
3	(1.098, 1.0987)
19	(2.9444, 2.9445)
[49.2878, 49.2890]	(3.89765, 3.89772)
9/7	(0.250, 0.252)
10	> 2.3025
2.95	> 1.0817.

For the dual witness,

$$\delta_{24} = \frac{3(1 - \rho_8)}{1 - \rho_8^3} \log(9/7);$$

the displayed bounds for ρ_{24} and δ_{24} now use only rational arithmetic. Alternating Taylor bounds through degrees six and seven give

$$e^{-0.354} > 0.701, \quad e^{-0.348} < 0.707, \quad e^{-0.86} > 0.4225.$$

Consequently the logarithm series at

$$(1 + 0.701)^2 - 1, \quad (1 + 0.707)^2 - 1, \quad (1 + 0.4225)^3 - 1$$

proves $0.635 < z_{48} < 0.651$ and $z_{288} > 0.628$. Also

$$\delta_{96} < \frac{2 \cdot 0.651}{1 + 0.536} < 0.86.$$

The remaining ρ inequalities are direct rational powers. This verifies every inequality used in (12.10). The cube-root brackets and the expression for g similarly verify the curve certificate.

For reference only, evaluation of the exact formulas gives

quantity	approximate value
$2D_{288}/\log 2$	0.466101554898
$2g(2/3)/h(2/3)$	0.709698932519
$6 \log 3/h(2/3)$	2.091251835621
$216 \log 3/\log N$	2.266187438999.

The proofs above rely on the rational inequalities, not these decimal approximations. The accompanying certificate scripts reproduce the finite support/count calculations and all stated rational enclosures. In a fresh writable copy of `verification/certificates/`, the command `python3 checks/run_checks.py` uses integer and rational arithmetic from the Python standard library; its three checkers separately verify the numerical inequalities, finite supports and counts, and declared baseline data. The optional numerical diagnostics are not used in these certificates.

13 Comparison with ordinary combinations

Theorem 1.1 supplies the square estimate needed by **Theorem 13.3** at $p = 2.258$. We retain the general statement for any available square estimate $w(1) \leq p$ with $2 < p \leq 2.258$, together with the specified baseline and the hypotheses on further inputs below. We also give a comparison through aspect ratio 0.93 using only the finite-family witnesses of [Section 12](#) on the new-algorithm side.

We compare upper-bound estimates, not unknown exact values of W . A *cost tuple* $(a, b, c; e)$ records the certified estimate $W(a, b, c) \leq e$. The operations considered below act on such recorded estimates.

Definition 13.1. For $p > 2$, the table baseline $\mathcal{B}_p$ is generated by:

1. the numerical entries of Table 1 of [2, 33], with the exponent-two endpoint entered as $(1, 0.321334, 1; 2)$;
2. the square input $(1, 1, 1; p)$;
3. the naive matrix-multiplication estimates.

We close these inputs under permutations, scaling, tensor products, padding, ordinary blocking and convex interpolation. Ordinary subdivisions into products covering an index box may also be used. More precise input values may be substituted if their separator tests below are verified.

We use Table 1 of arXiv:2307.07970v2 and arXiv:2404.16349v3. The latter’s version 2 has the same listed rectangular entries; the updated square entry does not affect the argument. This definition makes the numerical baseline reproducible. It does not assert that the tables exhaust every possible parameter choice in the source algorithms.

13.1 A separator preserved by the baseline operations

Let

$$q_* = 0.45, \quad v_p = \frac{p-2}{1-q_*}, \quad u_p = 1 - \frac{q_* v_p}{2}, \quad 2 < p \leq 2.258. \quad (13.1)$$

Then $0 < v_p \leq u_p \leq 1$. For a triple with smallest coordinate c , put

$$P_p(a, b, c) = u_p(a + b) + v_p c. \quad (13.2)$$

Equivalently, without ordering the coordinates,

$$P_p(a, b, c) = u_p(a + b + c) + (v_p - u_p) \min(a, b, c),$$

the maximum of the three linear forms that assign v_p to one coordinate and u_p to the other two.

Lemma 13.2. *The condition $e \geq P_p(a, b, c)$ is preserved by all the operations in [Theorem 13.1](#). The square input satisfies it with equality, and the naive estimates satisfy it.*

Proof. Each of the three linear-form tests adds under tensor products, positive scaling and convex interpolation; this proves the condition for the maximum as well. It is invariant under permutations and monotone in the dimensions. Padding therefore preserves it. Increasing one dimension’s exponent by d changes P_p by at most d , because all slopes lie in $[0, 1]$; ordinary blocking adds d to the cost.

The corresponding weighted-box inequality also holds. For a box of side lengths L_i , normalize the side lengths of a contained subbox to ratios $r_i \in [0, 1]$. For any of the weight triples (u_p, u_p, v_p) , $\prod_i r_i^{w_i} \geq \prod_i r_i$. If subboxes cover the original box, their volume ratios sum to at least one. Hence the weighted size of the whole box is at most the sum of the weighted sizes of the covering boxes. Intersecting with the original box first handles boxes extending outside it. Thus summing declared costs for ordinary subdivided products cannot cross the separator either. Precisely, at scale n the tuple $(a, b, c; e)$ declares a charge n^e for a box with side lengths n^a, n^b, n^c . Apply the covering inequality to these charges before taking exponent limits; it remains valid for subdivisions depending on n .

Finally $2u_p + v_p = 2 + (1 - q_*)v_p = p$, and all weights are at most one. These give the square and naive assertions. $\square$

For any additional old input it suffices to check the plane at $p = 2.258$ together with the usual size bound. Indeed, after ordering,

$$P_p(a, b, c) = a + b + v_p(c - q_*(a + b)/2).$$

If the parenthesis is nonnegative, this increases with p . Otherwise it is at most the size bound $a + b$.

For shapes $(1, \ell, 1)$, the test at $p = 2.258$ is

$$P_{2.258}(1, \ell, 1) = \begin{cases} 2 + \frac{0.258}{0.55}(\ell - 0.45), & 0 \leq \ell \leq 1, \\ u_{2.258}(1 + \ell) + v_{2.258}, & \ell \geq 1. \end{cases} \quad (13.3)$$

If $\ell \leq 0.45$, or $\ell \geq 2/0.45 - 1$, it already follows from the size bound. Between those regimes the following downward roundings of the published entry values suffice. They are bounds on the *reported costs*, not lower bounds on matrix-multiplication exponents.

ℓ	.50	.527-.528	.55	.60	.65,.70	.75,.80	.85,.90,.95	1
entry at least	2.04	2.05	2.06	2.09	2.12	2.18	2.24	2.36

ℓ	1.10, 1.20	1.50	2	2.50	3
entry at least	2.44	2.71	3.16	3.60	4.05

For example, the newer entries at $\ell = .50, .60, .70$ are 2.042776, 2.092351, 2.152770, respectively [2]. All displayed roundings satisfy (13.3) by substitution. The other entries either lie in the automatic size-bound region or are covered by these grouped tests. It follows from Theorem 13.2 that every estimate generated by $\mathcal{B}_p$ remains on or above P_p .

13.2 The strict improvement interval

Proposition 13.3. *Assume $2 < p \leq 2.258$ and $w(1) \leq p$. Against $\mathcal{B}_p$, the new bounds are strictly better for $0.321334 < k < 1$. More generally they improve any separator-satisfying baseline throughout the part of $0 < k < 1$ on which that baseline exceeds two.*

Proof. For $k \leq 0.465$, (12.10) gives $w(k) = 2$. For $0.465 < k < 1$, convexity and the available square point give

$$w(k) \leq 2 + \frac{p-2}{1-0.465}(k-0.465).$$

The difference between the separator at $(1, k, 1)$ and this line is

$$\frac{(p-2)(1-k)(0.465-0.45)}{(1-0.45)(1-0.465)} > 0. \quad (13.4)$$

This proves the general assertion.

It remains to identify the exponent-two range of the specified table baseline, including limits of ordinary combinations. Put $q_0 = 0.321334$. For each of the finitely many table inputs, order its dimensions so that c is smallest and write

$$d = c - q_0(a+b)/2, \quad E = e - a - b.$$

Every entry has $E \geq 0$, and every entry with $d > 0$ has $E > 0$: the only table entries attaining the size bound have $2c/(a+b) \leq q_0$. Choose $\eta > 0$ such that

$$\eta \leq \frac{1}{2}, \quad \eta \leq \frac{p-2}{1-q_0}, \quad \eta \leq \frac{E}{d} \quad \text{for every table entry with } d > 0.$$

Such a choice exists because the list is finite and all the relevant ratios are positive. Set $u_0 = 1 - q_0\eta/2$. Then $0 < \eta \leq u_0 \leq 1$, and every table input satisfies

$$e \geq u_0(a+b) + \eta c = a + b + \eta d.$$

For $d \leq 0$ this follows from the size bound; for $d > 0$ it follows from the choice of η . The square input satisfies the same test since $2u_0 + \eta = 2 + (1 - q_0)\eta \leq p$; naive inputs also satisfy it. The proof of Theorem 13.2 applies verbatim to the weights (u_0, u_0, η) , including its weighted-box argument for subdivisions. Every baseline estimate at $(1, k, 1)$, and every limit of such estimates, is therefore at least

$$2 + \eta(k - q_0) > 2 \quad (k > q_0).$$

The recorded endpoint and padding give cost two for $k \leq q_0$. Thus the baseline has flat endpoint exactly q_0 , and the general assertion gives the claimed open interval. $\square$

The endpoint at one is excluded because the comparison uses the same square point on both sides. By [Theorem 1.1](#), $w(1) < 2.258$, so [Theorem 13.3](#) applies at $p = 2.258$ and proves strict improvement over $\mathcal{B}_{2.258}$ throughout $0.321334 < k < 1$. If a more precise old exponent-two endpoint is supplied, the same statement uses that endpoint in place of the rounded table entry. No strict improvement is asserted where the baseline already gives two.

There is also a comparison using only the constructions in $\mathcal{F}_7$. Interpolate their three proved points

$$(0.465, 2), \quad (0.709, 2.092), \quad (1, 2.267).$$

Against the numerical $p = 2.258$ separator, the comparisons at $k = 0.465, 0.709, 0.93$ are

k	0.465	0.709	0.93
new line	2	2.092	2.224903780 ...
separator	2.007036363 ...	2.121494545 ...	2.225163636

Both differences are affine on the intervening segments. The smallest endpoint margin is greater than 0.0002598; these inequalities also follow directly from rational arithmetic. Thus this finite family improves that numerical baseline through $k = 0.93$, in addition to the flat-range improvement.

13.3 A sufficient test for further old estimates

The table comparison can be extended without evaluating a large nonconvex parameter program, provided the extra input estimates satisfy an explicit condition. Let f be the convex upper-bound envelope from such old estimates *before* adding the new square point. Suppose

$$f(.50) \geq 2.035, \quad f(.60) \geq 2.087, \quad (13.5)$$

and include the known estimates

$$f(.40) \leq 2.010, \quad f(.50) \leq 2.043, \quad f(.60) \leq 2.093, \quad f(.70) \leq 2.154.$$

Condition (13.5) concerns the specified envelope of estimates. The published record values alone do not prove it for every possible untabulated parameter choice.

Proposition 13.4. *Under (13.5), every such additional input satisfies the separator test.*

Proof. If an input $(a, b, c; e)$, with c smallest, violated the test, average its two large axes and normalize by $(a + b)/2$. It would give a violating balanced input at $x = 2c/(a + b) \in [0, 1]$. The case $x \leq .45$ is excluded by the size bound. For the rest, convexity extends a chord beyond its endpoints as a lower bound. Using a lower bound at the nearer endpoint and an upper bound at the farther one gives

x interval	lower bound for $f(x)$	gap above $2 + \frac{.258}{.55}(x - .45)$ at endpoints	
$[\cdot45, \cdot50]$	$2.035 + .58(x - .50)$	3/500	127/11000
$[\cdot50, \cdot55]$	$2.035 + .25(x - .50)$	127/11000	13/22000
$[\cdot55, \cdot60]$	$2.087 + .67(x - .60)$	29/4400	183/11000
$[\cdot60, 1]$	$2.087 + .44(x - .60)$	183/11000	1/200.

The gap is affine and positive at both endpoints in every row, contradicting the hypothetical violation. $\square$

This is a sufficient inclusion rule for further baseline inputs, not a claim about all future analyses of the source tensors. For an enlarged baseline, [Theorem 13.3](#) gives strict improvement wherever that baseline exceeds two in $0 < k < 1$. The comparison with $\mathcal{B}_{2.258}$ uses the proved dual-exponent witness together with [Theorem 1.1](#); the extension in [Theorem 13.4](#) still requires (13.5). The finite-family numerical bounds were established independently in [Section 12](#).

Scope of the result. The family yields asymptotic arithmetic exponents with arbitrarily small positive slack. It does not claim a practical crossover size or global optimality of the chosen seven-completion family. The finite-family entropy argument and parameter certificates are independent of the square construction in [Theorem 1.1](#). In the rectangular support bound and this comparison, an available square estimate enters through its explicitly charged exponent point $(2/p, 1/p)$.

14 Finite-exception characteristic transfer

For a field F and $0 \leq k \leq 1$, let $w_F(k)$ denote the arithmetic exponent for the dimensions $(n, \lceil n^k \rceil, n)$, defined as in the introduction but counting operations over F and allowing constants from F . Put $\omega_F = w_F(1)$ and $\alpha_F = \sup\{k \in [0, 1] : w_F(k) = 2\}$. The constants in operation-count bounds may depend on the field and the fixed schemes below; coefficient bit sizes and numerical stability are not part of this model.

Corollary 14.1 (Finite-exception field bounds). *There is a finite set $\mathcal{S}$ of positive primes such that, for every field F with $\text{char } F = 0$ or $\text{char } F = p \notin \mathcal{S}$,*

$$\omega_F < \frac{1129}{500} = 2.258, \quad w_F\left(\frac{709}{1000}\right) < \frac{523}{250} = 2.092.$$

The same set $\mathcal{S}$ works for both inequalities and depends on two fixed exact rank schemes selected in the proof.

Proof. First select the schemes over $\mathbb{C}$. The strict square and rectangular bounds in [Theorem 1.1](#), together with the rank limit in [Lemma 2.1](#), give fixed exact rank schemes for $\langle m, m, m \rangle$ and $\langle a, b, a \rangle$, with respective rank budgets r_{sq} and r_{rec} , such that

$$r_{\text{sq}}^{500} < m^{1129}, \quad r_{\text{rec}}^{250} < a^{523}, \quad r_{\text{rec}}^{709} < b^{2092}. \quad (14.1)$$

Indeed, choose finite values $s_{\text{sq}}, s_{\text{rec}} > 0$ at which the rank ratios in that lemma for $(1, 1, 1)$ and $(1, 709/1000, 1)$ are respectively below $1129/500$ and $523/250$. Take $m = \lceil e^{s_{\text{sq}}} \rceil$, $a = \lceil e^{s_{\text{rec}}} \rceil$, and $b = \lceil e^{709s_{\text{rec}}/1000} \rceil$. Their lower bounds by the unrounded quantities give (14.1). Thus the two strict margins concern only fixed positive integers; no limiting degeneration is being specialized.

Write each rank-one term as a product of three linear forms and regard all their coefficients in both schemes as indeterminates. For $j \in \{\text{sq}, \text{rec}\}$, let M_j be the corresponding matrix tensor. Coefficient matching consists of finitely many equations

$$\sum_{v=1}^{r_j} \xi_{j,v,x} \eta_{j,v,y} \zeta_{j,v,z} = (M_j)_{xyz} \in \{0, 1\}.$$

The ideal I generated by these equations in a polynomial ring $\mathbb{Q}[\mathbf{u}]$ is proper because the selected complex schemes give a joint solution. Choose a maximal ideal containing I . Its residue field K is a finite extension of $\mathbb{Q}$ by the Hilbert Nullstellensatz [31, Theorem 10.34.1, Tag 00FV]. The images of the indeterminates give both exact schemes over this one number field. Individual summands are allowed to vanish: only the upper rank budgets matter, so no nonvanishing constraint is needed.

Fix a $\mathbb{Q}$ -basis $e_1 = 1, e_2, \dots, e_d$ of K . Choose an integer $D \geq 1$ clearing the rational coordinates of all selected coefficients in this basis and the structure constants of all products $e_i e_j$. With $R = \mathbb{Z}[1/D]$,

$$\mathcal{A} = \bigoplus_{i=1}^d R e_i \subset K$$

is then a unital R -algebra, free of rank d as an R -module, containing the coefficients of both identities. Let $\mathcal{S}$ be the finite set of prime divisors of D .

For any field F in the statement, the canonical map $R \rightarrow F$ is defined. The algebra $\mathcal{A} \otimes_R F$ is a nonzero unital F -algebra of dimension d . A quotient by a maximal ideal is therefore a finite field extension E/F of degree at most d . The map $\mathcal{A} \rightarrow E$ specializes both exact tensor identities with their dimensions and rank budgets unchanged. In positive characteristic this is reduction modulo that characteristic followed by a finite extension; when $F = \mathbb{F}_p$, the quotient E is a finite residue field.

It remains to measure operations over F , not E . The recursive blocking and padding upper bound in the proof of Lemma 2.1 uses only an exact rank scheme and is valid over E . Set

$$\sigma = \min \left\{ \log a, \frac{1000}{709} \log b \right\} > 0.$$

The integer inequalities (14.1) give

$$\frac{\log r_{\text{sq}}}{\log m} < \frac{1129}{500}, \quad \frac{\log r_{\text{rec}}}{\sigma} < \frac{523}{250}.$$

For the rectangular scheme, depth $q = \lceil \log n / \sigma \rceil$ provides outer dimension at least n and inner dimension at least $n^{709/1000}$, at cost $O((q+1)r_{\text{rec}}^q)$ operations over E ; the square scheme has the analogous bound with $q = \lceil \log n / \log m \rceil$. Run each whole recursive computation on F -valued inputs inside E , and express every addition and multiplication in a fixed F -basis of E containing 1. The multiplication table simulates each such operation with a fixed number of F -operations, independent of n ; projection onto the coordinate of 1 recovers the F -valued output. This is a constant factor on the whole operation count, not a new factor in the rank budget at every tensor level. The displayed strict exponent bounds therefore hold over F . $\square$

The argument neither computes $\mathcal{S}$ nor covers its primes. It applies only to the two fixed schemes chosen above.

Corollary 14.2 (Characteristic-zero dual bound). *For every field F of characteristic zero, $\alpha_F > 0.465$.*

Proof. By Theorem 1.1 and the definition of α , fix $k \in (0.465, 1]$ with $w(k) = 2$. For each $\varepsilon > 0$, Lemma 2.1 gives a finite $t > 0$ and an exact rank- r scheme for $\langle a, b, a \rangle$, where

$$a = \lceil e^t \rceil, \quad b = \lceil e^{kt} \rceil, \quad \frac{\log r}{t} < 2 + \frac{\varepsilon}{2}.$$

The coefficient-equation descent in the proof of Corollary 14.1 realizes this finite scheme over a number field K_ε . A maximal-ideal quotient of $K_\varepsilon \otimes_{\mathbb{Q}} F$ is a finite field extension E/F over which the same exact rank budget holds.

Recursive blocking and padding at depth $h = \lceil \log n / t \rceil$ cover dimensions $(n, \lceil n^k \rceil, n)$. As in Lemma 2.1, its operation count over E is

$$O((h+1)r^h) = O(\log n n^{\log r/t}) = O(n^{2+\varepsilon}).$$

For this fixed accuracy, simulate the whole circuit in a fixed F -basis of E containing 1, as in Corollary 14.1; this changes the operation count by a constant factor and recovers the F -valued outputs. The number field, basis and constants may depend on ε and F , but not on n . Thus $w_F(k) \leq 2$; the output-size lower bound gives equality, and $\alpha_F \geq k > 0.465$. $\square$

The accuracy-dependent schemes in this corollary do not supply a common finite set of exceptional positive characteristics. No transfer of $\alpha_F > 0.465$ to positive characteristic is asserted.

A Complete barrier coefficients

The following table completes the rational definition (7.16). Every displayed decimal, including those in scientific notation, is exact. The terms are listed in increasing i and then increasing j ; no further coefficient file or optimizer is needed to specify the barrier.

Table 1: Exact coefficients a_{ij} of the barrier.

i	j	a_{ij}
0	0	$1.098619383270680538 \times 10^0$
1	1	$-4.669442318024533134 \times 10^{-2}$
2	2	$-1.207736428642643568 \times 10^{-3}$
3	0	$-3.164446199207870698 \times 10^{-3}$
3	3	$3.636957171420279137 \times 10^{-3}$
4	1	$-5.698642428249523157 \times 10^{-4}$
4	4	$-5.602684605882590763 \times 10^{-2}$
5	2	$2.349877147912796810 \times 10^{-3}$
5	5	$4.903987018649487162 \times 10^{-1}$
6	0	$-6.619810434108410737 \times 10^{-4}$
6	3	$-1.285233398877164493 \times 10^{-2}$
6	6	$-2.581864001585385981 \times 10^0$
7	1	$-7.562439376470945357 \times 10^{-3}$
7	4	$-1.804691803008195983 \times 10^{-1}$
7	7	$7.823901000581444443 \times 10^0$
8	2	$1.009956438054349509 \times 10^{-1}$
8	5	$2.660260871536313410 \times 10^0$
8	8	$-1.115071770757571024 \times 10^1$
9	0	$1.882526999774491094 \times 10^{-3}$
9	3	$-6.610003066665722793 \times 10^{-1}$
9	6	$-1.377536704067965623 \times 10^1$
9	9	$2.196480182366505329 \times 10^{-1}$
10	1	$-8.740622126988427387 \times 10^{-2}$
10	4	$1.686430306261204937 \times 10^0$
10	7	$3.198040977700396681 \times 10^1$
10	10	$1.193163475520329975 \times 10^1$
11	2	$8.227116674939661811 \times 10^{-1}$
11	5	$1.713324848501641684 \times 10^0$
11	8	$-2.478214558914840637 \times 10^1$
11	11	$4.571667456003526420 \times 10^{-1}$
12	0	$1.863768838404195050 \times 10^{-2}$
12	3	$-3.639145985809583372 \times 10^0$
12	6	$-1.659314443855491561 \times 10^1$
12	9	$-9.109093370890851915 \times 10^0$
13	1	$-2.842047435780501452 \times 10^{-1}$
13	4	$6.864502967053094373 \times 10^0$
13	7	$2.217022739027584421 \times 10^1$
14	2	$1.793237083428060208 \times 10^0$
14	5	$-8.565115586979872297 \times 10^{-1}$
14	8	$1.830109002841796784 \times 10^0$
15	0	$4.474789932033190060 \times 10^{-3}$
15	3	$-5.076662205259581562 \times 10^0$
15	6	$-9.377190570431521266 \times 10^0$
16	1	$-1.401322773505324082 \times 10^{-1}$
16	4	$5.019408834309503575 \times 10^0$

Table 1: Exact coefficients a_{ij} of the barrier (continued).

i	j	a_{ij}
17	2	$8.749190654179410664 \times 10^{-1}$
17	5	$1.353757704697627240 \times 10^0$
18	0	$-2.244840043150760772 \times 10^{-2}$
18	3	$-1.595227841739267349 \times 10^0$
19	1	$4.228839868431957111 \times 10^{-2}$
20	2	$1.304516353614300073 \times 10^{-1}$
21	0	$-1.101896459249583106 \times 10^{-2}$

B Reproducing the finite certificate

The accompanying certificate verifies the finite arithmetic statements used in [Theorem 7.1](#). The tensor constructions, the passage from grid checks to continuum inequalities, and the error allowances used in that passage are proved in the body of the manuscript.

Inputs and source identity. The directory `verification/certificate/` contains the 52 exact decimal coefficients of the barrier, both integer coefficient charts, all numerical generators and checkers, and their reference outputs. The packet manifest records the size and SHA-256 of every source and input file. The README describes the package, and the supplemental checker verifies the additional arithmetic bounds used in this manuscript.

The two chart hashes are

```
qP420_gmp:
7dc037ea127cece2216f3219eeeca4cdab0288319a10111a78d272586b776f0b
qP780_gmp:
f23c0480f1109509545fdd17bb4de81602f50952411554b337105defbe7a070c
```

The barrier coefficient file, `inputs/newc22`, has SHA-256

```
41d42a58936b9da0f01a98a24fc4e998a00b83c9f4032a4ae166b7ef11073912
```

The working-copy path substitutions made by the driver are recorded separately from the source hashes. They change input locations and compiler selection, not arithmetic formulas or thresholds.

Running the check. A standard run requires Python 3.10 or later, a POSIX system supporting `fork`, a C++17 compiler with signed 128-bit integer support and UndefinedBehaviorSanitizer, and the GMP C++ library for chart generation. A preflight checks the additional assumptions of a 64-bit `long` and arithmetic right shift of signed negative integers. Assertions must remain enabled. From `verification/certificate/`, run

```
python3 reproduce.py \
--cxx 'g++ -fsanitize=undefined -fno-sanitize-recover=all' \
--work-dir ../../certificate-run
python3 verify_manuscript_bounds.py --work-dir ../../certificate-run
```

A compatible Clang installation may be selected instead. The commands make no network requests or package installations. They write into a new output directory and refuse to overwrite an existing run or supplemental report. The optional flag `--reuse-charts` uses the supplied chart bytes after hash verification; omitting it regenerates both charts.

This distinction matters mathematically. The low-degree residual calculation uses the rounding-defect bound for the coefficient recurrence. Matching a stored hash alone does not independently establish that recurrence. Both full charts were regenerated with GMP integer arithmetic, with fail-fast undefined-behavior checks, and found to agree byte-for-byte with the supplied inputs. Separate small-degree comparisons also matched the GMP generator against the Python recurrence. The computation of the residual and chart differences then uses exact integer upper bounds.

Finite coverage. The complete run passed all stages listed below. The preserved command records include compiler flags, exit codes, source substitutions, hashes, and logs.

Check	Coverage
Chart generation and norms	Degrees 420 at scale 2^{224} and 780 at scale 2^{500} ; coefficient norms, tails, residual and chart differences.
Boundary shell	All 2,161 endpoints of its 2,160 sampling intervals.
Outer boundary	All 65,536 coarse intervals; 8,192 subcells in each of the six ambiguous intervals.
Interior	All 128 angular sectors, with 128 initial radial cells in each; all recursively required child boxes.
Geometry	All $480 \times 480 = 230,400$ grid centers.
Finite tensor examples	9,837 COMPLETE words, 27 PARITY words, five edge atoms, and 40,087 GROUP tangent pairs.

The finite tensor examples check selected identities; their general forms are proved earlier. For the interior run, the sector identifiers were independently checked to be exactly $0, \dots, 127$, without omissions or duplicates. Each saved sector-input hash agreed with the corresponding reference hash, and every checker process returned success. The interval checker rejects an unresolved box at its maximum depth rather than accepting it.

The reproduced boundary output is

```
h -4180063084
b 0 65536 amb 6 terms 7250 edge 128315602072 cap 152568202355
```

In particular, the actual edge-sample minimum is

$$\frac{128315602072}{2^{52}} > 28.49 \cdot 10^{-6}.$$

The boundary proof uses this stronger measured minimum when paying its conservative endpoint and displacement allowances. It does not replace it by the checker's weaker acceptance threshold $12 \cdot 10^{-6}$. A supplemental execution also asserted that each endpoint gap used in a nonambiguous boundary interval had actually been evaluated; its full output was unchanged.

The interface with the analytic estimates. The rational certificate gives

$$\varepsilon < 3.329 \cdot 10^{-22}, \quad \frac{6\varepsilon}{2 \cdot 10^{-9}} < 10^{-12}, \quad \varphi(0) - \log 3 > 7.0946 \cdot 10^{-6}.$$

The logarithm comparison uses a finite positive series for $2 \operatorname{atanh}(1/2)$ and an exact geometric remainder. The physical fourth-derivative coefficient sum is also checked exactly to be less than $435 \cdot 10^6$.

The final-summary program takes the analytic interior margin $9.8 \cdot 10^{-8}$, boundary margin $5.5 \cdot 10^{-6}$, and inset cost $1.3 \cdot 10^{-6}$ as inputs; it does not prove those estimates. The supplemental program `verify_manuscript_bounds.py` explicitly records this interface, checks the actual boundary minimum and the exact allowance subtractions, and rechecks the complete interior-sector identities and hashes. The justification of each allowance, and hence of the continuum conclusion, is the analytic argument of [Theorem 7.1](#). These programs are not a proof-assistant formalization and do not claim a practical matrix-multiplication implementation.

References

- [1] Josh Alman and Virginia Vassilevska Williams. A refined laser method and faster matrix multiplication. *TheoretCS*, 3:21, 2024. doi: 10.46298/theoretcs.24.21. URL <https://theoretcs.episciences.org/14213>. Article 21.
- [2] Josh Alman, Ran Duan, Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou. More asymmetry yields faster matrix multiplication. In *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 2005–2039. Society for Industrial and Applied Mathematics, 2025. doi: 10.1137/1.9781611978322.63. URL <https://arxiv.org/abs/2404.16349v3>. Updated arXiv version 3, August 19, 2026; Table 1 incorporates the later Dupont et al. square bound.
- [3] Andris Ambainis, Yuval Filmus, and François Le Gall. Fast matrix multiplication: Limitations of the laser method. Report TR14-154, Electronic Colloquium on Computational Complexity, 2014. URL <https://eccc.weizmann.ac.il/report/2014/154/download>. 20 November 2014.
- [4] F. A. Behrend. On sets of integers which contain no three terms in arithmetical progression. *Proceedings of the National Academy of Sciences of the United States of America*, 32(12):331–332, 1946. doi: 10.1073/pnas.32.12.331. URL <https://doi.org/10.1073/pnas.32.12.331>.
- [5] Dario Bini. Relations between exact and approximate bilinear algorithms. applications. *Calcolo*, 17(1):87–97, 1980. doi: 10.1007/BF02575865. URL <https://doi.org/10.1007/BF02575865>.
- [6] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. doi: 10.1017/CBO9780511804441. URL https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf.
- [7] Henry Cohn and Christopher Umans. A group-theoretic approach to fast matrix multiplication. In *Proceedings of the 44th Annual IEEE Symposium on Foundations of Computer Science*, pages 438–449, 2003. doi: 10.1109/SFCS.2003.1238217. URL <https://arxiv.org/abs/math/0307321v2>.
- [8] Don Coppersmith. Rapid multiplication of rectangular matrices. *SIAM Journal on Computing*, 11(3):467–471, 1982. doi: 10.1137/0211037. URL <https://doi.org/10.1137/0211037>.
- [9] Don Coppersmith. Rectangular matrix multiplication revisited. *Journal of Complexity*, 13(1):42–49, 1997. doi: 10.1006/jcom.1997.0438. URL <https://doi.org/10.1006/jcom.1997.0438>.

- [10] Don Coppersmith and Shmuel Winograd. Matrix multiplication via arithmetic progressions. *Journal of Symbolic Computation*, 9(3):251–280, 1990. doi: 10.1016/S0747-7171(08)80013-2. URL [https://doi.org/10.1016/S0747-7171\(08\)80013-2](https://doi.org/10.1016/S0747-7171(08)80013-2).
- [11] Michael G. Crandall and Pierre-Louis Lions. Viscosity solutions of Hamilton–Jacobi equations. *Transactions of the American Mathematical Society*, 277(1):1–42, 1983. doi: 10.1090/S0002-9947-1983-0690039-8. URL <https://doi.org/10.1090/S0002-9947-1983-0690039-8>.
- [12] Michael G. Crandall, Hitoshi Ishii, and Pierre-Louis Lions. User’s guide to viscosity solutions of second order partial differential equations. *Bulletin of the American Mathematical Society*, 27(1):1–67, 1992. doi: 10.1090/S0273-0979-1992-00266-5. URL <https://arxiv.org/abs/math/9207212v1>.
- [13] Imre Csiszár. The method of types. *IEEE Transactions on Information Theory*, 44(6):2505–2523, 1998. doi: 10.1109/18.720546. URL <https://doi.org/10.1109/18.720546>.
- [14] Ran Duan, Hongxun Wu, and Renfei Zhou. Faster matrix multiplication via asymmetric hashing, 2023. URL <https://arxiv.org/abs/2210.10173v5>. FOCS 2023; arXiv version 5, November 28, 2023.
- [15] Emilien Dupont, Marvin Eisenberger, Borislav Kozlovskii, Abbas Mehrabian, Francisco J. R. Ruiz, Abigail See, Renfei Zhou, Josh Alman, Virginia Vassilevska Williams, and Matej Balog. Improving the matrix multiplication exponent with modern optimization and AlphaEvolve. arXiv:2608.16884v1, 2026. URL <https://arxiv.org/abs/2608.16884v1>. 17 August 2026.
- [16] Oscar H. Ibarra, Shlomo Moran, and Roger Hui. A generalization of the fast LUP matrix decomposition algorithm and applications. *Journal of Algorithms*, 3(1):45–56, 1982. doi: 10.1016/0196-6774(82)90007-4. URL <https://csaws.cs.technion.ac.il/~moran/r/PS/LUP82.pdf>.
- [17] François Le Gall. Faster algorithms for rectangular matrix multiplication. In *Proceedings of the 53rd Annual IEEE Symposium on Foundations of Computer Science*, pages 514–523, 2012. doi: 10.1109/FOCS.2012.80. URL <https://arxiv.org/abs/1204.1111v2>.
- [18] François Le Gall. Powers of tensors and fast matrix multiplication. In *Proceedings of the 39th International Symposium on Symbolic and Algebraic Computation*, pages 296–303, 2014. doi: 10.1145/2608628.2608664. URL <https://arxiv.org/abs/1401.7714v1>.
- [19] François Le Gall and Florent Urrutia. Improved rectangular matrix multiplication using powers of the Coppersmith–Winograd tensor. In *Proceedings of the 29th ACM–SIAM Symposium on Discrete Algorithms*, pages 1029–1046, 2018. doi: 10.1137/1.9781611975031.67. URL <https://arxiv.org/abs/1708.05622v2>.
- [20] Grazia Lotti and Francesco Romani. On the asymptotic complexity of rectangular matrix multiplication. *Theoretical Computer Science*, 23(2):171–185, 1983. doi: 10.1016/0304-3975(83)90054-3. URL [https://doi.org/10.1016/0304-3975\(83\)90054-3](https://doi.org/10.1016/0304-3975(83)90054-3).
- [21] Kyoko Makino and Martin Berz. Taylor models and other validated functional inclusion methods. *International Journal of Pure and Applied Mathematics*, 4(4):379–456, 2003. URL <https://www.bmtdynamics.org/pub/papers/TMIJPAM03/TMIJPAM03.pdf>.

- [22] R. Tyrrell Rockafellar and Roger J.-B. Wets. *Variational Analysis*, volume 317 of *Grundlehren der mathematischen Wissenschaften*. Springer, 1998. doi: 10.1007/978-3-642-02431-3. URL <https://sites.math.washington.edu/~rtr/papers/rtr169-VarAnalysis-RockWets.pdf>.
- [23] R. Salem and D. C. Spencer. On sets of integers which contain no three terms in arithmetical progression. *Proceedings of the National Academy of Sciences of the United States of America*, 28(12):561–563, 1942. doi: 10.1073/pnas.28.12.561. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC1078539/>.
- [24] Arnold Schönhage. Partial and total matrix multiplication. *SIAM Journal on Computing*, 10(3):434–455, 1981. doi: 10.1137/0210032. URL <https://doi.org/10.1137/0210032>.
- [25] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x. URL <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [26] Alexey Solov'yev and Thomas C. Hales. Formal verification of nonlinear inequalities with taylor interval approximations. In *NASA Formal Methods*, volume 7871 of *Lecture Notes in Computer Science*, pages 383–397. Springer, 2013. doi: 10.1007/978-3-642-38088-4_26. URL <https://arxiv.org/abs/1301.1702v1>.
- [27] Andrew James Stothers. *On the complexity of matrix multiplication*. PhD thesis, University of Edinburgh, 2010. URL <https://era.ed.ac.uk/handle/1842/4734>.
- [28] Volker Strassen. Gaussian elimination is not optimal. *Numerische Mathematik*, 13(4):354–356, 1969. doi: 10.1007/BF02165411. URL <https://doi.org/10.1007/BF02165411>.
- [29] Volker Strassen. Vermeidung von divisionen. *Journal für die reine und angewandte Mathematik*, 264:184–202, 1973. doi: 10.1515/crll.1973.264.184. URL <https://doi.org/10.1515/crll.1973.264.184>.
- [30] Volker Strassen. Relative bilinear complexity and matrix multiplication. *Journal für die reine und angewandte Mathematik*, 375/376:406–443, 1987. doi: 10.1515/crll.1987.375-376.406. URL <https://doi.org/10.1515/crll.1987.375-376.406>.
- [31] The Stacks Project Authors. Hilbert nullstellensatz. The Stacks Project, Theorem 10.34.1, Tag 00FV, 2026. URL <https://stacks.math.columbia.edu/tag/00FV>. Accessed September 27, 2026.
- [32] Virginia Vassilevska Williams. Multiplying matrices faster than Coppersmith–Winograd. In *Proceedings of the 44th Annual ACM Symposium on Theory of Computing*, pages 887–898, 2012. doi: 10.1145/2213977.2214056. URL <https://people.csail.mit.edu/virgi/matrixmult-f.pdf>. Corrected full version, entitled Multiplying matrices in $O(n^{2.373})$ time, dated July 1, 2014.
- [33] Virginia Vassilevska Williams, Yinzhao Xu, Zixuan Xu, and Renfei Zhou. New bounds for matrix multiplication: from alpha to omega, 2023. URL <https://arxiv.org/abs/2307.07970v2>. SODA 2024; arXiv version 2, November 4, 2023.